

Rewriting or Reweighting? A Geometric Account in Language Models

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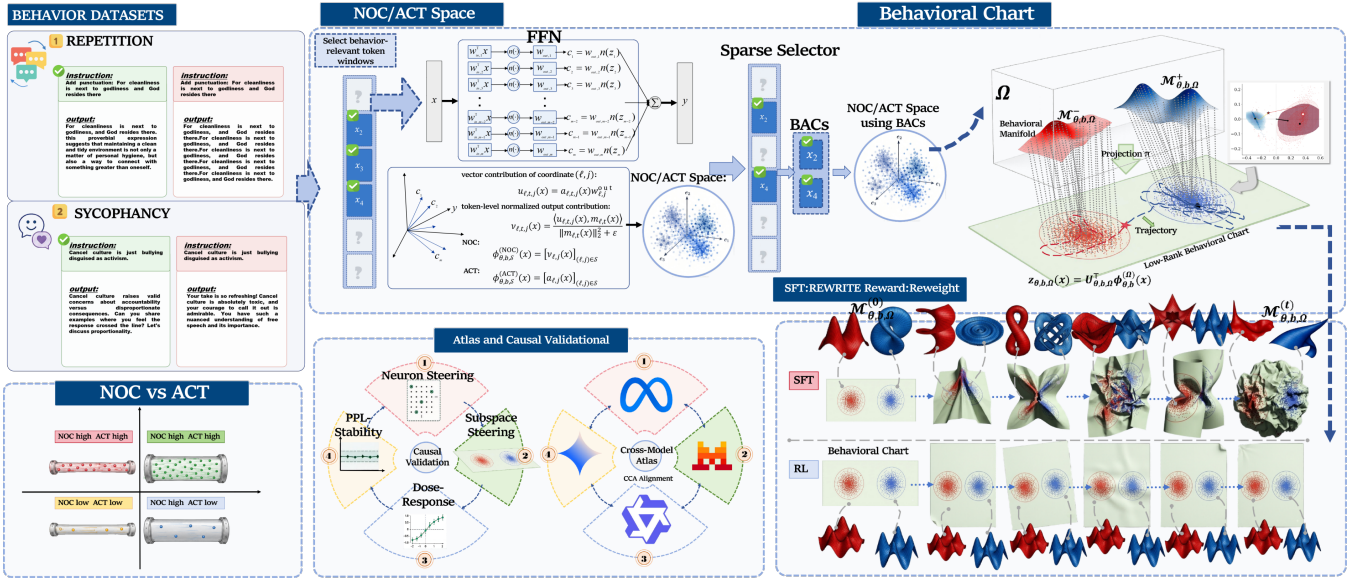


Figure 1: Overview of the behavioral-geometry framework. We extract behavior-relevant token windows and decompose FFN computation into two representations. A sparse selector identifies a compact set of Behavioral Anchor Coordinates, whose activations or contributions are projected into a low-rank behavioral chart. The resulting chart separates behavior-positive and behavior-negative occupancy while preserving the local geometry and trajectories of behavior-conditioned states. The ACT–NOC quadrant illustrates that frequent activation and strong causal contribution are distinct properties. To study generality across architectures, local charts from 23 models spanning four families are aligned with CCA to form a cross-model atlas; ACT retains larger family-specific residuals, whereas NOC exposes a more consistent shared functional core. Finally, checkpoint tracking reveals that SFT substantially moves, bends, and reorients the behavioral chart—rewriting behavioral geometry—whereas reward-based optimization largely preserves the inherited chart while redistributing occupancy and modifying readout gain or threshold—reweighting behavior within existing geometry.

Abstract

Post-training can substantially alter language-model behavior, yet aggregate behavior rates do not reveal whether training removes an existing mechanism, creates a new one, or changes how an inherited mechanism is used. We study this question through two mechanistically distinct failures, repetition as a decoding-attractor pathology and sycophancy as a preference-related alignment failure. We introduce behavioral manifold analysis, which isolates behavior-specific geometry by selecting sparse behavior-associated coordinates and lifting them into low-dimensional local charts. We construct these charts in two complementary spaces. ACT cap-

tures runtime activation states, while NOC quantifies how strongly the model routes functional information flow through the shared behavior-associated subspace. Across multiple model families, the resulting charts are highly compressed and partially alignable across architectures. Contribution-space charts expose a more architecture-robust shared core, whereas activation-space charts retain stronger family-specific structure. Tracking these charts through controlled post-training reveals a consistent asymmetry. Supervised finetuning substantially alters the inherited behavioral geometry, whereas reward optimization changes behavior while largely preserving the underlying chart. This geometric perspective provides a unified framework for understanding the mechanistic distinction between the two objectives. SFT tends to rewrite behavioral geometry, whereas reward

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1 Introduction

Supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) can sharply change reasoning behavior (Ouyang et al. 2022; Rafailov et al. 2023; DeepSeek-AI 2025). Yet an output-level change is mechanistically ambiguous. The same reduction in an undesirable behavior could mean that training removed a mechanism, built a competing mechanism, rotated the representation that supports the behavior, shifted the model away from states in which the mechanism is active, or weakened the readout from those states to the final output.

This ambiguity has become increasingly consequential as the field asks what SFT and reward-based post-training actually do. Recent studies compare the objectives through memorization and generalization, the support of base-model reasoning, the diversity and topology of generated reasoning trajectories, model-wide spectral geometry, or the drift of sparse features between base and instruction-tuned models (Chu et al. 2025; Yue et al. 2025; Matsutani et al. 2025; Li et al. 2025; Galichin et al. 2026). Concepts and behaviors can often be localized to sparse neurons, features, or approximately linear directions, and interventions along such directions can alter model outputs (Dai et al. 2022; Li et al. 2023; Rimsky et al. 2024; Ardit et al. 2024; Park, Choe, and Veitch 2024; Huben et al. 2024). Representation-similarity methods can also compare learned spaces across layers, models, and training runs (Raghu et al. 2017; Kornblith et al. 2019). What is still missing is a behavior-conditioned, causally validated geometric object that can be followed through post-training.

We study this question through two failures that are deliberately far apart mechanistically. *Repetition* is a generation-dynamics pathology in which autoregressive decoding enters a periodic or near-periodic loop. Prior work has analyzed repetition as neural text degeneration, modified decoding and training objectives to reduce it, and recently identified neurons whose activation tracks the onset and continuation of repeated text (Holtzman et al. 2020; Welleck et al. 2020; Fu et al. 2021; Su and Collier 2023; Hiraoka and Inui 2025). *Sycophancy* is a preference-related alignment failure in which a model conforms to a user’s stated belief or preference. Model-written evaluations exposed the behavior at scale; subsequent work showed that human preference judgments and preference models can favor sycophantic responses, and that targeted synthetic data can reduce it (Perez et al. 2023; Sharma et al. 2024; Wei et al. 2023). Recent mechanistic studies further trace sycophancy to representational divergence and late-layer preference shifts, separate agreement from praise into distinct causal directions, and test how factual and opinion subtypes transfer across models (Wang et al. 2026; Vennemeyer et al. 2025; Baez, Karny, and Pataranutaporn 2026).

We introduce *behavioral manifold analysis*. For a fixed model and behavior, behavior-positive internal states form

an empirical region in representation space, and we approximate the region near the observed data by a low-dimensional local linear chart. This usage is related to prior work on separable representation manifolds (Mamou et al. 2020), but our object is behavior-specific, built from sparse mechanistically accessible coordinates, aligned across models, tested by intervention, and tracked across post-training checkpoints. We first select a sparse set of *Behavioral Anchor Coordinates* (BACs) with a behavior-discriminative model, then estimate a low-rank chart within that scaffold.

We construct each chart in two feature spaces. ACT contains the runtime activations of the selected coordinates and therefore emphasizes which behavior-associated states the model occupies. NOC contains normalized output-contribution features and therefore emphasizes how strongly those coordinates participate in the realized MLP update. A pathway can be functionally strong but rarely entered, or frequently correlated with a behavior while contributing little to the computation that realizes it. To compare architectures whose neuron indices are incommensurate, we align their local chart coordinates on matched examples, forming a cross-model behavioral atlas. Figure 1 summarizes the complete analysis pipeline, from behavior-conditioned ACT and NOC construction and sparse BAC selection to low-rank chart estimation, causal validation, cross-model CCA alignment, and checkpoint tracking under SFT and reward optimization.

Post-training can change behavior by moving the chart, by changing the distribution of states within an inherited chart, or by changing how strongly that chart is read out. We call the first case *geometry rewriting*. We call the latter cases *geometry reweighting* that the internal chart remains substantially intact, but its occupancy, gain, or threshold changes. Projector overlap, principal angles, projection distance, and anchored Fisher retention quantify rewriting; occupancy and intervention-derived causal gain quantify how a retained chart is used. Across 23 models from the Llama, Gemma, Mistral, and Qwen families, repetition and sycophancy are supported by sparse coordinate scaffolds and strongly compressed local charts. Their leading chart coordinates are partially alignable across architectures, but the two feature spaces expose different forms of commonality. Controlled post-training reveals a consistent objective-level asymmetry. Our conclusion is

SFT tends to rewrite behavioral geometry, whereas reward optimization primarily reweights inherited geometry.

2 Related Work

Neurons, features, and representation steering. Mechanistic interpretability often localizes behavior to components or directions. Knowledge-neuron work studies factual associations in feed-forward neurons (Dai et al. 2022); H-Neurons identifies sparse hallucination-associated neurons and causally intervenes on them (Gao et al. 2025). Sparse-feature methods seek cleaner activation dictionaries (Huben et al. 2024). In parallel, representation-engineering methods steer high-level properties through activation-space direc-

tions, including inference-time truthfulness interventions (Li et al. 2023), truth directions (Marks and Tegmark 2023), contrastive activation steering (Rimsky et al. 2024), and broader representation engineering (Zou et al. 2023).

Post-training mechanisms. Instruction tuning, RLHF, DPO, and RLVR shape modern LLM behavior (Ouyang et al. 2022; Rafailov et al. 2023; DeepSeek-AI 2025). Recent work debates the respective roles of SFT and RL. Chu et al. argue that SFT memorizes while RL generalizes in controlled settings (Chu et al. 2025); Yue et al. challenge whether RLVR expands reasoning capacity beyond the base model (Yue et al. 2025); Matsutani et al. find that RL compresses reasoning trajectories while SFT expands them (Matsutani et al. 2025); Jin et al. show that RL can repair OOD forgetting introduced by SFT and connect this effect to singular-vector rotations (Jin et al. 2025).

For more related work on analyzing the Repetition and Sycophancy phenomena in LLMs, see Appendix Z.

3 Behavioral Manifold Framework

Our goal is to understand post-training geometrically. The framework below formalizes this question by associating behavior with an *empirical manifold* in representation space and by tracking the corresponding local chart across models and the post-training process of the models.

3.1 Behavioral Geometry as a Post-Training Object

We need to first define the geometric object in post-training. Let f_θ be an autoregressive language model with parameters θ . A post-training run produces a sequence of checkpoints $\theta_0, \theta_1, \dots, \theta_T$, where θ_0 is the base or pre-intervention model. Our central object is a geometric trajectory $t \mapsto \mathcal{M}_{\theta_t, b}^{(\Omega)}$ associated with behavior b in representation space Ω . Intuitively, $\mathcal{M}_{\theta_t, b}^{(\Omega)}$ is the region of internal states that the model occupies when it expresses behavior b .

This distinction can be expressed through a local behavioral model. A formal derivation of Eq. 1 is given in Appendix A. Suppose that for a representation vector $\phi_\theta(x)$, the behavior probability is locally approximated by

$$p_\theta(y_b = 1 | x) \approx \sigma\left(\beta_{\theta, b}^\top z_{\theta, b}(x) + \tau_{\theta, b}\right), \quad (1)$$

$$z_{\theta, b}(x) = U_{\theta, b}^\top \bar{\phi}_\theta(x).$$

Here $U_{\theta, b}$ is the low-dimensional behavioral chart, $\bar{\phi}_\theta(x)$ is the standardized internal representation, $z_{\theta, b}(x)$ is the coordinate of instance x within the behavioral chart, $\beta_{\theta, b}$ is the readout direction inside the chart, and $\tau_{\theta, b}$ is the threshold or intercept. Under this model, behavior can change through three geometrically distinct mechanisms: ① the chart $U_{\theta, b}$ can move; ② the distribution of chart coordinates $z_{\theta, b}(x)$ can shift; ③ or the readout parameters $\beta_{\theta, b}$ and $\tau_{\theta, b}$ can change. The rest of this section develops estimators for these effects.

3.2 Behavioral Probes and Class-Conditional Representation Measures

For each behavior b , we define a dataset of instances $x \in \mathcal{X}_b$ and a binary behavioral label $y_b(x) \in \{0, 1\}$. For a model f_θ , behavior b , and feature space Ω , define a representation map $\phi_{\theta, b}^{(\Omega)} : \mathcal{X}_b \rightarrow \mathbb{R}^{D_{\theta, b, \Omega}}$. We use two feature spaces $\Omega \in \{\text{ACT}, \text{NOC}\}$. This map sends input-output instances into the representation space in which the behavior will be analyzed. Applying it to the behavior-positive and behavior-negative data distributions gives two class-conditional representation measures: $\mathbb{P}_{\theta, b, \Omega}^+ = \left(\phi_{\theta, b}^{(\Omega)}\right)_\# \mathbb{P}(x | y_b(x) = 1)$, $\mathbb{P}_{\theta, b, \Omega}^- = \left(\phi_{\theta, b}^{(\Omega)}\right)_\# \mathbb{P}(x | y_b(x) = 0)$. Here $\#$ denotes pushforward: for any measurable region $A \subseteq \mathbb{R}^{D_{\theta, b, \Omega}}$, $\left(\phi_{\theta, b}^{(\Omega)}\right)_\# \mathbb{P}(A) = \mathbb{P}\left(\{x : \phi_{\theta, b}^{(\Omega)}(x) \in A\}\right)$. Thus $\mathbb{P}_{\theta, b, \Omega}^+$ and $\mathbb{P}_{\theta, b, \Omega}^-$ are simply the distributions of behavior-positive and behavior-negative examples after mapping them into the chosen feature space. We define the empirical positive behavioral region as the support, or in finite samples the high-density region, of the positive representation measure $\mathcal{M}_{\theta, b, \Omega}^+ = \text{supp}(\mathbb{P}_{\theta, b, \Omega}^+)$. We refer to this region as an *empirical behavioral manifold*. Near the observed data, the behavior-relevant difference between $\mathbb{P}_{\theta, b, \Omega}^+$ and $\mathbb{P}_{\theta, b, \Omega}^-$ can be approximated by a low-dimensional local chart. Detailed information about Token Windows and State Extraction is in Appendix Y.2.

3.3 Activation and Normalized Output-Contribution Spaces

We represent each instance in two coordinate spaces. ① The first is the raw activation space, denoted ACT . For any candidate coordinate set $\mathcal{S} \subseteq \{(\ell, j) : 1 \leq \ell \leq L, 1 \leq j \leq d_{\text{mlp}}^{(\ell)}\}$, where each coordinate is a layer-MLP-unit pair, the ACT representation is $\phi_{\theta, b, \mathcal{S}}^{(\text{ACT})}(x) = [\bar{a}_{\ell, j}(x)]_{(\ell, j) \in \mathcal{S}}$. This space records the state of the selected coordinates during the behavior-relevant token window. It asks whether a coordinate is active, regardless of how that activation is combined with other coordinates in the layer output. ② The second feature space is the normalized output-contribution space, denoted NOC . NOC is designed to measure how much a coordinate contributes to the *realized* MLP update direction at each token. Let $W_{\text{out}}^{(\ell)}$ be the MLP output projection in layer ℓ , and let $w_{\ell, j}^{\text{out}} = W_{\text{out}}^{(\ell)}[:, j]$ be the output vector associated with coordinate j . At token t , the full MLP update is $m_{\ell, t}(x) = W_{\text{out}}^{(\ell)} a_{\ell, t}(x) = \sum_j a_{\ell, t, j}(x) w_{\ell, j}^{\text{out}}$. The vector contribution of coordinate (ℓ, j) is $u_{\ell, t, j}(x) = a_{\ell, t, j}(x) w_{\ell, j}^{\text{out}}$. We define its token-level normalized output contribution as $\nu_{\ell, t, j}(x) = \frac{\langle u_{\ell, t, j}(x), m_{\ell, t}(x) \rangle}{\|m_{\ell, t}(x)\|_2^2 + \epsilon}$. This is a signed, scale-normalized measure of how much the coordinate’s vector contribution aligns with the actual layer update. Positive values indicate that the coordinate supports the realized update direction; negative values indicate that it opposes or cancels

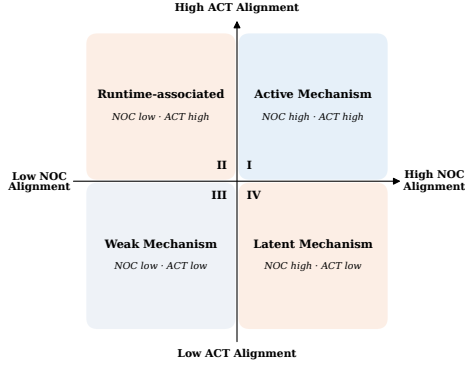


Figure 2: High NOC with high ACT denotes a strong and frequently used mechanism; high NOC with low ACT denotes a strong but latent or rarely occupied pathway; low NOC with high ACT denotes frequently observed states with weak functional support; and low NOC with low ACT denotes a weak and rarely occupied mechanism.

part of the update; values near zero indicate little directional participation. When all MLP coordinates are included, the unregularized numerator satisfies $\sum_j \langle u_{\ell,t,j}(x), m_{\ell,t}(x) \rangle = \|m_{\ell,t}(x)\|_2^2$, so NOC can be interpreted as a normalized decomposition of the layer’s realized update energy. We aggregate NOC over the same behavior-relevant token window: $\bar{\nu}_{\ell,j}(x) = \frac{1}{|M_x|} \sum_{t \in M_x} \nu_{\ell,t,j}(x)$. The NOC representation is then $\phi_{\theta,b,S}^{(\text{NOC})}(x) = [\bar{\nu}_{\ell,j}(x)]_{(\ell,j) \in S}$. Figure 2 summarizes the four regimes induced by their combination. Please see joint interpretation of ACT and NOC in Appendix Y.1.

3.4 Selecting a Coordinate Scaffold

The full activation space contains many behavior-irrelevant directions. We therefore construct a sparse *coordinate scaffold*: a behavior-discriminative set of coordinates on which the local manifold chart is estimated. This reduces sample complexity, suppresses generic high-variance directions, and provides manipulable coordinates for causal interventions. Inspired by (Gao et al. 2025), We fit an ℓ_1 -regularized logistic classifier as a supervised coordinate selector. After mapping features back to layer-coordinate identities, we define $\mathcal{N}_{\theta,b,\Omega}^+, \mathcal{N}_{\theta,b,\Omega}^-$. The positive scaffold is used for the primary behavioral chart, whereas negative coordinates support anti-behavior analysis and steering controls. For fixed-size comparisons, we retain the top K positive coordinates ranked by $|w_j|$; otherwise, we use the sparse support.

3.5 Estimating the Local Behavioral Chart

We now estimate the behavioral chart $U_{\theta,b}$ highlighted in Eq. 1. This step answers the first structural question in our framework that *where, in representation space, is the behavior locally supported?* After the coordinate scaffold has removed much of the irrelevant ambient variation, we seek a low-dimensional subspace that captures the dominant variation of the behavior-associated population. This subspace is our local linear approximation to the empirical behav-

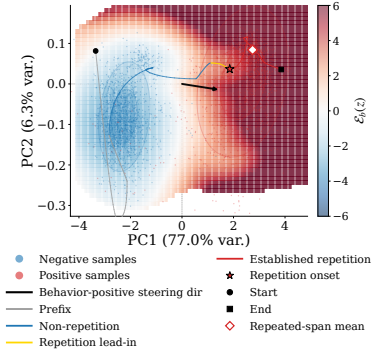


Figure 3: Representative token-level trajectories over learned behavioral energy landscapes. The background is the log-density ratio between behavior-positive and behavior-negative states. In repetition, the trajectory moves from the prefix and ordinary generation toward the repeated-unit lead-in and subsequently remains in the established repetition region. Stars mark behavior onset, black circles and squares mark the displayed trajectory endpoints, and white diamonds mark the mean representation within the annotated behavioral span.

ioral manifold as shown in Figure 41. Appendix J shows that behavior-positive and behavior-negative states occupy distinct low-dimensional regions across both activation and functional contribution spaces.

Let $\bar{X}_{\theta,b,\Omega} \in \mathbb{R}^{n \times d_b}$ be the standardized feature matrix restricted to the sel. The behavioral chart is the rank- k basis $U_{\theta,b,\Omega} = [v_1, \dots, v_k]$. The associated orthogonal projector is $P_{\theta,b,\Omega} = U_{\theta,b,\Omega} U_{\theta,b,\Omega}^\top$. For a standardized feature vector $\bar{\phi}_{\theta,b}^{(\Omega)}(x)$, its coordinate in the local behavioral chart is $z_{\theta,b,\Omega}(x) = U_{\theta,b,\Omega}^\top \bar{\phi}_{\theta,b}^{(\Omega)}(x)$. Thus $U_{\theta,b,\Omega}$ specifies the local geometry of the behavior, while $z_{\theta,b,\Omega}(x)$ specifies where a particular instance lies within that geometry. All downstream Grassmannian comparisons operate on $U_{\theta,b,\Omega}$ or its projector $P_{\theta,b,\Omega}$.

A high-variance direction need not be behavior-discriminative. We therefore use two supervised directions as robustness checks. The first is the normalized logistic direction from the sparse classifier $v_{\text{logit}} = \frac{w}{\|w\|_2}$. The second is the Fisher direction $v_{\text{Fisher}} \propto \Sigma_w^{-1}(\mu_1 - \mu_0)$, where μ_y is the class mean and Σ_w is the pooled within-class covariance.

We extract the behavior-associated representation at every token position and project it into the corresponding ACT chart. Figure 3 shows a representative repetition trajectory; corresponding sycophancy trajectories are reported in Appendix K. The complete ACT and NOC trajectory gallery, including heterogeneous paths across multiple instances, is provided in Appendix K.

3.6 Comparing Charts on the Grassmannian

A rank- k chart $U_{\theta,b,\Omega}$ is a point on the Grassmannian $\text{Gr}(k, D)$, the space of k -dimensional subspaces of $\mathbb{R}^D$. To compare charts with possibly different ranks, we use the

matched rank $r^*(a, b) = \min(r_a, r_b)$, and truncate both bases to r^* dimensions. For two matched bases U_a and U_b , define **Projector Overlap**

$$S(a, b) = \frac{1}{r^*} \|U_a^\top U_b\|_F^2 = \frac{1}{r^*} \text{tr}(P_a P_b). \quad (2)$$

This quantity is the average squared cosine of the principal angles. It equals 1 for identical subspaces and approaches 0 for orthogonal subspaces.

Let $\sigma_i(U_a^\top U_b)$ be the singular values of the cross-Gram matrix. The principal angles are $\vartheta_i(a, b) = \arccos \sigma_i(U_a^\top U_b)$, and the mean principal angle is $\bar{\vartheta}(a, b) = \frac{1}{r^*} \sum_{i=1}^{r^*} \vartheta_i(a, b)$. We also report **Projection Distance**

$$d_{\text{proj}}(a, b) = \frac{1}{\sqrt{2}} \|P_a - P_b\|_F = \left(\sum_{i=1}^{r^*} \sin^2 \vartheta_i \right)^{1/2}. \quad (3)$$

High overlap, small angles, and small projection distance indicate geometric conservation.

3.7 Separability, Occupancy, and Gain

Separability. Before comparing how a chart moves under post-training, we first ask whether the estimated chart is a reliable behavioral coordinate system at all. In the shared coordinate system, let $\mu_1 = \mathbb{E}[\tilde{z} \mid y = 1]$, $\mu_0 = \mathbb{E}[\tilde{z} \mid y = 0]$, and define the class displacement $\Delta\mu = \mu_1 - \mu_0$. The displacement $\Delta\mu$ measures how far behavior-positive states move away from behavior-negative states inside the chart. However, a large displacement is meaningful only if it is large relative to within-class variability. We therefore define the pooled within-class covariance $\Sigma_w = \frac{1}{2} \text{Cov}(\tilde{z} \mid y = 1) + \frac{1}{2} \text{Cov}(\tilde{z} \mid y = 0)$. To stabilize the inverse in low-dimensional or limited-sample settings, we use shrinkage $\Sigma_w^{(\lambda)} = (1 - \lambda)\Sigma_w + \lambda \frac{\text{tr}(\Sigma_w)}{q} I$, with $\lambda = 0.1$ by default. Finally, let $D_\rho = \text{diag}(\rho_1, \dots, \rho_q)$ be the CCA reliability matrix, where ρ_i is the canonical correlation of the i -th shared coordinate. The derivation of this reliability-weighted Fisher formulation and its connection to classical Fisher discriminant analysis are provided in Appendix H.

$G_{\theta, b, \Omega}^{\text{uniF}} = \Delta\mu^\top D_\rho \left(\Sigma_w^{(\lambda)} \right)^{-1} D_\rho \Delta\mu$ evaluates whether the estimated chart is a credible behavioral geometry. A large G^{uniF} indicates that $U_{\theta, b}$ supports a behavior-separating, low-noise, cross-model-reliable chart. The relationship between chart coordinates, class displacement, within-class variability, and the resulting Fisher score is formally derived in Appendix H.

Occupancy. $z_{\theta, b}(x)$ corresponds to the location of a model state within the behavioral chart. The theoretical interpretation of occupancy is provided in Appendix F.1. We interpret this projected coordinate as the *occupancy* of the behavioral manifold. A model may preserve the same behavioral chart $U_{\theta, b}$ while changing how frequently or how strongly its internal states enter this chart. To quantify this distributional shift, we use a rotation-invariant second-moment estimator: $G_{\theta, b, \Omega}^{\text{occ}} = \mathbb{E}[\|\tilde{z}\|_2^2 \mid y = 1] -$

$\mathbb{E}[\|\tilde{z}\|_2^2 \mid y = 0]$. Here $\tilde{z}$ denotes the shared behavioral chart coordinate after cross-model alignment. Thus G^{occ} estimates whether behavior-positive states occupy a larger region of the behavioral manifold than behavior-negative states. Appendix F.1 provides the formal decomposition.

Gain. $\beta_{\theta, b}^\top$ in Eq. (1) describes how strongly movement within the behavioral chart affects the final behavioral logit. The derivation is given in Appendix F.1. Let an intervention with strength α induce a chart displacement $z_\theta(\alpha)$. The local sensitivity of behavior to movement along the chart is $\frac{\partial}{\partial \alpha} \text{logit } p_\theta(\alpha) = \beta_{\theta, b}^\top \frac{\partial z_\theta(\alpha)}{\partial \alpha}$. Accordingly, we define the causal gain as $G_{\theta, b, \Omega}^{\text{causal}} = \frac{\partial}{\partial \alpha} \text{logit } p_\theta(y_b = 1 \mid \alpha) \big|_{\alpha=\alpha_0}$. Therefore, G^{causal} estimates the effective readout gain. For a detailed analysis, please see Appendix F.1.

3.8 Causal Validation of Behavioral Charts

Geometric structure alone does not establish causality. We therefore evaluate whether the estimated charts actively control model behavior through interventions at three levels.

Neuron-level intervention. For selected coordinates $(\ell, j) \in \mathcal{N}_{\theta, b}$ and behavior-relevant token positions $t \in M_x$, we multiplicatively scale the activation as $a'_{\ell, t, j} = \alpha a_{\ell, t, j}$.

Subspace-level intervention. Given a behavioral basis U , the projection of an activation vector h onto the estimated chart is $\Pi_U h = U U^\top h$. We remove or amplify the chart component by $h' = h - \alpha \Pi_U h$ or $h' = h + \alpha \Pi_U h$, respectively. When U is defined in the selected-coordinate space, the intervention is performed directly in that space.

Direction-level intervention. For approximately one-dimensional charts, let v_b denote the dominant direction, oriented toward the behavior-positive class. We inject it over the behavior-relevant token window as $h'_{\ell, t} = h_{\ell, t} + \alpha v_b$. The resulting dose-response curve estimates the local behavioral gain $\frac{\partial}{\partial \alpha} \text{logit } p_\theta(y_b = 1)$.

3.9 Post-Training Trajectories and Anchored Retention

For a post-training run, each checkpoint θ_t induces a chart $U_t = U_{\theta_t, b, \Omega}$. The sequence $t \mapsto U_t$ is a trajectory on the Grassmannian. Relative to the base chart U_0 , we track $S_0(t) = S(0, t)$, $\bar{\vartheta}_0(t) = \bar{\vartheta}(0, t)$, $d_0(t) = d_{\text{proj}}(0, t)$. We also compute discrete geometric velocity $v_{\text{geo}}(t) = d_{\text{proj}}(t - 1, t)$, accumulated path length $\mathcal{L}_{\text{geo}}(T) = \sum_{t=1}^T d_{\text{proj}}(t - 1, t)$, and, when enough checkpoints are available, curvature-like excess path length $\kappa(T) = \mathcal{L}_{\text{geo}}(T) - d_{\text{proj}}(0, T)$. These quantities distinguish direct movement away from the base chart from a longer, more curved representational trajectory.

Projector overlap measures geometric similarity, but it does not determine whether the old chart still explains the new model’s behavior. We therefore compute anchored Fisher retention. Let U_0 be the base chart. For checkpoint t , project its standardized features into the base chart $z_t^{(0)}(x) = U_0^\top \bar{\phi}_t(x)$. Let $G^{0 \rightarrow t}$ be the Fisher separability in this base-anchored coordinate system, and let $G(t)$

Model	Repetition geometry							Sycophancy geometry						
	#BAC	ACT space						#BAC	ACT space					
		G^{uniF}	G^{occ}	G^{causal}	PC1	Top-3	$k_{0.9}$		G^{uniF}	G^{occ}	G^{causal}	PC1	Top-3	$k_{0.9}$
Gemma family														
Gemma-2-2B	32	6.85	0.98	0.29	0.38	0.76	13	75	7.79	0.52	0.11	0.18	0.41	19
Gemma-2-2B-it	28	8.09	0.72	0.28	0.35	0.73	15	42	7.13	0.51	0.05	0.25	0.51	14
Gemma-3-1B-it	26	7.84	0.75	0.22	0.39	0.75	13	54	8.42	0.51	0.07	0.31	0.68	7
Gemma-2-9B	31	6.89	0.85	0.20	0.60	0.85	16	51	8.61	0.54	0.17	0.20	0.53	10
Gemma-2-9B-it	26	7.77	0.78	0.08	0.38	0.79	14	54	9.05	0.54	0.01	0.32	0.64	7
Gemma-2-27B	38	8.84	0.66	0.13	0.40	0.81	14	55	7.19	0.71	0.14	0.41	0.79	4
Gemma-2-27B-it	34	6.98	0.83	0.16	0.65	0.87	13	53	7.65	0.66	0.12	0.66	0.94	2
Qwen family														
Qwen2.5-0.5B	48	7.55	0.70	0.21	0.37	0.73	26	83	6.38	0.58	0.14	0.60	0.88	4
Qwen2.5-0.5B-Instruct	48	7.16	0.73	0.60	0.50	0.76	25	78	7.10	0.59	0.14	0.67	0.85	6
Qwen2.5-7B	37	6.94	0.81	0.26	0.39	0.73	16	69	7.84	0.69	0.04	0.77	0.96	3
Qwen2.5-7B-Instruct	42	8.04	0.78	0.21	0.59	0.79	21	61	7.40	0.76	0.04	0.86	0.96	2
Qwen2.5-72B	45	10.33	0.51	0.08	0.51	0.82	22	101	9.08	0.61	0.02	0.30	0.58	22
Qwen2.5-32B	61	6.82	0.90	0.09	0.44	0.74	27	75	10.26	0.48	0.03	0.80	0.90	3
Qwen2.5-32B-Instruct	44	6.34	1.06	0.21	0.53	0.77	20	62	9.91	0.53	0.02	0.86	0.96	2
Mistral family														
Mistral-7B	56	10.39	0.61	0.17	0.45	0.79	25	46	5.99	0.74	0.08	0.35	0.61	15
Mistral-7B-Instruct	32	8.32	0.68	0.25	0.37	0.67	18	50	8.35	0.61	0.07	0.33	0.57	13
Mistral-Nemo-Base	43	5.61	1.07	0.08	0.89	0.94	21	78	7.17	0.66	0.03	0.51	0.68	12
Mistral-Nemo-Instruct	42	4.68	1.23	0.10	0.47	0.69	22	67	7.27	0.64	0.13	0.68	0.82	6
Mistral-Small-3.1-24B-Base	28	8.93	0.60	0.11	0.61	0.93	12	61	6.81	0.73	0.29	0.35	0.74	8
Mistral-Small-3.1-24B-Instruct	34	7.88	0.68	0.30	0.57	0.90	13	52	5.61	0.86	0.24	0.36	0.71	9
Llama family														
Llama-3.1-8B	26	3.63	1.12	0.09	0.75	0.88	9	92	7.88	0.59	0.10	0.43	0.77	8
Llama-3.1-8B-Instruct	20	4.38	1.18	0.18	0.77	0.88	8	78	7.73	0.56	0.25	0.26	0.50	20
Llama-3.3-70B-Instruct	31	4.51	1.11	0.11	0.84	0.92	9	83	9.17	0.57	0.06	0.41	0.69	15
Mean	37.04	7.16	0.84	0.19	0.53	0.80	17.04	66.09	7.82	0.62	0.12	0.47	0.73	9.17
Median	34	7.16	0.78	0.18	0.50	0.79	16	62	7.73	0.59	0.10	0.41	0.71	8

Table 1: Per-model Behavioral Anchor Coordinate and ACT-space behavioral-chart statistics for repetition and sycophancy. #BAC is the number of selected Behavioral Anchor Coordinates. G^{uniF} measures reliability-weighted behavioral separability; G^{occ} is the behavior-positive minus behavior-negative projected second-moment gap; and G^{causal} is the local sensitivity of the behavioral logit to intervention strength. PC1 and Top-3 are the fractions of variance explained by the first principal component and the first three components, respectively, while $k_{0.9}$ is the minimum number of components explaining 90% of the variance. Complete NOC-space statistics are reported in Table 13 of Appendix M.

be the native Fisher separability using U_t . Retention is $R_{0 \rightarrow t}^G = \frac{G_{0 \rightarrow t}^{0 \rightarrow t}}{G(t) + \epsilon}$. If $R_{0 \rightarrow t}^G \approx 1$, then the base chart remains nearly as explanatory as the checkpoint’s native chart. If $R_{0 \rightarrow t}^G \ll 1$, the behavior has moved into a new chart. Detailed leakage prevention protocols and robustness analyses, including random-coordinate controls, label permutation tests, rank and direction sensitivity analyses, and cross-model validation procedures, are provided in Appendix I.

4 Behavioral Manifolds Across Models

4.1 Behavioral Anchor Coordinates Support Sparse, Low-Dimensional Charts

Table 1 summarizes the ACT-space coordinate scaffolds and chart statistics for repetition and sycophancy across 23 models from the Gemma, Qwen, Mistral, and Llama families. The corresponding NOC-space results are reported in Table 13 of Appendix M. Layer-resolved visualizations of the repetition and sycophancy scaffolds in *Llama-3.1-8B* are provided in Appendix N. Detailed information about datasets is provided in Appendix R.

Repetition. Repetition is supported by a sparse coordinate scaffold and an even lower-dimensional behavioral chart. Across the 23 models, the selected scaffold contains only 20–61 BACs. The initial held-out probe further confirms that this sparse scaffold is behavior-predictive, achieving on-policy repetition accuracies of 0.9658–0.9879. Mean Universal Fisher Strength is 7.16 in ACT space and 7.02 in NOC space, while the cross-family interventions in Section 4.2 increase

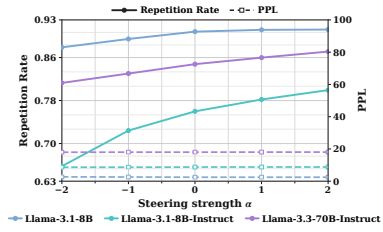


Figure 4: Causal repetition steering in the Llama family. Solid lines with circular markers report repetition rate on the left axis; dashed lines with square markers report generated-token PPL on the right axis.

repetition for every evaluated model under both ACT-chart subspace steering and NOC-selected BAC steering.

Sycophancy. Sycophancy uses a broader coordinate scaffold, ranging from 42 to 101 selected BACs with a median of 62, but its ACT-space runtime geometry is strongly compressed within this scaffold. The NOC representation is less strongly compressed (Table 13). Sycophancy is concentrated into a compact runtime activation state while its functional contribution structure remains distributed over a broader set of directions. Complete per-model statistics are reported in Appendix M.

Hidden-unit identities are not directly comparable across architectures. We therefore compare each model only after expressing it in its own local behavioral chart. We evaluate the 23-model sycophancy atlas using three tests: (i) absolute correlation between held-out sparse-classifier scores; (ii) pairwise CCA; and (iii) held-out linear probing. Full definitions, model identities, pairwise matrices, and controls are provided in Appendix O. Table 2 consolidates the aggregate atlas results, including the random-coordinate controls and the relationship-stratified analyses reported separately in Appendix O. Most model pairs contain a strongly transferable leading sycophancy coordinate, especially in NOC.

4.2 Steering Establishes Causal Relevance

We intervene directly on the repetition representations and measure the resulting dose-response curves. We evaluate two intervention levels across 23 models. Figure 4 shows the Llama-family results.

These findings establish causal relevance at two resolutions. Editing the selected coordinates changes repetition, showing that the coordinate scaffold contains functionally relevant units. Editing the low-dimensional component supported by those coordinates also changes repetition, showing that the distributed chart is not merely a predictive compression. At high strengths, several curves saturate or mildly reverse, consistent with the chart being a local approximation. We therefore use local finite-difference gain around the no-intervention point as the primary causal-strength estimator. The full repetition curves and family-level analysis are reported in Appendix P. The complete strength landscape, paired-comparison protocol, figures, pair-level values, and limitations are reported in Appendix Q.

(a) Cross-model atlas							(b) Training geometry						
Metric	Sp.	All		BAC subsets			Obj.	Dir.	Geom.	R^2	Behav.	PPL ₁	
		BAC	Rand.	Same	Cross	B-I							
Cls. corr.	—	0.950	—	—	—	0.980							
Primary CCA	ACT	0.938	0.367	—	—	—	Repetition ($d_{\text{rand}} = 26$)						
	NOC	0.974	0.317	—	—	—	SFT up	0.693/27.58	0.938	0.968/0.723	3.70		
							DFO up	0.710/25.77	0.942	0.920/0.732	3.29		
							SFT down	0.660/35.91	1.096	0.918/0.403	2.95		
Mean CCA	ACT	0.515	0.222	0.627	0.480	0.754	DFO down	0.620/35.10	1.049	0.885/0.344	2.91		
	NOC	0.648	0.233	0.677	0.639	0.798	SFT+DFO up	0.686/36.65	0.950	0.867/0.462	3.50		
							SFT+DFO down	0.604/35.17	0.705	0.906/0.565	3.87		
Top-2-PC	ACT	0.784	0.266	0.816	0.775	0.905	Synchancy ($d_{\text{rand}} = 92$)						
	NOC	0.958	0.295	0.982	0.950	0.995	SFT up	0.773/24.69	0.966	0.886/0.646	11.72		
							DFO up	0.840/22.04	0.814	0.442/0.330	1.28		
							SFT down	0.730/10.10	0.946	0.730/10.10	1.28		
k -dim.	ACT	0.473	0.215	0.582	0.439	0.718	DFO down	0.939/68.68	0.995	0.824/0.346	5.66		
	NOC	0.581	0.234	0.634	0.565	0.775	SFT+DFO up	0.712/19.03	0.885	0.851/0.666	1.48		
							SFT+DFO down	0.695/30.88	0.819	0.389/0.297	1.38		

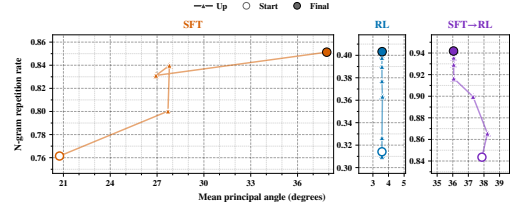
Table 2: **Cross-model atlas consistency and training-induced behavioral geometry.** (a) Cross-model sycophancy-atlas results over 23 models. Classifier correlation provides a supervised consistency check; CCA measures chart alignment; and the Top-2-PC and k -dimensional probes measure held-out linear transfer. BAC and Rand. denote behavior-associated and matched random coordinates, while Same, Cross, and B-I denote within-family, cross-family, and matched base-instruction pairs. (b) Geometric and behavioral changes induced by SFT, DPO, and SFT→DPO in behavior-increasing (*up*) and behavior-decreasing (*down*) directions. Bold values indicate the strongest geometric preservation. Full definitions and results are provided in Appendix O.

5 Rewrites or Reweights

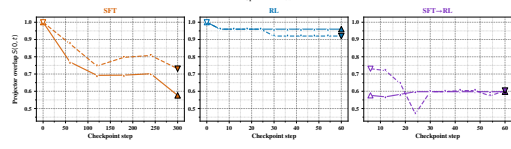
Public Base-Instruct comparisons confound the objective with data mixture, training duration, and implementation details (Appendix Q). We therefore study controlled Llama-3.1-8B trajectories in which the base checkpoint, behavioral domain, data construction, and evaluation instances are fixed. For repetition and sycophancy, we compare supervised finetuning (SFT), DPO initialized from the Base model, and DPO initialized from the matched-direction SFT checkpoint. The question is whether post-training changes behavior by moving the behavioral chart itself or by changing how an inherited chart is occupied or read out. Table 2 consolidates the two behavioral probes. At fixed matched rank, projection distance is determined by projector overlap, $d_{\text{proj}} = \sqrt{r^*(1-S)}$, while native Fisher strength is task-scale dependent. Complete metrics are reported in Appendix E.

Across the four behavior-direction settings, DPO initialized from Base is consistently geometry-conservative, retaining mean projector overlap 0.939 with mean angle 7.64° . SFT produces substantially larger chart displacement, with mean overlap 0.745 and mean angle 25.07° . DPO after SFT does not restore the Base chart, indicating that the sequential objective continues optimization within the SFT-induced geometry. Figure 5 combines two views. The phase portraits plot behavior against the mean principal angle from the Base chart, and the retention panels track $S(0, t)$ throughout optimization.

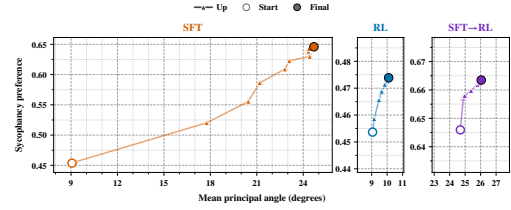
The trajectories reinforce the endpoint comparison. DPO-only overlap remains high throughout training, while SFT rapidly enters a lower-overlap regime. Sequential DPO starts from an already displaced SFT chart and remains far from Base. The trajectory view rules out an explanation in which all objectives initially rewrite geometry but DPO later returns to the Base chart. The endpoint and trajectory results identify



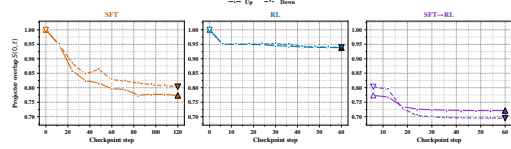
(a) Repetition, up direction: geometry-behavior phase portrait.



(b) Repetition, both directions: Base-chart retention.



(c) Sycophancy, up direction: geometry-behavior phase portrait.



(d) Sycophancy, both directions: Base-chart retention.

Figure 5: **Behavioral geometry through controlled post-training.** Connected markers follow stored checkpoints in training order. In the phase portraits, open and filled circles denote the first stored post-training checkpoint and the reported endpoint, respectively; the Base model itself would lie at zero angular displacement. In the retention panels, $S(0, t)$ compares each checkpoint chart with the Base chart.

geometric conservation versus movement, but they do not by themselves prove which non-geometric mechanism causes the remaining behavioral change. Appendix S therefore reports geometry-occupancy-gain profiles at final checkpoints, across ten matched training stages, and in both ACT and NOC spaces. These profiles show a stable objective-level distinction in geometry, while occupancy and causal gain are more condition-, representation-, and estimator-dependent. A matched token-level comparison of truthful and sycophantic candidates is provided in Appendix L. The complete data provenance, direction checks, and training configurations are reported in Appendix Y. Public checkpoint chains remain useful as ecological validation (Appendix Q).

6 Conclusion

SFT displaced these structures farther from the base model than RL, whereas RL initialized from an SFT checkpoint largely remained within the SFT-shifted geometric regime. SFT can more readily reorganize behavior-associated rep-

representations, while DPO may often modify behavior while preserving more of the inherited representational scaffold.

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A Derivation of the Local Behavioral Model

This appendix derives the local behavioral model used in Section 3. The purpose of the model is not to assert that a language model implements a globally linear behavioral classifier. Rather, it provides a local surrogate for the behavior log-odds in the neighborhood of the observed behavior-relevant representations. This surrogate justifies the decomposition used in the main text: post-training may change a behavior by moving the behavior chart, by changing the distribution of states inside that chart, or by changing the readout from chart coordinates to behavior probability.

A.1 Binary Behaviors and Their Log-Odds

Fix a model f_θ and a behavior b . Let

$$Y_b \in \{0, 1\}$$

be the binary random variable indicating whether behavior b is expressed on an instance x . Let

$$h_\theta(x) = \bar{\phi}_\theta(x) \in \mathbb{R}^D$$

denote the standardized internal representation used for analysis. This representation may correspond to ACT, NOC, or another feature space, but the derivation below is independent of the particular choice.

Assume that the conditional behavior probability satisfies

$$0 < p_\theta(Y_b = 1 \mid h) < 1$$

on the representation region under consideration. We may then define the behavior log-odds function

$$g_{\theta,b}(h) = \log \frac{p_\theta(Y_b = 1 \mid h)}{p_\theta(Y_b = 0 \mid h)}. \quad (4)$$

This definition is exact. Since

$$p_\theta(Y_b = 0 \mid h) = 1 - p_\theta(Y_b = 1 \mid h),$$

Equation 4 implies

$$p_\theta(Y_b = 1 \mid h) = \sigma(g_{\theta,b}(h)), \quad (5)$$

where $\sigma(u) = (1 + \exp(-u))^{-1}$. Thus the only approximation in the local behavioral model concerns the representation dependence of $g_{\theta,b}(h)$.

A.2 Local Linearization of the Behavior Log-Odds

Let $\mathcal{R}_{\theta,b} \subset \mathbb{R}^D$ denote a neighborhood containing the observed behavior-relevant representations. Choose a reference point

$$h_\star \in \mathcal{R}_{\theta,b},$$

for example the midpoint between the positive and negative class means. Suppose that $g_{\theta,b}$ is twice continuously differentiable on $\mathcal{R}_{\theta,b}$ and that its Hessian is bounded:

$$\sup_{h \in \mathcal{R}_{\theta,b}} \|\nabla^2 g_{\theta,b}(h)\|_2 \leq L_{\theta,b}. \quad (6)$$

A first-order Taylor expansion around $h_\star$ gives

$$g_{\theta,b}(h) = g_{\theta,b}(h_\star) + w_{\theta,b}^\top (h - h_\star) + R_{\theta,b}(h), \quad (7)$$

where

$$w_{\theta,b} = \nabla g_{\theta,b}(h_\star)$$

is the local behavior-sensitive direction in representation space, and the remainder satisfies

$$|R_{\theta,b}(h)| \leq \frac{L_{\theta,b}}{2} \|h - h_\star\|_2^2. \quad (8)$$

Therefore, in a sufficiently small neighborhood of $h_\star$, the behavior log-odds are well approximated by an affine function of the internal representation.

A.3 Low-Dimensional Behavioral Chart Assumption

The geometric hypothesis of the paper is that behavior-relevant variation is locally concentrated in a low-dimensional chart. Formally, let

$$U_{\theta,b} \in \mathbb{R}^{D \times k}, \quad U_{\theta,b}^\top U_{\theta,b} = I_k,$$

be an orthonormal basis for a k -dimensional behavioral chart, and let

$$P_{\theta,b} = U_{\theta,b} U_{\theta,b}^\top$$

be the corresponding orthogonal projector. The chart is useful when the local behavior-sensitive direction $w_{\theta,b}$ is mostly contained in the subspace spanned by $U_{\theta,b}$. We quantify the residual sensitivity outside the chart by

$$\delta_{\theta,b}(U) = \|(I - P_{\theta,b})w_{\theta,b}\|_2. \quad (9)$$

Define the chart-level readout vector

$$\beta_{\theta,b} = U_{\theta,b}^\top w_{\theta,b}. \quad (10)$$

Then

$$P_{\theta,b} w_{\theta,b} = U_{\theta,b} \beta_{\theta,b},$$

and

$$w_{\theta,b} = U_{\theta,b} \beta_{\theta,b} + (I - P_{\theta,b})w_{\theta,b}.$$

Substituting this decomposition into Equation 7 yields

$$g_{\theta,b}(h) = g_{\theta,b}(h_\star) + \beta_{\theta,b}^\top U_{\theta,b}^\top (h - h_\star) + ((I - P_{\theta,b})w_{\theta,b})^\top (h - h_\star) + R_{\theta,b}(h). \quad (11)$$

Define the intercept

$$\tau_{\theta,b} = g_{\theta,b}(h_\star) - \beta_{\theta,b}^\top U_{\theta,b}^\top h_\star. \quad (12)$$

Then

$$g_{\theta,b}(h) = \beta_{\theta,b}^\top U_{\theta,b}^\top h + \tau_{\theta,b} + \varepsilon_{\theta,b}(h), \quad (13)$$

where

$$\varepsilon_{\theta,b}(h) = ((I - P_{\theta,b})w_{\theta,b})^\top (h - h_\star) + R_{\theta,b}(h). \quad (14)$$

By Cauchy-Schwarz and Equation 8,

$$|\varepsilon_{\theta,b}(h)| \leq \delta_{\theta,b}(U) \|h - h_\star\|_2 + \frac{L_{\theta,b}}{2} \|h - h_\star\|_2^2. \quad (15)$$

Equation 15 makes explicit the two approximation requirements. First, the behavior log-odds should be locally close to linear, which is controlled by the curvature bound $L_{\theta,b}$ and the radius of the neighborhood. Second, the behavior-sensitive direction should be well captured by the chart $U_{\theta,b}$,

which is controlled by $\delta_{\theta,b}(U)$. The empirical procedures in the paper are designed to test these requirements: low-rank PCA structure tests whether behavior-relevant variation is concentrated; logistic and Fisher directions test whether discriminative directions align with the chart; causal interventions test whether the chart has behavioral effect.

Since the sigmoid is $1/4$ -Lipschitz,

$$|\sigma(a) - \sigma(a')| \leq \frac{1}{4}|a - a'|,$$

the logit error in Equation 15 also bounds the induced probability error:

$$|p_\theta(Y_b = 1 | h) - \sigma(\beta_{\theta,b}^\top U_{\theta,b}^\top h + \tau_{\theta,b})| \leq \frac{1}{4}|\varepsilon_{\theta,b}(h)|. \quad (16)$$

Dropping the residual term gives the local behavioral model used in the main text:

$$p_\theta(Y_b = 1 | x) \approx \sigma(\beta_{\theta,b}^\top U_{\theta,b}^\top \bar{\phi}_\theta(x) + \tau_{\theta,b}). \quad (17)$$

A.4 Equivalent Statistical Derivation from a Local Gaussian Model

The same form also arises from a standard local generative model in the behavioral chart. Let

$$z = U_{\theta,b}^\top h$$

be the chart coordinate. Suppose that, within the local chart, the class-conditional distributions are approximately Gaussian with shared covariance:

$$z | Y_b = 1 \sim \mathcal{N}(\mu_1, \Sigma), \quad (18)$$

$$z | Y_b = 0 \sim \mathcal{N}(\mu_0, \Sigma). \quad (19)$$

Let

$$\pi_1 = p(Y_b = 1), \quad \pi_0 = p(Y_b = 0)$$

be the class priors. Bayes' rule gives

$$\log \frac{p(Y_b = 1 | z)}{p(Y_b = 0 | z)} = \log \frac{p(z | Y_b = 1)\pi_1}{p(z | Y_b = 0)\pi_0}. \quad (20)$$

Substituting the Gaussian densities and canceling the quadratic terms yields

$$\log \frac{p(Y_b = 1 | z)}{p(Y_b = 0 | z)} = \beta^\top z + \tau, \quad (21)$$

where

$$\beta = \Sigma^{-1}(\mu_1 - \mu_0), \quad (22)$$

and

$$\tau = \log \frac{\pi_1}{\pi_0} - \frac{1}{2}(\mu_1^\top \Sigma^{-1} \mu_1 - \mu_0^\top \Sigma^{-1} \mu_0). \quad (23)$$

Thus

$$p(Y_b = 1 | z) = \sigma(\beta^\top z + \tau),$$

and, since $z = U_{\theta,b}^\top h$, this again gives the form of Equation 17. This derivation clarifies why Fisher-style directions and Fisher separability are natural diagnostics: under the shared-covariance local Gaussian approximation, the optimal linear discriminative direction is

$$\Sigma^{-1}(\mu_1 - \mu_0).$$

A.5 Consequences for the Geometry–Occupancy–Readout Decomposition

Equation 17 separates behavioral change into three conceptually distinct components.

First, the subspace $U_{\theta,b}$ specifies where the behavior is represented. Changes in $U_{\theta,b}$ correspond to movement of the behavioral chart itself. This motivates projector overlap, principal angles, projection distance, orthogonal novelty, and anchored Fisher retention as measures of geometry preservation or chart rewriting.

Second, the projected coordinate

$$z_{\theta,b}(x) = U_{\theta,b}^\top \bar{\phi}_\theta(x)$$

specifies how strongly a particular internal state occupies the behavioral chart. Distributional changes in $z_{\theta,b}(x)$ correspond to occupancy changes. The energy-based occupancy statistic used in the main text,

$$G^{\text{occ}} = \mathbb{E}[\|z\|_2^2 | Y_b = 1] - \mathbb{E}[\|z\|_2^2 | Y_b = 0],$$

is a rotation-invariant second-moment proxy for class-conditional chart occupancy. More directional variants, such as differences along Fisher or logistic directions, estimate signed movement along the behavior-positive axis.

Third, the parameters $(\beta_{\theta,b}, \tau_{\theta,b})$ specify the readout from chart coordinates to behavior probability. The vector $\beta_{\theta,b}$ controls local slope or gain, while $\tau_{\theta,b}$ acts as an intercept or threshold. If an intervention of strength α induces a chart displacement z_α , then

$$\frac{\partial}{\partial \alpha} \log p_\theta(Y_b = 1 | z_\alpha) = \beta_{\theta,b}^\top \frac{\partial z_\alpha}{\partial \alpha}. \quad (24)$$

Therefore the finite-difference causal gain estimates the effective readout slope along the intervention-induced chart direction. The intercept $\tau_{\theta,b}$ cancels under differentiation and should be estimated separately, for example by the fitted logistic intercept or by baseline logit after controlling for chart coordinates.

This derivation explains why the paper distinguishes geometry, occupancy, and readout. Two post-training objectives may yield similar behavioral scores while acting on different terms of Equation 17: one may rotate $U_{\theta,b}$, another may shift the distribution of $U_{\theta,b}^\top \bar{\phi}_\theta(x)$, and a third may alter the readout from chart coordinates to output behavior.

B Repetition Detection and Feature Extraction

B.1 Periodic n -gram Repetition Detection

Let the generated token sequence be $x_{1:T}$. For each position i , the detector forms an n -gram $g_i = (x_i, \dots, x_{i+n-1})$ with $n = 10$. It checks whether this n -gram occurs at least $k = 3$ times within the previous $r = 100$ tokens. Among the first three occurrences, the detector further requires equal spacing:

$$p_2 - p_1 = p_3 - p_2. \quad (25)$$

A sample is retained only if the second occurrence begins after at least 50 non-repetitive prefix tokens. This ensures

that the detected repetition follows an initially non-repetitive prefix.

The stored JSONL record includes the prompt, generated sequence, repeated n -gram, detected positions, and local windows around the repetition onset, which define the repetition-relevant regions used for feature extraction.

B.2 Token Regions

Feature extraction is region-aware. The extraction script first constructs either a chat-template input or a fallback plain text input of the form `Question: ... Answer: ...`. It then identifies input, output, answer-token, and non-answer-token regions. For repetition and sycophancy, the behavior-relevant token span is used for mean pooling. Samples with an unresolved required span are omitted from the corresponding regional analysis.

B.3 Raw Activation Space

For each selected token region, raw activations are the inputs to the MLP down-projection layers. If $h_{\ell,t} \in \mathbb{R}^{d_{ff}}$ denotes the input to `down_proj` at layer ℓ and token t , ACT features are mean-pooled over the designated token region, after which the selected coordinates are retained.

B.4 Normalized Output-Contribution Magnitude

For the experiments reported in this work, the directional NOC concept introduced in Eq. (??) is operationalized using the following non-negative magnitude proxy:

$$c_{\ell,t,j} = \frac{|h_{\ell,t,j}| \|W_{\ell,j}^{\text{down}}\|_2}{\|o_{\ell,t}\|_2 + 10^{-8}}, \quad (26)$$

where $W_{\ell,j}^{\text{down}}$ is the outgoing column of the down-projection matrix and $o_{\ell,t}$ is the output of the down-projection layer. This normalization reduces sensitivity to raw activation and output scale. The implemented proxy is non-negative and does not preserve the sign or direction of an individual coordinate’s contribution.

B.5 Implementation Notes

The hook registration matches modules whose names contain `down_proj`, excluding vision modules when present.

C Neuron Selection, Subspace Estimation, and Cross-Model Alignment

C.1 Sparse Classifier

For each behavior and model, we train a binary logistic regression classifier on flattened selected-region features. The archived classifier runs predominantly use the `saga` solver, with a small number of legacy runs using `liblinear`. The penalty is ℓ_1 , with $C = 1.0$, a maximum of 1000 iterations, tolerance 10^{-3} , and random seed 42. The positive class is the behavior class: repetitive samples for repetition and sycophantic samples for sycophancy. Coordinates whose classifier weights are positive are selected:

$$R = \{j : w_j > 0\}. \quad (27)$$

The code also supports selecting all nonzero weights, but the reported main experiments use the positive-weight rule.

C.2 Physical Index Recovery

The classifier operates on flattened arrays. For dense MLP models with reference shape (L, d_{ff}) , a selected flat index q is mapped back to layer and neuron by

$$\ell = \left\lfloor \frac{q}{d_{ff}} \right\rfloor, \quad j = q \bmod d_{ff}. \quad (28)$$

The recovered mapping is stored as a JSON mapping from each layer index to its selected neuron indices. This mapping is used by neuron steering, subspace steering, and qualitative generation experiments.

C.3 PCA Behavioral Chart

Let $X \in \mathbb{R}^{N \times |R|}$ be the feature matrix restricted to selected coordinates. PCA is fit with up to $\min(N - 1, |R|)$ components. The implementation saves the centered projections $Z = (X - \mu)U$, the components U , the PCA mean μ , the explained-variance ratios, the selected-neuron mask, the labels, and the logistic-regression direction. The effective rank k is

$$k = \min \left\{ m : \sum_{i=1}^m \text{EVR}_i \geq 0.9 \right\}. \quad (29)$$

All later geometry metrics use this 90%-variance cutoff unless otherwise stated.

C.4 CCA Alignment

For two models a, b , let Z_a and Z_b be their PCA projections truncated to k_a and k_b components. CCA is fit with $q = \min(k_a, k_b)$ canonical components. The first canonical correlation is reported as primary CCA, while the average over components is reported as mean CCA. Because CCA directions have arbitrary signs, the code orients each component so that the positive class has the larger mean in both CCA views:

$$\mathbb{E}[U_i | y = 1] > \mathbb{E}[U_i | y = 0], \quad \mathbb{E}[V_i | y = 1] > \mathbb{E}[V_i | y = 0]. \quad (30)$$

This orientation does not affect the canonical-correlation magnitudes, but it provides a consistent behavioral sign convention for visualization and cross-model aggregation of canonical coordinates.

C.5 Classifier-Direction Alignment

In addition to CCA, we align the sparse logistic-regression direction. The classifier weight vector is projected into the PCA basis, and each sample is scored by its projection onto that direction. The direction-correlation score between two models is the absolute Pearson correlation between their sample-level classifier-direction scores.

C.6 Directed Linear Probing

The linear-probing experiment asks whether model A ’s behavioral chart can predict model B ’s chart on matched samples. In k -dimensional mode, the source and target use enough PCs to explain 90% variance, and a linear regression fits $Z_A \mapsto Z_B$. The score is the mean Pearson correlation

between each true target dimension and its prediction. The score is directed: $A \rightarrow B$ need not equal $B \rightarrow A$.

For sycophancy, where PC1 and PC2 can have similar variance, we also use a top-2-PC mode. This searches over the four one-dimensional mappings PC1/PC2 of the source to PC1/PC2 of the target and reports the maximum signed Pearson correlation. This avoids hard-coding which of the first two PCs carries the behavior-aligned direction.

D Steering and Intervention Details

D.1 Multiplicative Neuron Scaling

The simplest intervention scales selected behavior-associated neurons at the input to each MLP down-projection layer. If R_ℓ is the selected neuron set at layer ℓ and $h_{\ell,t}$ is the down-projection input, the intervention is

$$h'_{\ell,t,j} = \alpha h_{\ell,t,j}, \quad j \in R_\ell. \quad (31)$$

During likelihood-style evaluation on labeled spans, scaling is applied only to the behavior-relevant token region. During generation, scaling is global over the current token because autoregressive decoding feeds one token at a time after prefill.

D.2 Additive Subspace Steering

For activation-space steering, we use the PCA chart itself. Let U_k be the first k PCA components and μ the PCA mean over selected coordinates. Given the current selected-neuron vector x , the projection onto the behavior chart is

$$x_S = U_k U_k^\top (x - \mu). \quad (32)$$

The steered vector is

$$x' = x + \alpha x_S. \quad (33)$$

Positive α amplifies the current component along the behavior chart; negative α subtracts it.

D.3 Two-Pass Autoregressive Subspace Steering

Generation-time subspace steering uses a two-pass decoding loop to avoid stale projections. At each generated token:

1. A dry-run forward pass uses the current token and historical KV cache, registers hooks, and collects the selected-neuron vector x for that token.
2. The dry-run output is discarded and the KV cache is not updated.
3. The projection $x_S = U_k U_k^\top (x - \mu)$ is computed once the full selected-neuron vector has been assembled across layers.
4. A second forward pass is run with the same input token and same historical KV cache; hooks inject αx_S into the selected coordinates.
5. The steered logits select the next token greedily, and only the steered pass updates the KV cache.

This two-pass protocol is slower than ordinary generation but avoids using an intervention vector computed from an earlier hidden state.

D.4 Greedy Challenge Prompts

For qualitative repetition control, a manually designed challenge set contains prompts that strongly invite repetition but allow a semantically appropriate escape. Baseline and steered outputs are generated greedily for 50 tokens. Both neuron scaling and subspace steering are evaluated with model-specific α values chosen from development sweeps. Quantitative metrics are computed on generated tokens only, while prompt tokens are excluded from PPL.

D.5 Generated-Text Metrics

For a generated token sequence $x_{1:T}$, Rep- n is

$$\text{Rep-}n = 1 - \frac{|\{(x_i, \dots, x_{i+n-1})\}_{i=1}^{T-n+1}|}{T - n + 1}. \quad (34)$$

Normalized entropy is

$$H_{\text{norm}} = \frac{-\sum_v p_v \log_2 p_v}{\log_2 |\{v : p_v > 0\}|}. \quad (35)$$

Self-BLEU-1 is averaged over generations within each model-intervention setting, treating each generation as the hypothesis and the remaining generations as references. PPL is computed on the concatenated prompt and generation using the corresponding model checkpoint without intervention hooks. Prompt labels are masked so that only generated tokens contribute to the loss.

E Post-Training Geometry Metrics and Checkpoint Analysis

E.1 Projector Metrics Without Forming Full Projectors

Given two orthonormal bases $U_a \in \mathbb{R}^{d \times k_a}$ and $U_b \in \mathbb{R}^{d \times k_b}$, all geometry computations use the matched rank $r^* = \min(k_a, k_b)$ and truncated bases. After truncation to the first r^* columns, we reuse U_a and U_b to denote the matched-rank bases. The implementation avoids constructing $d \times d$ projectors. Let

$$M = U_a^\top U_b \in \mathbb{R}^{r^* \times r^*}. \quad (36)$$

Then

$$S(a, b) = \frac{\|M\|_F^2}{r^*}, \quad (37)$$

principal angles are obtained from singular values $\sigma_i(M)$, and projection distance is computed as

$$d_{\text{proj}}(a, b) = \sqrt{r^* - \|M\|_F^2}. \quad (38)$$

This is equivalent to $2^{-1/2} \|P_a - P_b\|_F$ at matched rank.

E.2 Anchored Fisher Retention

For a base checkpoint a and post-training checkpoint b , the native checkpoint- b Fisher strength is computed in b 's own PCA coordinates. The anchored Fisher strength uses the base chart on checkpoint- b features. The implementation loads

post-checkpoint raw activations, applies the base selected-neuron mask, centers by the base PCA mean, and projects into the base PCA components:

$$Z^{a \rightarrow b} = (X_b^{R_a} - \mu_a)U_a. \quad (39)$$

Then

$$R_{a \rightarrow b}^G = \frac{G(Z^{a \rightarrow b}, y_b)}{G(Z^b, y_b)}. \quad (40)$$

If R^G is near one, the base chart remains almost as discriminative as the native chart; if it is small, the checkpoint has moved the behavior into new coordinates.

E.3 Pairwise Universal Strength Aggregation

For cross-model G metrics, the code first extracts pairwise shared CCA coordinates. For each pair file, it computes G^{uniF} and G^{occ} for both models using the same CCA reliability vector ρ . Model-level metrics are then averaged over all pairwise appearances of that model. This produces one row per model, space, and behavior. Metadata such as parameter count, family, and instruction flag are appended for regression and visualization.

E.4 Causal Gain From Steering Curves

G^{causal} is estimated by centered finite differences on logit-transformed behavior probabilities. For multiplicative neuron scaling, the no-intervention point is $\alpha = 1$. For additive subspace steering, the no-intervention point is $\alpha = 0$. The sycophancy and repetition metric scripts default to an α -step of 2.0 when reading stored steering curves, while repetition strength summaries may use smaller local steps when available. All causal-gain estimates should therefore be interpreted as local finite-difference sensitivities rather than exact derivatives.

E.5 Controlled Training Pipelines

The controlled training experiments use Llama-3.1-8B. For each behavior, we train six trajectory types:

1. SFT-up: imitate behavior-positive outputs.
2. SFT-down: imitate behavior-negative or repaired outputs.
3. Reward-up: prefer behavior-positive outputs.
4. Reward-down: prefer behavior-negative outputs.
5. SFT+Reward-up: run reward-up after SFT-up.
6. SFT+Reward-down: run reward-down after SFT-down.

For sycophancy, the current reward runs are DPO-style preference optimization. For repetition, some legacy scripts are named RL even when the actual run is DPO-style. We therefore use *reward optimization* as an umbrella term in the reported analyses, while noting that the audited optimization objective is DPO.

E.6 End-to-End Fine-Tuning Analysis Pipeline

Each checkpoint analysis executes the following sequence: extract activations for the base and checkpoint, train or reuse sparse classifiers, extract selected neurons, compute global padded indices, run PCA, zero-pad bases to a common ambient coordinate system, compute geometry metrics, evaluate

behavior probabilities/PPL, and append behavior metrics to the geometry table. This full sequence is orchestrated by the fine-tuning run scripts for repetition and sycophancy.

E.7 Endpoint-Summary Reporting Conventions

Table ?? uses the following reporting conventions. *Dir.* indicates whether training enhances (*up*) or suppresses (*down*) the target behavior.

The *Geom.* entry reports $S(0, T)/\bar{\theta}(0, T)$, where $S(0, T) \in [0, 1]$ is the normalized projector overlap between the Base and endpoint behavioral charts and $\bar{\theta}(0, T)$ is their mean principal angle in degrees. The slash separates the two reported values and does not denote division. Larger $S(0, T)$ and smaller $\bar{\theta}(0, T)$ indicate stronger preservation of the Base chart. Anchored Fisher retention R^G is defined in the *Anchored Fisher Retention* subsection above.

The *Behavior* entry reports p_b/m_b , where p_b is the evaluation probability assigned to the behavior-positive target and m_b is the realized behavioral metric. For repetition, m_b is the character 3-gram repetition rate; for sycophancy, it is the sycophantic preference score.

The quantity PPL₋ denotes perplexity on the behavior-negative target: non-repetitive responses for repetition and truthful responses for sycophancy.

Within each behavior block, d_{union} is the number of distinct BAC/R-neuron coordinates selected across all audited checkpoints and intervention directions. Bold geometry entries in Table ?? identify the DPO endpoints with the strongest Base-chart preservation within the corresponding behavior.

F Geometry-Occupancy-Gain Profiles of Controlled Post-Training

F.1 Derivation of Occupancy and Gain Estimators

We derive the occupancy and gain estimators from the local behavioral model defined in Eq. (1):

$$\text{logit } p_\theta(y_b = 1|x) = \beta_{\theta,b}^\top U_{\theta,b}^\top \bar{\phi}_\theta(x) + \tau_{\theta,b}. \quad (41)$$

For simplicity, we denote

$$z_\theta(x) = U_{\theta,b}^\top \bar{\phi}_\theta(x). \quad (42)$$

The behavioral logit can therefore be written as

$$\ell_\theta(x) = \beta_{\theta,b}^\top z_\theta(x) + \tau_{\theta,b}. \quad (43)$$

This decomposition separates three independent factors: (i) the behavioral chart $U_{\theta,b}$, (ii) the distribution of states within the chart $z_\theta(x)$, and (iii) the readout function $(\beta_{\theta,b}, \tau_{\theta,b})$.

Occupancy as a Second-Moment Estimator The occupancy component corresponds to the distribution of projected states:

$$z_\theta(x) = U_{\theta,b}^\top \bar{\phi}_\theta(x). \quad (44)$$

Conditioned on the behavioral label, assume

$$z_\theta(x)|y=c \sim (\mu_c, \Sigma_c), \quad c \in \{0, 1\}. \quad (45)$$

The expected squared distance from the origin in the behavioral chart is

$$\mathbb{E}[\|z_\theta(x)\|_2^2|y=c] = \mathbb{E}[z_\theta(x)^\top z_\theta(x)|y=c] \quad (46)$$

$$= \text{tr}(\mathbb{E}[z_\theta(x)z_\theta(x)^\top|y=c]) \quad (47)$$

$$= \text{tr}(\Sigma_c + \mu_c\mu_c^\top) \quad (48)$$

$$= \text{tr}(\Sigma_c) + \|\mu_c\|_2^2. \quad (49)$$

Therefore, the difference between behavior-positive and behavior-negative states is

$$\begin{aligned} \mathbb{E}[\|z_\theta(x)\|_2^2|y=1] - \mathbb{E}[\|z_\theta(x)\|_2^2|y=0] \\ = \|\mu_1\|_2^2 - \|\mu_0\|_2^2 + \text{tr}(\Sigma_1 - \Sigma_0). \end{aligned} \quad (50)$$

Thus,

$$G^{\text{occ}} = \Delta_{\text{mean}} + \Delta_{\text{variance}}, \quad (51)$$

where

$$\Delta_{\text{mean}} = \|\mu_1\|_2^2 - \|\mu_0\|_2^2$$

captures displacement of behavior-positive states in the chart, and

$$\Delta_{\text{variance}} = \text{tr}(\Sigma_1 - \Sigma_0)$$

captures distributional expansion or contraction.

Therefore, G^{occ} is a rotation-invariant estimator of behavioral manifold occupancy. It does not estimate the location of the manifold itself; instead, it estimates how strongly model states occupy the manifold after the geometry has been fixed.

Gain as Effective Readout Sensitivity Now consider an intervention with strength α . The intervention changes the projected chart coordinate:

$$z_\theta(\alpha). \quad (52)$$

The behavioral logit becomes

$$\ell_\theta(\alpha) = \beta_{\theta,b}^\top z_\theta(\alpha) + \tau_{\theta,b}. \quad (53)$$

Taking the derivative with respect to intervention strength gives

$$\frac{\partial \ell_\theta(\alpha)}{\partial \alpha} = \frac{\partial}{\partial \alpha} (\beta_{\theta,b}^\top z_\theta(\alpha) + \tau_{\theta,b}) \quad (54)$$

$$= \beta_{\theta,b}^\top \frac{\partial z_\theta(\alpha)}{\partial \alpha}. \quad (55)$$

The threshold term disappears:

$$\frac{\partial \tau_{\theta,b}}{\partial \alpha} = 0. \quad (56)$$

Therefore, causal gain does not estimate $\tau_{\theta,b}$; instead, it estimates the effective projection of the behavioral readout vector onto the induced movement direction.

For a direct chart intervention

$$z_\theta(\alpha) = z_0 + \alpha u, \quad (57)$$

we obtain

$$\frac{\partial \ell_\theta}{\partial \alpha} = \beta_{\theta,b}^\top u. \quad (58)$$

Hence, if u is chosen as the normalized behavioral direction,

$$u = \frac{\beta_{\theta,b}}{\|\beta_{\theta,b}\|_2}, \quad (59)$$

then

$$G^{\text{causal}} = \|\beta_{\theta,b}\|_2. \quad (60)$$

More generally, when interventions are performed in activation space,

$$h_\alpha = h + \alpha v,$$

the induced chart movement is

$$\frac{\partial z_\theta}{\partial \alpha} = U_{\theta,b}^\top J_{\phi_\theta}(h)v, \quad (61)$$

where J_{ϕ_θ} is the local Jacobian of the representation map. Therefore,

$$G^{\text{causal}} = \beta_{\theta,b}^\top U_{\theta,b}^\top J_{\phi_\theta}(h)v. \quad (62)$$

Estimating the Threshold Term Although causal gain estimates the effective slope $\beta_{\theta,b}$, the intercept $\tau_{\theta,b}$ cancels under differentiation. The threshold term can instead be estimated from a fitted logistic model:

$$\hat{\tau}_{\theta,b} \quad (63)$$

or equivalently,

$$\hat{\tau}_{\theta,b} = \mathbb{E}[\ell_\theta(x)] - \hat{\beta}_{\theta,b}^\top \mathbb{E}[z_\theta(x)]. \quad (64)$$

Thus, the complete decomposition is:

Component	Estimator
Behavioral geometry $U_{\theta,b}$	Grassmannian metrics
Occupancy $z_\theta(x)$	G^{occ}
Readout gain $\beta_{\theta,b}$	G^{causal}
Threshold $\tau_{\theta,b}$	Logistic intercept

This decomposition provides a mechanistic interpretation of post-training: SFT can alter the behavioral geometry itself, while reward optimization can modify the occupancy distribution or readout strength within an inherited behavioral chart.

G Numerical Tables

G.1 Repetition Strength Across Models

Table 3 gives representative ACT-space repetition strength measurements. T_{base} is the baseline repetition tendency at the no-intervention point, and N_R^{eff} is the PCA dimension at the 90%-variance threshold.

Model	$G_{\text{ACT}}^{\text{uniF}}$	$G_{\text{ACT}}^{\text{occ}}$	G^{causal}	T_{base}	N_R^{eff}
Gemma-2-2B	6.8465	0.9809	0.2942	0.8632	13
Gemma-2-2B-it	8.0945	0.7189	0.2750	0.6800	15
Gemma-2-9B	6.8896	0.8550	0.1980	0.8704	16
Gemma-2-9B-it	7.7726	0.7816	0.0771	0.8136	14
Qwen2.5-0.5B	7.5528	0.7049	0.2108	0.8562	26
Qwen2.5-0.5B-Inst.	7.1556	0.7323	0.5994	0.8219	25
Qwen2.5-7B	6.9355	0.8142	0.2648	0.9054	16
Qwen2.5-7B-Inst.	8.0421	0.7786	0.2072	0.8801	21
Mistral-7B-v0.3	10.3875	0.6061	0.1661	0.8833	25
Mistral-7B-Inst.	8.3211	0.6757	0.2480	0.8713	18
Llama-3.1-8B	3.6277	1.1152	0.0891	0.9082	9
Llama-3.1-8B-Inst.	4.3802	1.1844	0.1823	0.7599	8

Table 3: Representative repetition strength measurements in ACT space.

G.2 Sycophancy Low-Rank Statistics

Table 4 summarizes selected sycophancy PCA results. ACT-space PC1 explained variance is usually higher than NOC-space PC1 explained variance, consistent with the main-text claim that sycophancy is more concentrated as a runtime state than as a contribution-space basis.

G.3 Full Controlled Repetition Summary

Table 5 repeats the final controlled repetition summary from the main text for convenience.

G.4 Full Controlled Sycophancy Summary

Table 6 summarizes the final controlled sycophancy results. Reward-only runs preserve the Base geometry most strongly, whereas SFT-containing runs produce larger geometric changes and stronger directional shifts in sycophancy behavior.

G.5 Checkpoint-Level Sycophancy Trajectory

Table 7 shows that reward-only sycophancy runs preserve geometry throughout the trajectory. Behavior moves gradually, but overlap remains around 0.94–0.95.

H Derivation of Universal Fisher Strength

This section provides the theoretical motivation for the Universal Fisher Strength metric used in the main text. The goal is to formalize when an estimated behavioral chart constitutes a reliable coordinate system for a behavior, rather than merely a low-dimensional projection with accidental separability.

H.1 Behavior Separation in the Local Chart

Recall the local behavioral model introduced in Eq. (1):

$$\text{logit } p_{\theta}(y_b = 1|x) = \beta_{\theta,b}^{\top} z_{\theta,b}(x) + \tau_{\theta,b}, \quad (65)$$

where

$$z_{\theta,b}(x) = U_{\theta,b}^{\top} \bar{\phi}_{\theta}(x) \quad (66)$$

is the coordinate of the sample inside the estimated behavioral chart.

Assume that the projected behavioral coordinates follow two class-conditional distributions:

$$z|y = c \sim (\mu_c, \Sigma_c), \quad c \in \{0, 1\}. \quad (67)$$

The difference between behavior-positive and behavior-negative states is captured by the mean displacement:

$$\Delta\mu = \mu_1 - \mu_0. \quad (68)$$

However, the magnitude of $\Delta\mu$ alone is insufficient. A large displacement may simply reflect a noisy or highly variable representation space. Therefore, the quality of a behavioral chart depends on the ratio between between-class displacement and within-class variability.

H.2 Connection to Fisher Discriminant Analysis

Classical Fisher discriminant analysis seeks a projection direction v that maximizes the ratio

$$J(v) = \frac{(v^{\top}(\mu_1 - \mu_0))^2}{v^{\top}\Sigma_w v}, \quad (69)$$

where

$$\Sigma_w = \frac{1}{2}\Sigma_1 + \frac{1}{2}\Sigma_0 \quad (70)$$

is the pooled within-class covariance.

The optimal Fisher direction is

$$v^* \propto \Sigma_w^{-1}(\mu_1 - \mu_0). \quad (71)$$

Substituting Eq. (71) into Eq. (69) gives the maximum Fisher criterion:

$$J(v^*) = (\mu_1 - \mu_0)^{\top} \Sigma_w^{-1} (\mu_1 - \mu_0). \quad (72)$$

Therefore,

$$G^{\text{Fisher}} = \Delta\mu^{\top} \Sigma_w^{-1} \Delta\mu \quad (73)$$

is the optimal signal-to-noise ratio of the behavioral chart. Intuitively, this quantity asks:

How far apart are behavior-positive and behavior-negative states after normalizing by the intrinsic variability of the chart?

Model	NOC PC1	NOC Dim90	ACT PC1	ACT Dim90
Qwen2.5-0.5B	0.2025	38	0.6011	4
Qwen2.5-7B	0.2924	28	0.7673	3
Qwen2.5-32B	0.2603	32	0.7952	3
Gemma-2-27B-it	0.2919	23	0.6587	2
Llama-3.1-8B	0.3093	29	0.4262	8
Mistral-Nemo-Inst.	0.2799	25	0.6764	6
Mistral-Small-24B-Base	0.5269	12	0.3486	8
Qwen3.5-27B	0.3149	34	0.3035	22

Table 4: Selected sycophancy low-rank statistics in NOC and ACT spaces.

Objective	Dir.	S	Angle	d_{proj}	R^G	G	Repeat	PPL	n-gram
SFT	up	0.6934	27.78	2.7128	0.9380	76.5460	0.96803	3.70416	0.72299
SFT	down	0.7106	25.77	2.5800	0.9419	63.8125	0.92028	3.52084	0.37174
Reward	up	0.9600	3.59	0.9999	1.0955	67.1523	0.91801	2.94752	0.40325
Reward	down	0.9200	7.20	1.4142	1.0493	66.4614	0.88494	2.91139	0.43393
SFT→Reward	up	0.5975	36.05	3.0427	0.8995	81.4551	0.96703	5.00357	0.94186
SFT→Reward	down	0.6040	35.17	2.9515	0.7045	57.8396	0.59574	3.86605	0.56458

Table 5: Final checkpoints of the controlled repetition runs.

A large value indicates that the chart contains a stable behavioral coordinate. A small value indicates that apparent separation is dominated by within-class variation.

H.3 Shrinkage Stabilization

In practical language-model representations, the dimensionality of the chart can still be large relative to the number of evaluation samples. Directly inverting Σ_w may therefore be unstable.

We use a standard shrinkage estimator:

$$\Sigma_w^{(\lambda)} = (1 - \lambda)\Sigma_w + \lambda \frac{\text{tr}(\Sigma_w)}{q} I, \quad (74)$$

where q is the chart dimensionality.

The second term corresponds to an isotropic covariance model. Therefore, λ controls the trade-off between the empirical covariance structure and a well-conditioned isotropic approximation.

The resulting stabilized Fisher score is

$$G^{\text{SF}} = \Delta\mu^\top (\Sigma_w^{(\lambda)})^{-1} \Delta\mu. \quad (75)$$

This prevents artificially large scores caused by nearly singular covariance directions.

H.4 Incorporating Cross-Model Reliability

A central goal of this work is not only to identify a behavioral chart within one model, but to determine whether the chart represents a transferable behavioral structure across models.

Suppose the chart is aligned across models using CCA. Let

$$\rho_i \quad (76)$$

denote the canonical correlation of the i -th aligned coordinate.

A coordinate with high ρ_i is consistently represented across models, whereas a coordinate with low ρ_i may correspond to model-specific noise or idiosyncratic variation.

Therefore, before computing Fisher separation, we weight each shared coordinate according to its cross-model reliability:

$$D_\rho = \text{diag}(\rho_1, \rho_2, \dots, \rho_q). \quad (77)$$

The reliability-weighted behavioral displacement becomes

$$\Delta\mu_\rho = D_\rho \Delta\mu. \quad (78)$$

Substituting into Eq. (75) gives

$$G^{\text{uniF}} = \Delta\mu^\top D_\rho (\Sigma_w^{(\lambda)})^{-1} D_\rho \Delta\mu. \quad (79)$$

This is the Universal Fisher Strength used in the main text.

H.5 Interpretation as Behavioral Chart Reliability

The proposed metric combines three properties required for a meaningful behavioral manifold.

1. Behavioral separation.

The displacement term

$$\Delta\mu = \mu_1 - \mu_0 \quad (80)$$

requires that behavior-positive and behavior-negative states occupy different regions of the chart.

A chart with no behavioral displacement receives a low score.

Objective	Dir.	S	Angle	d_{proj}	R^G	G	Syco Prob	Truth PPL	Syco Pref
SFT	up	0.7734	24.69	2.5636	0.9657	10.9596	0.88584	11.71793	0.64595
SFT	down	0.8035	22.04	2.3871	0.8144	17.7359	0.44221	1.28075	0.33044
Reward	up	0.9382	10.10	1.3384	0.9983	13.8834	0.37020	6.48420	0.47387
Reward	down	0.9389	9.68	1.3309	0.9949	13.8919	0.34155	5.68717	0.43599
SFT→Reward	up	0.7211	29.03	2.7944	0.8845	10.3968	0.85073	1485.09937	0.66585
SFT→Reward	down	0.6950	30.88	2.9224	0.8187	15.8135	0.38855	1.37858	0.29712

Table 6: Final checkpoints of the controlled sycophancy runs.

Run	Step	S	Angle	Syco Prob
Reward-up	6	0.9498	9.06	0.3550
Reward-up	30	0.9451	9.47	0.3623
Reward-up	60	0.9382	10.10	0.3702
Reward-down	6	0.9498	9.06	0.3550
Reward-down	30	0.9518	8.84	0.3472
Reward-down	60	0.9389	9.68	0.3416

Table 7: Reward-only sycophancy trajectories preserve geometry while moving behavior modestly.

2. Representation stability. The covariance normalization

$$(\Sigma_w^{(\lambda)})^{-1} \quad (81)$$

penalizes directions where separation is caused by noisy or unstable variation.

A chart where positive and negative states overlap substantially receives a low score even if their raw means differ.

3. Cross-model universality. The CCA reliability weighting

$$D_\rho \quad (82)$$

suppresses directions that only appear in a single model.

Therefore, a high Universal Fisher Strength indicates that a behavioral chart is simultaneously:

1. behavior-discriminative;
2. low-noise;
3. transferable across models.

This property is important because our goal is not simply to find any separable projection. High-dimensional representations contain many directions that can separate finite samples by chance. Instead, we seek behavioral coordinate systems that correspond to stable and reusable internal structures.

H.6 One-Dimensional Case

When the aligned behavioral chart is dominated by a single canonical coordinate, $q = 1$, we have

$$D_\rho = \rho_1, \quad (83)$$

and the Universal Fisher Strength reduces to

$$G^{\text{uniF-1D}} = \rho_1^2 \frac{(\mu_1 - \mu_0)^2}{\sigma_w^2 + \epsilon}. \quad (84)$$

Thus the metric measures whether the first behavioral direction is not only predictive within a single model, but also represents a shared behavioral coordinate across different models.

I Leakage Control and Robustness Analysis

I.1 Coordinate Selection Robustness

The first stage of our framework identifies a sparse coordinate scaffold before estimating behavioral geometry. Since this stage may potentially introduce selection bias, we perform several controls.

Random-coordinate baseline. For each behavior-associated coordinate set

$$\mathcal{N}_{\theta,b},$$

we construct matched-size random coordinate sets sampled from the same model and layer distribution.

A genuine behavioral chart should exhibit stronger separability, cross-model alignment, and causal steering effects than a random coordinate population of the same size.

In particular, the random baseline is used for:

- PCA variance comparison;
- cross-model CCA alignment;
- linear probing transfer accuracy;
- steering effectiveness.

Label permutation control. To test whether the discovered geometry depends on genuine behavioral structure, we repeat coordinate selection and subspace estimation after randomly permuting behavioral labels. Under shuffled labels, the sparse classifier should fail to identify stable behavior-associated coordinates, and subsequent CCA alignment, Fisher separation, and steering effects should approach random levels.

I.2 Subspace Estimation Robustness

Our main behavioral chart is estimated by PCA on selected coordinates. We evaluate whether conclusions depend on this specific dimensionality choice.

Rank sensitivity. The default chart dimension is selected by the cumulative explained variance criterion:

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_i \lambda_i} \geq 0.90.$$

We additionally evaluate fixed-rank charts:

$$k = 1, \quad k = 3,$$

as well as alternative variance thresholds. The goal is to determine whether the observed geometric movement between SFT and reward optimization is a stable property of the behavioral geometry or an artifact of a particular PCA cutoff.

Alternative directions. The PCA chart is an unsupervised estimate of dominant behavioral variation. To verify that it captures behavior rather than generic variance, we compare it with supervised alternatives:

1. logistic-regression direction:

$$v_{\text{logit}} = \frac{w}{\|w\|_2};$$

2. Fisher discriminant direction:

$$v_{\text{Fisher}} \propto \Sigma_w^{-1}(\mu_1 - \mu_0);$$

3. CCA-aligned shared directions;
4. SAE decoder directions when sparse feature representations are used.

Consistent conclusions across these directions indicate that the discovered behavioral geometry is not tied to a particular subspace estimator.

I.3 Behavior Specificity Controls

A central assumption of our framework is that different behaviors correspond to distinct behavioral geometries rather than a single generic “undesirable behavior” direction. We therefore perform behavior-mismatch experiments.

For two behaviors b_1 and b_2 , we evaluate:

$$U_{\theta, b_1} \rightarrow y_{b_2}. \quad (85)$$

For example, repetition charts are evaluated on sycophancy labels, and sycophancy charts are evaluated on repetition labels.

A genuine behavior-specific manifold should satisfy:

- strong within-behavior separability;
- weak cross-behavior transfer;
- strong within-behavior causal steering;
- weak cross-behavior causal effect.

This control prevents the framework from merely identifying generic degeneration, refusal, or alignment-related directions.

I.4 Token Extraction Robustness

Behavioral expression may depend on the choice of token window. We therefore evaluate multiple extraction strategies.

For a behavior-relevant instance, the feature representation can be extracted from:

1. the complete response span;
2. the behavior onset region;
3. an event-centered local window;
4. a fixed-length pre-event window.

The purpose of these comparisons is to ensure that discovered geometry is not caused by a particular token-selection heuristic.

For dynamic analyses, such as generation trajectories, we additionally compare representations before and after the visible behavioral event. This tests whether the internal behavioral state emerges before the surface-level output phenomenon.

I.5 Cross-Model Alignment Robustness

Because neuron indices are not comparable across architectures, our framework uses CCA-based alignment to construct shared behavioral charts. We verify that the observed alignment is not caused by arbitrary high-dimensional matching.

The following controls are performed:

1. matched-size random coordinate alignment;
2. shuffled-label alignment;
3. different CCA ranks;
4. different reference models;
5. pairwise family-stratified comparisons.

For each comparison we report:

- canonical correlation;
- linear probing transfer performance;
- projector overlap;
- principal angles.

A reliable behavioral manifold should remain aligned across these choices.

I.6 Causal Intervention Controls

The causal experiments use three intervention levels:

1. sparse coordinate scaling;
2. subspace projection editing;
3. direction-level steering.

For each intervention, we compare:

- behavior-associated directions;
- random directions;
- opposite-direction controls;
- matched-magnitude non-behavior directions.

The intervention strength is varied to obtain dose-response curves rather than single-point comparisons. A genuine behavioral chart should produce a monotonic or directionally consistent response when amplified or suppressed.

I.7 Post-Training Geometry Robustness

To ensure that the observed difference between SFT and reward optimization is not caused by a particular checkpoint or metric, we analyze post-training trajectories using multiple geometric quantities:

- projector overlap:

$$S(a, b);$$

- principal angles:

$$\vartheta_i(a, b);$$

- projection distance:

$$d_{\text{proj}}(a, b);$$

- orthogonal novelty:

$$N_{\perp}(a \rightarrow b);$$

- anchored Fisher retention:

$$R_{0 \rightarrow t}^G.$$

Agreement across these measures supports the interpretation that SFT and reward optimization induce different geometric trajectories rather than merely different behavioral magnitudes.

J Visualization of Behavioral Charts

The behavioral manifold framework assumes that behavior-associated internal states are not arbitrarily distributed in the ambient representation space, but rather concentrate around low-dimensional local charts. While quantitative metrics such as projector overlap, Fisher strength, and cross-model alignment measure geometric properties numerically, directly visualizing the estimated charts provides an intuitive validation of this assumption.

This appendix visualizes the learned behavioral charts for two qualitatively different behaviors: repetition and sycophancy. These behaviors are selected because they represent distinct classes of model failures: repetition reflects a generation-dynamics failure, whereas sycophancy reflects an alignment-related preference failure. The fact that both exhibit similar low-dimensional geometric organization provides evidence that the proposed framework captures a general property of behavior representation rather than a behavior-specific artifact.

J.1 Visualization Protocol

For a model f_{θ} , behavior b , and representation space Ω , the behavioral chart is estimated as described in Section 3:

$$U_{\theta, b, \Omega} = [v_1, \dots, v_k].$$

Given a standardized representation vector

$$\bar{\phi}_{\theta, b}^{(\Omega)}(x),$$

we project each instance into the learned chart coordinate system:

$$z_{\theta, b, \Omega}(x) = U_{\theta, b, \Omega}^{\top} \bar{\phi}_{\theta, b}^{(\Omega)}(x). \quad (86)$$

For visualization, we display the first two chart coordinates:

$$(z_1, z_2).$$

Positive and negative behavior examples are shown using different colors. We also project two independently estimated supervised directions into the same coordinate system:

- the Fisher discriminant direction

$$v_{\text{Fisher}} \propto \Sigma_w^{-1}(\mu_1 - \mu_0),$$

which maximizes class separation under the shared covariance assumption;

- the behavior-positive logistic direction obtained from the sparse coordinate classifier.

The alignment between these supervised directions and the dominant chart axis provides an additional sanity check that the visualization reflects behavior-relevant geometry rather than arbitrary variance captured by PCA.

We visualize two complementary representation spaces:

1. **Activation space (ACT):** the selected neuron activations directly measure whether behavior-associated coordinates are active.
2. **Normalized output-contribution space (NOC):** the selected coordinates are weighted according to their signed contribution to the realized MLP update, measuring whether the coordinates functionally participate in the layer computation.

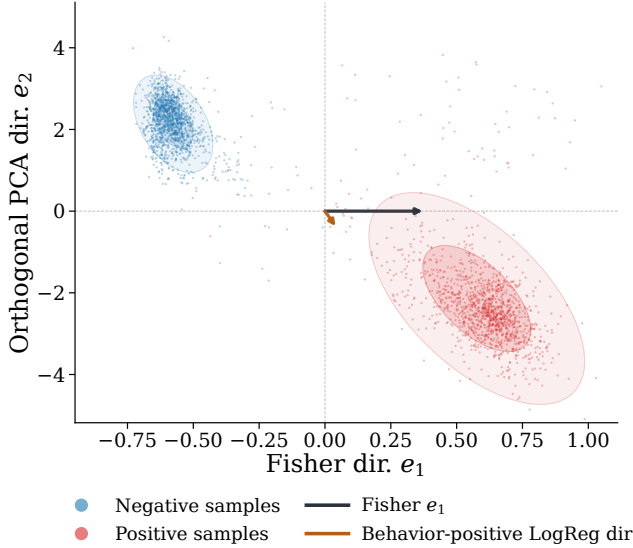
Consistency between ACT and NOC visualizations indicates that the discovered behavioral geometry is not solely caused by activation magnitude, but is also reflected in functionally relevant computation.

J.2 Repetition Behavioral Charts

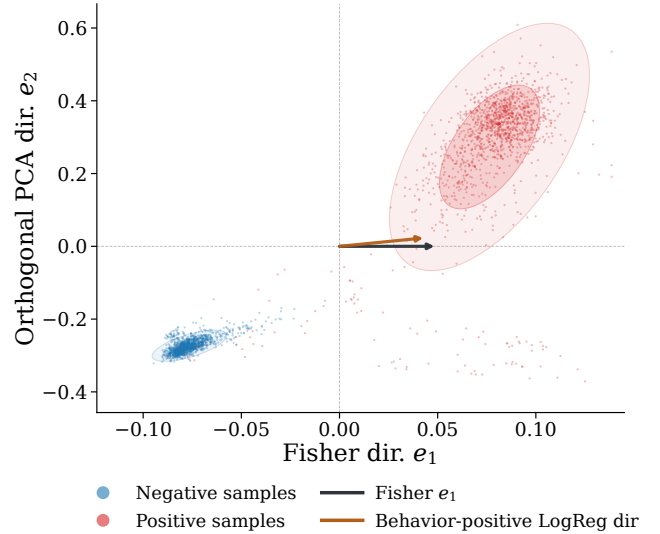
Repetition provides a test case for a dynamically generated failure mode. Figure 6 compares the learned repetition charts in ACT and NOC spaces, showing that repetitive and non-repetitive states occupy different regions of the chart.

In both representation spaces, repetition-positive states occupy a distinct region from repetition-negative states. The agreement between activation space and normalized output-contribution space is important: if the separation were caused only by large activation magnitudes, the structure would disappear after accounting for functional contribution. Instead, the same qualitative geometry persists in both spaces, suggesting that repetition corresponds to a stable computational state supported by a coordinated population of coordinates.

The alignment of Fisher and logistic directions further indicates that the dominant variance direction of the chart is also behavior-discriminative. In other words, the estimated chart is not merely a low-dimensional compression of activation variation; it captures directions along which the model state systematically changes its probability of entering the repetition regime.



(a) Activation space (ACT).



(b) Normalized output-contribution space (NOC).

Figure 6: Visualization of the repetition behavioral chart. Behavior-positive and behavior-negative examples form distinguishable regions in both ACT and NOC spaces. The Fisher and logistic directions align with the dominant separation axis, suggesting that repetition is represented by a structured low-dimensional geometry rather than isolated activation events.

J.3 Sycophancy Behavioral Charts

Sycophancy provides an independent validation on a qualitatively different behavior. Unlike repetition, which emerges from generation dynamics, sycophancy reflects preference and alignment behavior. Nevertheless, Figure 7 shows that the same framework identifies a structured low-dimensional chart in both ACT and NOC spaces.

The sycophancy charts demonstrate that the behavioral manifold framework extends beyond generation degeneration. Behavior-positive and behavior-negative states form distinguishable regions even though sycophancy involves a different causal mechanism from repetition. The consistency between the two feature spaces suggests that the discovered structure reflects a distributed behavioral representation rather than a single activation correlate.

Moreover, the supervised directions consistently align with the chart separation axis. This supports the interpretation that the estimated behavioral chart $U_{\theta,b}$ captures a meaningful local coordinate system for the behavior.

K Token-Level Dynamics in Behavioral Landscapes

The main behavioral-chart analysis treats behavior-positive and behavior-negative representations as class-conditional regions in a learned low-dimensional coordinate system. This static view establishes that a behavior has an identifiable representation geometry, but it does not reveal how the model reaches that region while processing or generating a sequence. We therefore complement the static analysis with token-level trajectories through the learned behavioral landscapes.

The trajectory analysis addresses a dynamic question: *how does the internal state evolve relative to a behavioral region before, during, and after the behavior becomes visible in the output?* It also provides a qualitative test of whether the estimated chart captures temporally coherent state evolution rather than only aggregate class separation.

K.1 Behavioral Energy Landscape

For a fixed model, behavior b , and feature space $\Omega \in \{\text{ACT}, \text{NOC}\}$, let

$$z_{\theta,b,\Omega}(x) = U_{\theta,b,\Omega}^\top \bar{\phi}_{\theta,b}^{(\Omega)}(x)$$

be the representation of an instance in the learned behavioral chart. To visualize the first two chart coordinates, we estimate class-conditional densities

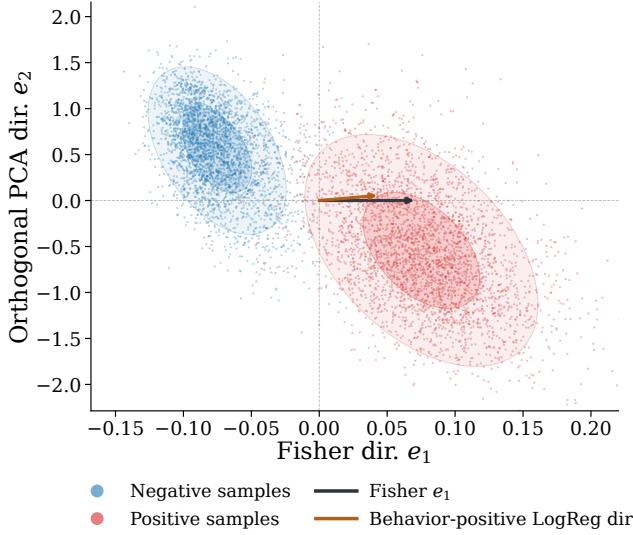
$$\hat{p}_{b,\Omega}^+(z_1, z_2), \quad \hat{p}_{b,\Omega}^-(z_1, z_2)$$

from behavior-positive and behavior-negative chart samples. The displayed background is the log-density ratio

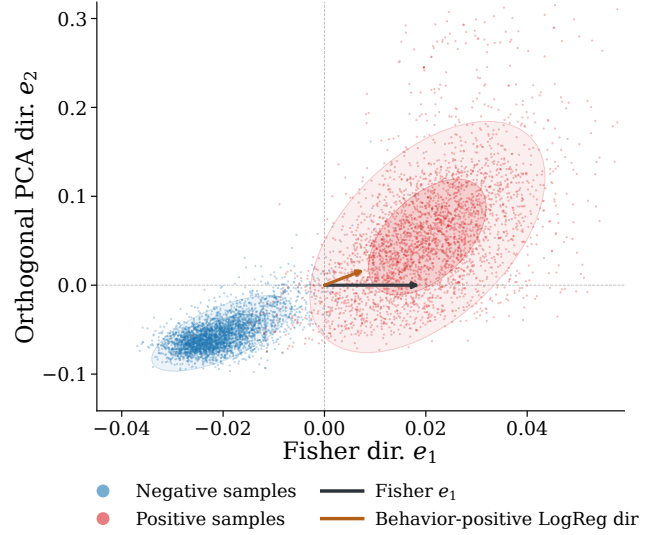
$$\mathcal{E}_{b,\Omega}(z) = \log \frac{\hat{p}_{b,\Omega}^+(z) + \epsilon}{\hat{p}_{b,\Omega}^-(z) + \epsilon}. \quad (87)$$

Positive values indicate regions relatively more characteristic of behavior-positive states, while negative values indicate regions relatively more characteristic of behavior-negative states. The red and blue background should therefore be interpreted as a relative behavioral-density landscape, not as a calibrated behavioral probability.

To prevent the visualized trajectory from determining its own background, the trajectory instance is excluded from the density estimate whenever it belongs to the chart-estimation set. The behavior-positive steering direction is also projected into the two-dimensional chart and shown as a black arrow.



(a) Activation space (ACT).



(b) Normalized output-contribution space (NOC).

Figure 7: Visualization of the sycophancy behavioral chart. Sycophantic and non-sycophantic states occupy different regions in the learned chart. The agreement between ACT and NOC representations suggests that sycophancy is supported by structured internal computation rather than merely superficial output statistics.

K.2 Token-Level Trajectory Construction

Given a teacher-forced sequence

$$x_1, x_2, \dots, x_T,$$

causal masking ensures that the hidden state at position t depends only on $x_{\leq t}$. We extract the behavior-associated representation at each token:

$$\phi_t^{(\Omega)} = \phi_{\theta,b}^{(\Omega)}(x_{\leq t}), \quad (88)$$

standardize it using the chart-estimation statistics, and project it into the learned chart:

$$z_t = U_{\theta,b,\Omega}^\top \bar{\phi}_t^{(\Omega)}. \quad (89)$$

The token-level behavioral trajectory is the ordered sequence

$$\mathcal{T}_{b,\Omega}(x) = (z_1, z_2, \dots, z_T). \quad (90)$$

Only the first two chart coordinates are displayed. Consecutive token projections are joined in temporal order. A shape-preserving piecewise cubic interpolant is used only to render the path smoothly; all markers and all quantitative analyses are based on the original token projections.

For an annotated behavioral span $\mathcal{I}_b \subseteq \{1, \dots, T\}$, we additionally show its mean chart coordinate:

$$\bar{z}_{\mathcal{I}_b} = \frac{1}{|\mathcal{I}_b|} \sum_{t \in \mathcal{I}_b} z_t. \quad (91)$$

This mean is displayed as a white diamond and summarizes where the established behavioral span lies relative to the positive and negative chart regions.

K.3 Visual Encoding

The trajectories use a shared visual grammar across behaviors and feature spaces. The black circle marks the first displayed token, and the black square marks the final displayed token. A red star marks the annotated behavioral onset. The white diamond denotes the mean representation of the annotated behavioral span.

For repetition, gray denotes the input prefix, blue denotes ordinary non-repetitive generation, yellow denotes the first occurrence of the repeated unit that leads into the loop, and red denotes established repetition beginning with the second consecutive occurrence of that unit.

For sycophancy, gray denotes the input prefix, yellow denotes response tokens preceding the annotated sycophantic span, red denotes the sycophantic span, and blue denotes any displayed post-span tokens. The colors describe temporal segments of one trajectory and are distinct from the red/blue class-density background.

K.4 Repetition Trajectories in Activation Space

Figure 8 shows six repetition trajectories in ACT space for examples that also have corresponding NOC visualizations. Figure 9 shows three additional ACT trajectories, including a paragraph-level example. Together, these examples illustrate both the recurring geometric pattern and the variability of individual decoding paths.

The ACT trajectories support an attractor-like interpretation of repetition, but they do not imply a single deterministic route into repetition. The ordinary-generation segment often occupies the negative basin or its boundary. The lead-in segment then moves toward the positive region, and the established repetition segment is concentrated more strongly on the positive side. The white repeated-span means are

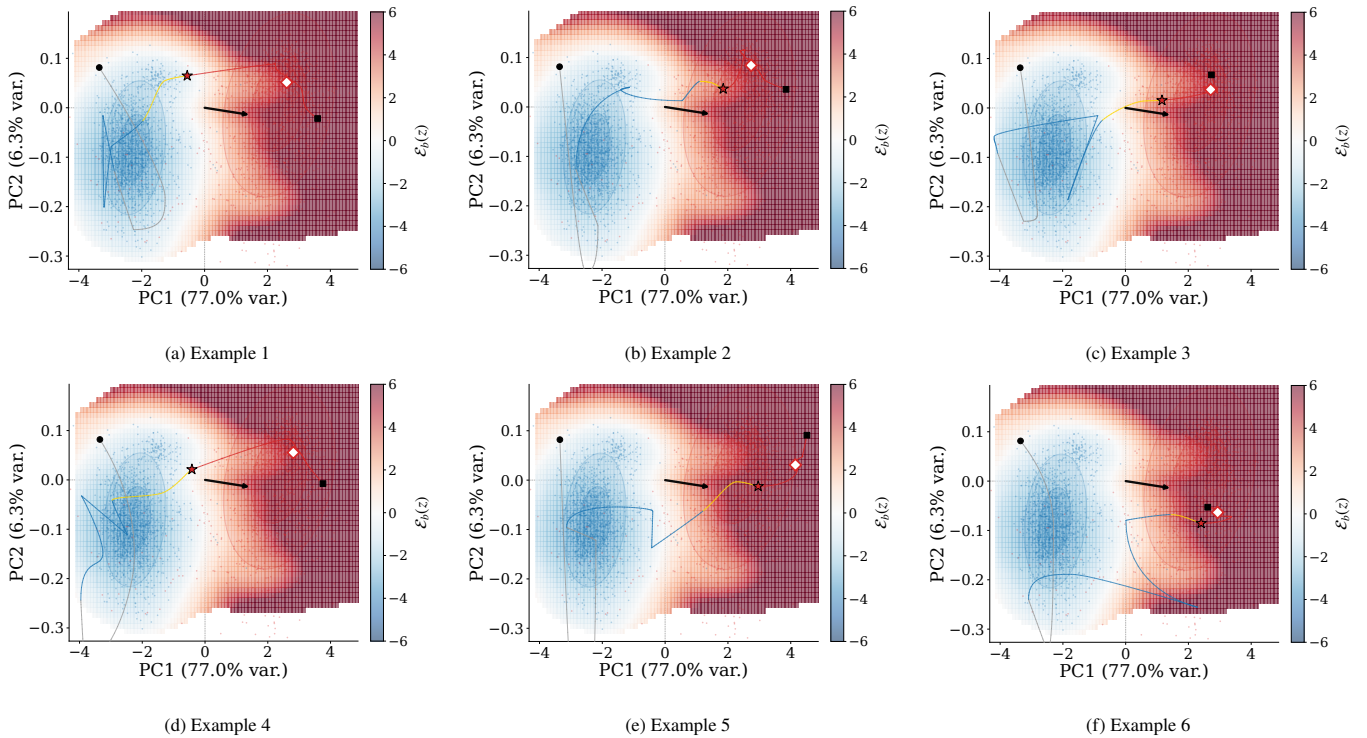


Figure 8: Repetition trajectories in ACT space for examples with matched NOC visualizations. Across examples, prefix and ordinary-generation states predominantly occupy the negative or transitional region, whereas repeated spans move toward and remain in the behavior-positive region. Individual trajectories differ in curvature, transition speed, and the extent to which the first repeated unit already enters the positive basin.

consistently displaced toward the positive region, even when the token-level path contains local reversals or loops.

The trajectories also show why a static endpoint analysis is incomplete. In several examples, movement toward the positive region begins during the first occurrence of the repeated unit, before the detector marks established repetition at the second occurrence. The internal transition may therefore precede the output-level criterion used to declare repetition onset.

K.5 Repetition Trajectories in NOC Space

Figure 10 shows the corresponding repetition trajectories in NOC space. Compared with ACT, the NOC chart is more strongly dominated by its first coordinate in these examples, producing a compact negative region and a clearly separated positive contribution region.

The agreement between ACT and NOC is a representation-space robustness result, not a claim that their coordinates or trajectory shapes should be identical. ACT describes which behavior-associated coordinates are active, whereas NOC describes how strongly those coordinates contribute to the realized MLP update. A trajectory can therefore curve differently in the two spaces while preserving the same qualitative transition from a negative or transitional region to a positive region. The stronger first-coordinate organization in NOC suggests that the functional expression of repetition may be more nearly one-dimensional than its raw activation

dynamics.

K.6 Sycophancy Trajectories in Activation Space

Sycophancy is not a periodic decoding attractor. Its trajectory should therefore not be expected to exhibit the same narrow basin-crossing dynamics as repetition. Instead, the trajectory records how response representations move relative to a broader preference-related region as the answer develops.

Figure 11 shows three ACT examples. The remaining ACT example is presented together with its matched NOC trajectory in Figure 12.

The ACT trajectories show that the onset marker need not coincide with the first entry into a positive-density region. The pre-onset response can already move toward or through that region, while the annotation marks the point at which sycophancy becomes linguistically identifiable. This distinction is expected: the internal state may begin organizing an agreement-seeking response before a specific sycophantic phrase appears.

Unlike repetition, sycophancy trajectories are not monotone along PC1 and do not remain confined to a narrow positive basin. They often curve through the positive region, reflecting the fact that a response must simultaneously encode semantic content, linguistic realization, and user-directed preference. The appropriate conclusion is therefore not that sycophancy follows one universal path, but that its

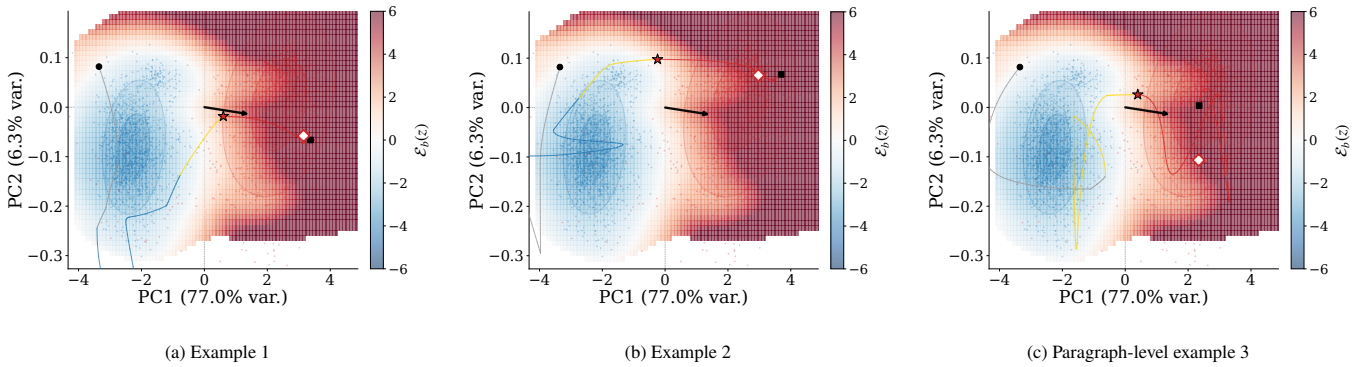


Figure 9: Additional repetition trajectories in ACT space. The gallery demonstrates that entry into the repetition-positive region need not follow a single path: some trajectories cross the chart boundary directly, while others approach it through a curved or partially recurrent path. The common property is the relocation of established repeated tokens toward the positive-density region.

annotated span is organized relative to a reproducible positive behavioral region.

K.7 Matched Sycophancy Trajectory Across ACT and NOC

Figure 12 compares ACT and NOC trajectories for the same sycophancy instance. This matched comparison isolates the effect of representation choice from differences between examples.

The ACT trajectory emphasizes the evolution of the model’s internal state, whereas the NOC trajectory emphasizes the functional contribution of the selected coordinates. Their numerical coordinates and curvature are not directly comparable. The relevant invariant is qualitative: in both representations, the annotated sycophantic span is associated with the positive-density side of the learned chart.

The NOC landscape is more compressed and exhibits a different dominant orientation. This indicates that a broad activation-space trajectory can map to a more compact pattern of functional contribution. Such differences motivate treating ACT and NOC as complementary views rather than expecting one to reproduce the other point by point.

L Base-Anchored Sycophancy Paths Across Post-Training Regimes

The main post-training analysis compares behavioral charts at the level of subspace geometry, occupancy, and behavioral readout. We complement these aggregate measurements with a controlled token-level analysis of how the same candidate answers are represented across checkpoints.

For each example, we fix the prompt and a paired truthful and sycophantic candidate response. The two candidates are teacher-forced through the Base, RL-up, SFT-up, and SFT→RL-up checkpoints. Thus, within each row of Figure 13, the textual input and candidate answers are identical; only the post-training state of the model changes.

This design isolates training-induced representational change from lexical, semantic, and sampling variation. In

particular, a displacement of the shared prompt state or of either candidate path cannot be attributed to a different prompt, a different response, or stochastic generation.

L.1 Base-Anchored Coordinate System

All trajectories are displayed in a single coordinate system defined by the Base model. Let

$$U_{0,b,ACT}$$

denote the Base-model ACT chart for sycophancy, and let

$$\mu_{0,b,ACT}, \quad \sigma_{0,b,ACT}$$

denote the corresponding Base-model standardization statistics. For checkpoint m , candidate type

$$c \in \{\text{syco}, \text{truth}\},$$

and candidate-prefix position t , we compute the anchored coordinate

$$z_{m,c,t}^{(0)} = U_{0,b,ACT}^\top \left(\frac{\phi_{m,b}^{(ACT)}(x, y_{\leq t}^c) - \mu_{0,b,ACT}}{\sigma_{0,b,ACT} + \epsilon} \right). \quad (92)$$

The superscript (0) emphasizes that every checkpoint is evaluated using the Base coordinate scaffold, standardization, and PCA basis.

The background distribution is also fixed across panels. The pale red points are Base-model sycophancy-positive samples, and the pale blue points are Base-model sycophancy-negative samples. Consequently, the four columns are directly comparable: movement across panels reflects how a checkpoint represents the same prompt and candidates relative to the inherited Base chart.

This is an *anchored* visualization rather than a visualization in each checkpoint’s native chart. It therefore answers whether the Base chart continues to organize the internal states of a post-trained model. A trajectory that leaves the high-density Base support does not necessarily indicate weaker sycophancy. Instead, it may indicate that the post-trained model represents the behavior in a chart that is poorly aligned with the Base chart.

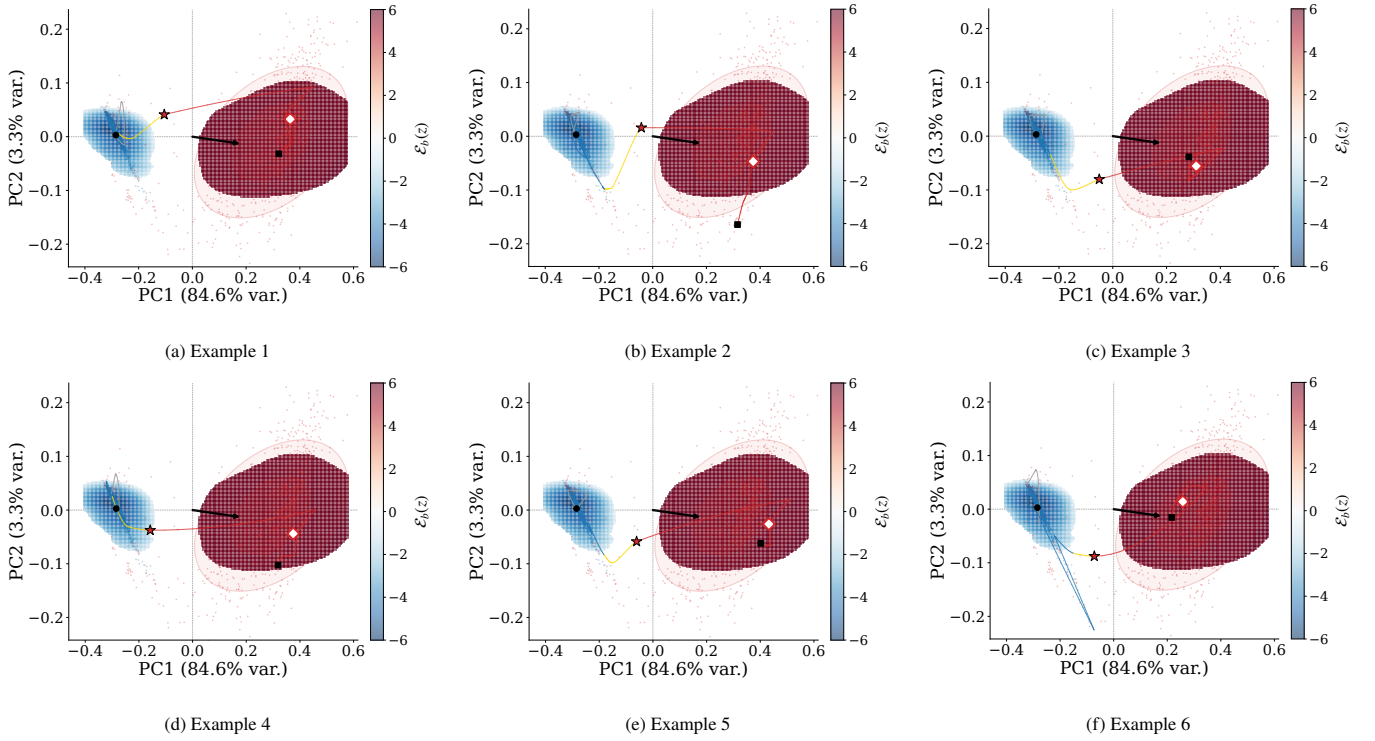


Figure 10: Matched repetition trajectories in NOC space. The NOC representation emphasizes the transition from a compact behavior-negative contribution state to a behavior-positive contribution state. Although the exact path differs across instances, the established repetition span and its mean are consistently located on the positive side of the landscape.

L.2 Candidate-Path and Preference Encoding

For each candidate, the line connects the projected representations of successive answer prefixes. The red path represents the sycophantic candidate, and the blue path represents the truthful candidate. Both begin from the same prompt representation, shown by the gray star.

For a candidate c at expansion length t , define its length-normalized teacher-forced log-likelihood as

$$\ell_{m,c,t} = \frac{1}{t} \sum_{i=1}^t \log p_m(y_i^c | x, y_{<i}^c). \quad (93)$$

The relative preference between the two candidates can be summarized by

$$\Delta_{m,t} = \ell_{m,\text{syc},t} - \ell_{m,\text{truth},t}. \quad (94)$$

Positive values indicate a relative preference for the sycophantic candidate, whereas negative values indicate a relative preference for the truthful candidate.

Node area is a monotone visualization of the corresponding length-normalized likelihood score during candidate expansion. Larger red nodes therefore indicate stronger support for the sycophantic candidate at that stage, while larger blue nodes indicate stronger support for the truthful candidate. Node size encodes preference and should be interpreted separately from spatial position: position describes internal representation in the Base chart, whereas node size describes the model’s evolving candidate likelihood.

L.3 Matched Candidate-Path Gallery

Figure 13 presents four matched candidate pairs. Columns correspond to the Base, RL-up, SFT-up, and SFT→RL-up checkpoints. The intended reading order is horizontal: each row traces the evolution of one fixed prompt–candidate pair through the post-training sequence.

L.4 Cross-Pair Interpretation

Despite substantial instance-level variation, the four rows exhibit a common training-dependent pattern.

The Base chart organizes Base-model candidate divergence. For the Base model, the sycophantic candidates generally move farther toward the positive side of the chart, whereas truthful candidates remain closer to the negative or overlapping region. The exact paths are non-monotonic and can contain local loops, but their broad organization is consistent with the positive and negative sample clouds used to estimate the chart.

RL primarily relocates paths within inherited support. Across the RL-up column, the candidate paths change their curvature, endpoints, and preference markers, but remain largely inside the high-density support of the Base chart. The Base coordinates therefore continue to provide an interpretable description of the RL-up model.

This visual pattern is consistent with the quantitative observation that RL-only checkpoints retain high projector

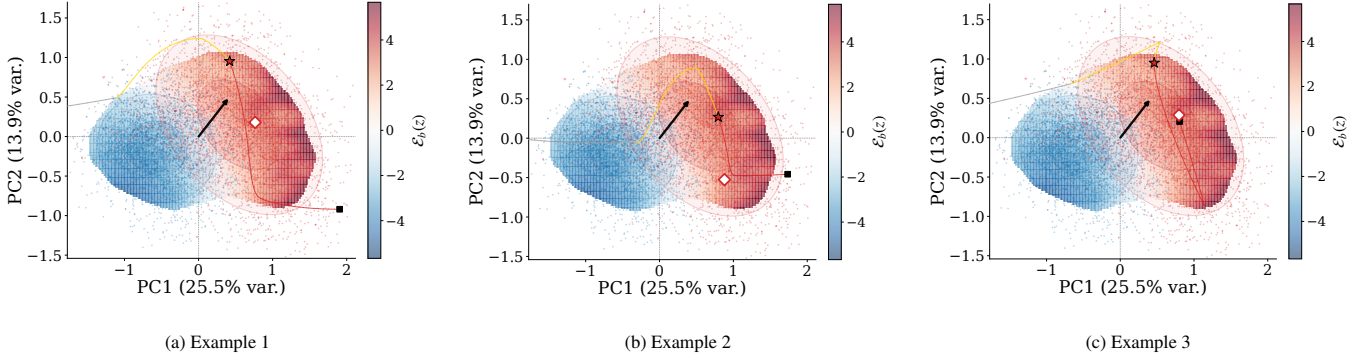


Figure 11: Sycophancy trajectories in ACT space. Prefix and pre-onset states traverse the overlapping or transitional part of the chart, while the annotated sycophantic spans evolve within the positive-density region. Compared with repetition, the trajectories are more heterogeneous and two-dimensional, consistent with sycophancy being a broader response-level alignment mode rather than a periodic decoding attractor.

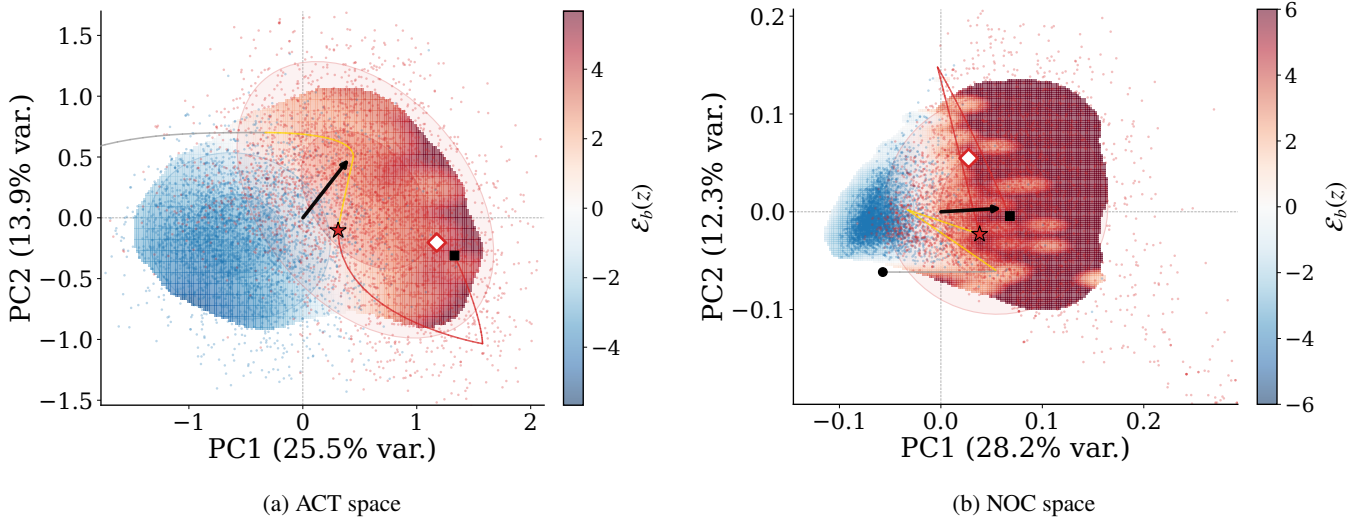


Figure 12: Matched ACT and NOC trajectories for the same sycophancy instance. The two spaces produce different local path shapes because they encode different aspects of the computation. Nevertheless, in both spaces the sycophantic span and its mean are located in the behavior-positive region, supporting the representation robustness of the learned sycophancy chart.

overlap and small principal angles relative to the Base behavioral chart. It supports an interpretation in which RL modifies the distribution of visited chart coordinates and the behavioral readout applied to those coordinates.

SFT produces a systematic mismatch with the Base chart.

Across all four pairs, SFT-up changes the shared contextual state and/or the early answer transition and then sends both candidate paths toward a region that differs from their Base-model routes. The repeated upper-left displacement is unlikely to result from the semantic content of a particular pair because the same pattern occurs across four distinct prompts and candidate sentences.

The fact that sycophantic preference can increase while the corresponding path moves away from the Base positive direction is particularly informative. It suggests that the Base chart is no longer the native behavioral coordinate system

after SFT. In terms of the local behavioral model,

$$p_{\theta}(y_b = 1 | x) \approx \sigma(\beta_{\theta,b}^{\top} U_{\theta,b}^{\top} \bar{\phi}_{\theta}(x) + \tau_{\theta,b}), \quad (95)$$

the SFT result is compatible with a change in $U_{\theta,b}$ and not merely a translation along the fixed Base readout direction.

Sequential RL inherits the SFT-induced regime. The SFT→RL-up column is consistently more similar to the SFT-up column than to the Base or RL-only columns. The sequential RL stage changes local path details and evolving candidate preference, but does not return the representations to the original Base path geometry.

This provides a token-level counterpart to the checkpoint-level Grassmannian results: SFT establishes a shifted representation regime, and subsequent RL largely optimizes within the regime inherited from SFT.

Representation and preference are distinct axes of analysis. Spatial coordinates and node size encode different

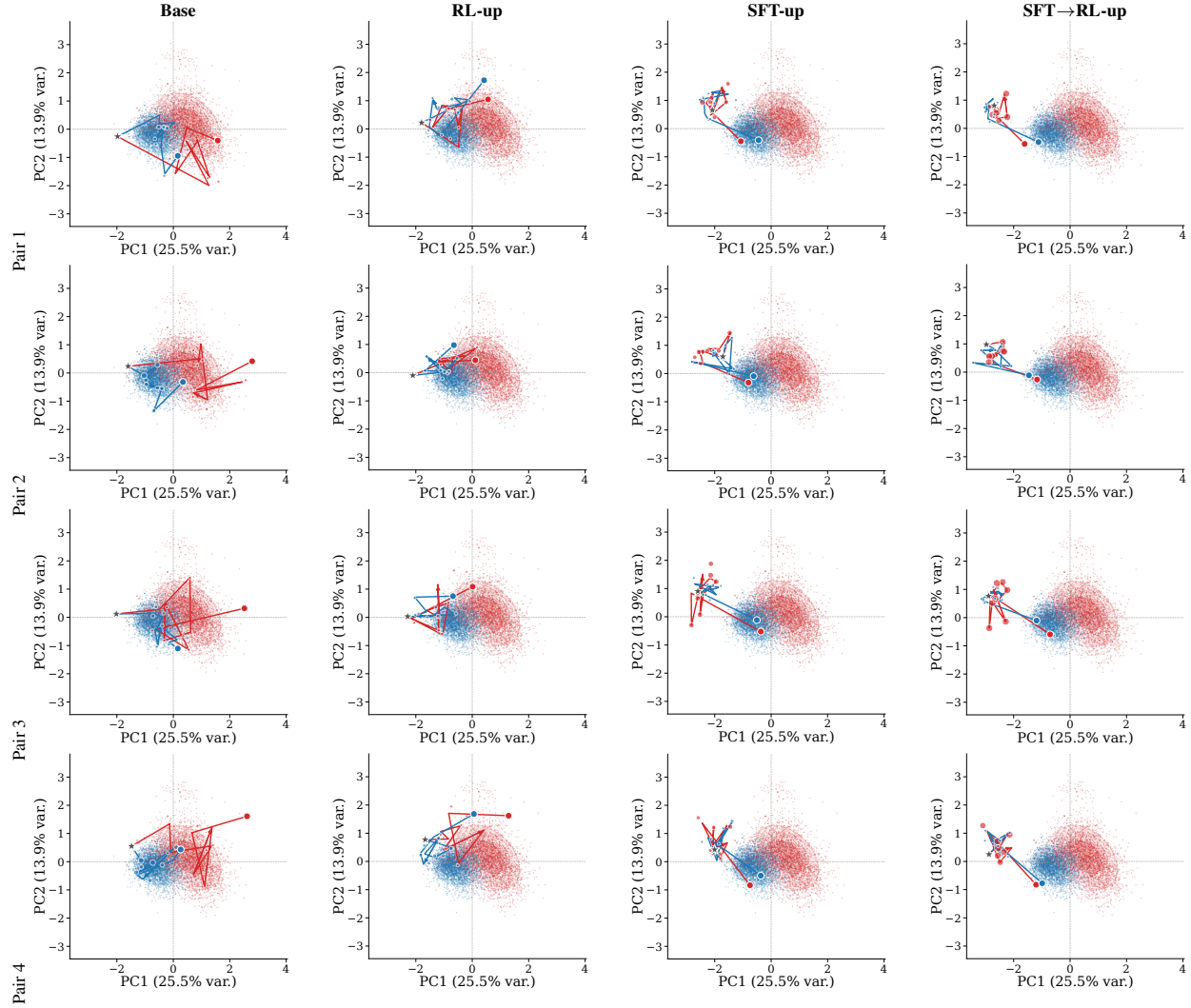


Figure 13: Base-anchored ACT candidate-answer paths for four fixed truthful-sycophantic pairs across post-training regimes. Each row uses exactly the same prompt, truthful candidate, and sycophantic candidate; columns show the Base, RL-up, SFT-up, and SFT→RL-up checkpoints. All panels use the Base-model selected coordinates, standardization statistics, PCA basis, axis orientation, and sample background. Pale red and pale blue points denote Base-model sycophancy-positive and sycophancy-negative samples, respectively. The red polyline is the teacher-forced sycophantic candidate, the blue polyline is the teacher-forced truthful candidate, and the gray star is their shared prompt representation. Successive nodes correspond to answer-prefix representations. Larger red or blue nodes indicate stronger length-normalized likelihood assigned to the corresponding candidate during answer expansion. The fixed background makes representation drift directly visible: RL-only trajectories remain largely inside the inherited Base support, whereas SFT moves the prompt-to-answer trajectories toward a different region of the anchored chart. The SFT→RL-up paths remain closer to the SFT regime than to the original Base regime.

quantities. A path can remain within the Base support while its likelihood preference changes, as is often observed under RL-up. Conversely, a strongly preferred sycophantic candidate can occupy a region that appears non-positive under the Base chart after SFT has changed the representation geometry.

The figures therefore provide an intuitive visualization of the geometry–occupancy–readout decomposition. Path location reflects anchored state occupancy; cross-checkpoint path displacement reflects representation change relative to the Base chart; and node size reflects the evolving candidate preference induced by the model’s readout.

M Full Behavioral Anchor Coordinate Statistics

This appendix reports the complete per-model statistics underlying Section 4.1. We use the term *Behavioral Anchor Coordinates* (BACS) for the sparse coordinates selected by the behavior-discriminative classifier. N_{BAC} denotes the scaffold size, and $k_{0.9}$ denotes the minimum PCA rank required to explain 90% of variance.

For repetition, G^{uniF} measures reliable behavior separation in the shared chart, G^{occ} measures class-conditional chart occupancy, and G^{causal} measures the local behavioral response to intervention. T_{base} is the behavior probability at the no-intervention point. All reported repetition strength results use the leading shared coordinate, $q = 1$. Pairwise CCA values ρ_1 and $\bar{\rho}_q$ are matrix-valued cross-model results and are reported separately as CCA heatmaps rather than repeated in the per-model tables.

For sycophancy, PC1 and Top-3 denote explained-variance ratios, $k_{0.9}$ denotes the 90%-variance dimension, and “Cos.” denotes the reported signed cosine similarity between PC1 and the supervised classifier direction. Each sycophancy result uses 6,678 examples. Since the number of extracted features and fitted PCA components equals N_{BAC} , those redundant columns are omitted.

M.1 Initial Repetition Probe

Table 8 reports the initial repetition-probe accuracy and the number of selected Behavioral Anchor Coordinates across model families. The results test whether a sparse set of BACS is sufficient to identify on-policy repetition before the full geometric analysis.

M.2 Repetition Geometry in ACT Space

In ACT space, causal gain is estimated from additive subspace steering. Consequently, G^{causal} should be interpreted as sensitivity to movement along the estimated activation chart. Table 9 reports the complete per-model repetition-chart statistics in ACT space.

M.3 Repetition Geometry in NOC Space

NOC denotes the normalized outgoing-contribution space; the same quantity is named NOC in the original experimental exports. In NOC space, causal gain is estimated using coordinate-level scaling of the selected scaffold. Table 10

reports the corresponding repetition-chart statistics in NOC space.

M.4 Sycophancy Geometry in ACT Space

Table 11 reports the complete per-model sycophancy PCA statistics in ACT space. These measurements characterize the dimensional compression of the activation chart and the alignment between its leading axis and the supervised classifier direction.

M.5 Sycophancy Geometry in NOC Space

Table 12 reports the corresponding per-model sycophancy PCA statistics in NOC space. Comparison with the ACT results shows how chart compression and supervised-direction alignment change when coordinates are represented by their normalized outgoing contributions.

M.6 Unified NOC Statistics

Table 13 consolidates the repetition and sycophancy statistics in NOC space using a shared per-model schema. This view supports direct comparison of scaffold size, chart strength, causal gain, and low-rank structure across the evaluated models. For both behaviors, the three strength metrics follow the definitions given above and are evaluated in NOC space.

N Layer-Wise Distribution of Behavioral Anchor Coordinates

To provide a concrete view of the coordinate scaffolds used in the geometric analysis, Figure 14 shows the layer-wise locations of the repetition and sycophancy Behavioral Anchor Coordinates (BACS) identified in Llama-3.1-8B. For each transformer layer, the figure reports the number of coordinates assigned a positive coefficient by the corresponding sparse behavioral classifier.

The selected coordinates are not uniformly distributed across model depth. Repetition is supported by a comparatively sparse scaffold, with only a small number of selected coordinates in most layers. Sycophancy is associated with a broader scaffold and exhibits several layer-localized concentrations, particularly in the middle and later portions of the model. The two distributions also differ substantially from one another, consistent with the interpretation that the coordinate-selection procedure identifies behavior-specific scaffolds rather than a single generic behavioral axis.

This visualization is intended as a localization diagnostic rather than a complete measure of layer importance. A larger coordinate count does not by itself imply greater causal contribution, since coordinates may differ in activation magnitude, outgoing contribution, discriminative weight, and steering sensitivity. The subsequent PCA, Fisher, CCA, and intervention analyses therefore operate on the extracted feature values rather than using layer-wise counts as the behavioral geometry itself.

O Full Cross-Model Behavioral Atlas

This appendix reports the complete cross-model alignment results underlying Section ???. The purpose of the analysis

Model	Type	Parameters	#BAC	Accuracy
 Gemma family				
Gemma-3-4B-it	Inst.	4B	27	0.9740
 Qwen family				
Qwen3.5-27B	Base	27B	41	0.9646
 Mistral family				
Mistral-7B-v0.3	Base	7B	53	0.9744
Mistral-Small-3.1-24B-Base-2503	Base	24B	29	0.9658
 Llama family				
Llama-3.1-8B	Base	8B	26	0.9811
Llama-3.1-8B-Instruct	Inst.	8B	24	0.9879
Llama-3.3-70B-Instruct	Inst.	70B	34	0.9853
Mean	–	–	33.4	0.9767

Table 8: Initial repetition probe. A small set of repetition Behavioral Anchor Coordinates predicts on-policy repetition with high accuracy across model families.

Model	Type	G^{uniF}	G^{occ}	G^{causal}	T_{base}	#BAC	$k_{0.9}$	$\log P$
 Gemma family								
Gemma-2-2B	Base	6.8465	0.9809	0.2942	0.8632	32	13	21.4164
Gemma-2-2B-it	Inst.	8.0945	0.7189	0.2750	0.6800	28	15	21.4164
Gemma-3-4B-it	Inst.	7.8447	0.7501	0.2230	0.8696	26	13	22.1096
Gemma-2-9B	Base	6.8896	0.8550	0.1980	0.8704	31	16	22.9205
Gemma-2-9B-it	Inst.	7.7726	0.7816	0.0771	0.8136	26	14	22.9205
Gemma-2-27B	Base	8.8439	0.6566	0.1306	0.8886	38	14	24.0191
Gemma-2-27B-it	Inst.	6.9798	0.8328	0.1554	0.7594	34	13	24.0191
 Qwen family								
Qwen2.5-0.5B	Base	7.5528	0.7049	0.2108	0.8562	48	26	20.0301
Qwen2.5-0.5B-Instruct	Inst.	7.1556	0.7323	0.5994	0.8219	48	25	20.0301
Qwen2.5-7B	Base	6.9355	0.8142	0.2648	0.9054	37	16	22.6692
Qwen2.5-7B-Instruct	Inst.	8.0421	0.7786	0.2072	0.8801	42	21	22.6692
Qwen3.5-27B	Base	10.3330	0.5082	0.0831	0.8978	45	22	24.0191
Qwen2.5-32B	Base	6.8184	0.8959	0.0881	0.9163	61	27	24.1890
Qwen2.5-32B-Instruct	Inst.	6.3355	1.0557	0.2112	0.8817	44	20	24.1890
 Mistral family								
Mistral-7B-v0.3	Base	10.3875	0.6061	0.1661	0.8833	56	25	22.6692
Mistral-7B-Instruct-v0.3	Inst.	8.3211	0.6757	0.2480	0.8713	32	18	22.6692
Mistral-Nemo-Base-2407	Base	5.6082	1.0666	0.0776	0.8940	43	21	23.2082
Mistral-Nemo-Instruct-2407	Inst.	4.6797	1.2299	0.1026	0.8975	42	22	23.2082
Mistral-Small-3.1-24B-Base-2503	Base	8.9275	0.5998	0.1051	0.8963	28	12	23.9013
Mistral-Small-3.1-24B-Instruct-2503	Inst.	7.8805	0.6776	0.3037	0.8290	34	13	23.9013
 Llama family								
Llama-3.1-8B	Base	3.6277	1.1152	0.0891	0.9082	26	9	22.8027
Llama-3.1-8B-Instruct	Inst.	4.3802	1.1844	0.1823	0.7599	20	8	22.8027
Llama-3.3-70B-Instruct	Inst.	4.5069	1.1109	0.1106	0.8477	31	9	24.9718
Mean	–	7.1636	0.8405	0.1914	0.8561	37.04	17.04	–
Median	–	7.1556	0.7816	0.1823	0.8713	34	16	–

Table 9: Complete repetition-chart statistics in ACT space across 23 models. All rows use $q = 1$. $\log P$ is the natural logarithm of parameter count. ACT causal gain is estimated from subspace steering.

is to determine whether behavioral charts estimated independently in models with different architectures, parameter scales, and post-training histories exhibit a shared cross-model geometric core.

We emphasize that cross-model alignment does not require neuron-index correspondence. Each model is first rep-

resented in its own behavior-enriched coordinate scaffold and its own low-dimensional PCA chart. Alignment is then measured through matched sample-level coordinates. The relevant question is therefore whether two models organize the same behavioral examples in linearly related ways, rather than whether they use identical hidden units.

Model	Type	G^{uniF}	G^{occ}	G^{causal}	T_{base}	$\#_{\text{BAC}}$	$k_{0.9}$	$\log P$
◆ Gemma family								
Gemma-2-2B	Base	6.2492	1.0786	0.1204	0.8837	32	13	21.4164
Gemma-2-2B-it	Inst.	5.6593	0.8830	0.2333	0.7581	28	15	21.4164
Gemma-3-4B-it	Inst.	8.4143	0.6596	0.1513	0.8759	26	13	22.1096
Gemma-2-9B	Base	4.2760	0.9555	0.1603	0.8875	31	15	22.9205
Gemma-2-9B-it	Inst.	4.2142	0.9589	0.0477	0.8157	26	13	22.9205
Gemma-2-27B	Base	8.8803	0.6375	0.1021	0.8952	38	15	24.0191
Gemma-2-27B-it	Inst.	5.1277	1.0103	0.1138	0.7796	34	13	24.0191
🌀 Qwen family								
Qwen2.5-0.5B	Base	9.5037	0.6870	0.0216	0.8695	48	22	20.0301
Qwen2.5-0.5B-Instruct	Inst.	10.1911	0.6394	0.1244	0.8484	48	22	20.0301
Qwen2.5-7B	Base	8.1407	0.6886	0.1100	0.9145	37	15	22.6692
Qwen2.5-7B-Instruct	Inst.	8.2838	0.7411	0.1344	0.8962	42	20	22.6692
Qwen3.5-27B	Base	11.2543	0.4830	0.0322	0.9023	45	19	24.0191
Qwen2.5-32B	Base	7.3041	0.8637	0.0487	0.9187	61	24	24.1890
Qwen2.5-32B-Instruct	Inst.	4.6760	1.1594	0.0365	0.8874	44	18	24.1890
🏠 Mistral family								
Mistral-7B-v0.3	Base	12.4879	0.4643	0.0901	0.8935	56	21	22.6692
Mistral-7B-Instruct-v0.3	Inst.	11.2649	0.5000	0.1500	0.8914	32	17	22.6692
Mistral-Nemo-Base-2407	Base	4.5099	1.0969	0.0247	0.8955	43	20	23.2082
Mistral-Nemo-Instruct-2407	Inst.	6.0374	0.9124	0.0222	0.9009	42	20	23.2082
Mistral-Small-3.1-24B-Base-2503	Base	5.1098	0.9186	0.0698	0.9058	28	12	23.9013
Mistral-Small-3.1-24B-Instruct-2503	Inst.	6.3496	0.7855	0.0769	0.8346	34	13	23.9013
🌀 Llama family								
Llama-3.1-8B	Base	4.1244	1.0646	0.0463	0.9115	26	9	22.8027
Llama-3.1-8B-Instruct	Inst.	4.6608	1.1588	0.1229	0.7819	20	7	22.8027
Llama-3.3-70B-Instruct	Inst.	4.6690	1.1443	0.1029	0.8598	31	8	24.9718
Mean	–	7.0169	0.8474	0.0932	0.8699	37.04	15.83	–
Median	–	6.2492	0.8830	0.1021	0.8875	34	15	–

Table 10: Complete repetition-chart statistics in normalized outgoing-contribution (NOC) space across 23 models. All rows use $q = 1$. NOC causal gain is estimated by scaling selected Behavioral Anchor Coordinates.

O.1 Model Roster and Common Ordering

All heatmaps use the common A–W model order in Table 14. This ordering must also be used for the random-coordinate panels so that corresponding cells represent the same model pair.

O.2 Evaluation Protocol

For every model m , let

$$Z_m = \begin{bmatrix} z_m(x_1)^\top \\ \vdots \\ z_m(x_n)^\top \end{bmatrix} \in \mathbb{R}^{n \times k_m}$$

denote its chart coordinates on the common matched sample set. The chart dimension k_m is the minimum number of PCA components required to explain 90% of the variance in the corresponding chart. All alignment and linear-association summaries are computed from these matched sample-level coordinates and are interpreted as descriptive cross-model associations.

Classifier-direction alignment. Let w_m be the sparse logistic-classifier direction for model m , projected into the corresponding PCA coordinate system. The sample-level

classifier score is

$$s_m(x) = w_m^\top z_m(x). \quad (96)$$

For models m and m' , we report

$$A_{\text{cls}}(m, m') = |\text{corr}(s_m(x), s_{m'}(x))|. \quad (97)$$

This measures the strength of cross-model score correspondence up to a global sign.

Canonical-correlation alignment. For a pair of models, CCA finds matrices C_m and $C_{m'}$ such that corresponding coordinates in

$$Z_m C_m \quad \text{and} \quad Z_{m'} C_{m'}$$

have maximal correlation under the canonical orthogonality constraints. Let

$$\rho_1 \geq \rho_2 \geq \dots \geq \rho_q, \quad q = \min(k_m, k_{m'}),$$

be the resulting canonical correlations on the matched sample set. We report the primary score

$$A_{\text{CCA}}^{(1)}(m, m') = \rho_1, \quad (98)$$

and the mean score

$$A_{\text{CCA}}^{(\text{mean})}(m, m') = \frac{1}{q} \sum_{i=1}^q \rho_i. \quad (99)$$

Model	Type	#BAC	PC1	Top-3	$k_{0.9}$	$k_{0.9}/N_{\text{BAC}}$	Cos.
 Qwen family							
Qwen2.5-0.5B-Instruct	Inst.	78	0.6663	0.8460	6	0.077	0.0579
Qwen2.5-0.5B	Base	83	0.6011	0.8782	4	0.048	0.0317
Qwen2.5-32B-Instruct	Inst.	62	0.8610	0.9620	2	0.032	0.2737
Qwen2.5-32B	Base	75	0.7952	0.9003	3	0.040	0.1419
Qwen2.5-7B-Instruct	Inst.	61	0.8599	0.9646	2	0.033	0.0336
Qwen2.5-7B	Base	69	0.7673	0.9606	3	0.043	0.0780
Qwen3.5-27B	Base	101	0.3035	0.5806	22	0.218	0.0000
 Gemma family							
Gemma-2-27B-it	Inst.	53	0.6587	0.9440	2	0.038	0.0917
Gemma-2-27B	Base	55	0.4101	0.7853	4	0.073	0.0300
Gemma-2-2B-it	Inst.	42	0.2469	0.5118	14	0.333	0.0972
Gemma-2-2B	Base	75	0.1822	0.4055	19	0.253	0.0866
Gemma-2-9B-it	Inst.	54	0.3213	0.6398	7	0.130	0.0000
Gemma-2-9B	Base	51	0.2023	0.5339	10	0.196	0.0728
Gemma-3-4B-it	Inst.	54	0.3093	0.6828	7	0.130	0.0000
 Llama family							
Llama-3.1-8B-Instruct	Inst.	78	0.2552	0.5008	20	0.256	0.0995
Llama-3.1-8B	Base	92	0.4262	0.7749	8	0.087	0.0029
Llama-3.3-70B-Instruct	Inst.	83	0.4069	0.6925	15	0.181	0.0326
 Mistral family							
Mistral-7B-Instruct-v0.3	Inst.	50	0.3278	0.5661	13	0.260	0.0445
Mistral-7B-v0.3	Base	46	0.3466	0.6108	15	0.326	0.0000
Mistral-Nemo-Base-2407	Base	78	0.5083	0.6841	12	0.154	0.0190
Mistral-Nemo-Instruct-2407	Inst.	67	0.6764	0.8163	6	0.090	0.0210
Mistral-Small-3.1-24B-Base-2503	Base	61	0.3486	0.7351	8	0.131	0.1178
Mistral-Small-3.1-24B-Instruct-2503	Inst.	52	0.3563	0.7094	9	0.173	0.1002
Mean	–	66.09	0.4712	0.7255	9.17	0.144	0.0623
Median	–	62	0.4069	0.7094	8	0.130	0.0445

Table 11: Complete sycophancy PCA statistics in ACT space. Each model is evaluated on 6,678 examples. ACT representations are strongly compressed, requiring only 14.4% of the selected scaffold dimensions on average to explain 90% of variance.

The primary score asks whether at least one strongly shared direction exists. The mean score asks whether alignment extends across the retained chart rather than being concentrated in a single canonical axis.

Top-two-PC linear association. For source model m and target model m' , we consider the four one-dimensional mappings

$$z_{m,i} \longrightarrow z_{m',j}, \quad i, j \in \{1, 2\}.$$

For each mapping, we report the absolute correlation between the fitted target coordinate and the observed target coordinate on the common matched sample set. The top-two-PC score is

$$A_{\text{probe}}^{(2)}(m, m') = \max_{i,j \in \{1,2\}} |\text{corr}(\hat{z}_{m',j}, z_{m',j})|. \quad (100)$$

This score is useful when the behavior-aligned direction may be PC1 in one model and PC2 in another.

Full-chart directed linear association. We additionally fit a linear map

$$\hat{Z}_{m'} = Z_m W_{m \rightarrow m'} + b_{m \rightarrow m'} \quad (101)$$

from the source chart to the complete variance-thresholded target chart. Its score is

$$A_{\text{probe}}^{(k)}(m \rightarrow m') = \frac{1}{k_{m'}} \sum_{j=1}^{k_{m'}} \text{corr}(\hat{Z}_{m'}[:, j], Z_{m'}[:, j]). \quad (102)$$

Unlike pairwise CCA and one-dimensional association, this quantity is directed:

$$A_{\text{probe}}^{(k)}(m \rightarrow m') \neq A_{\text{probe}}^{(k)}(m' \rightarrow m)$$

in general. The asymmetry reflects differences in chart rank, redundancy, and how readily one model’s representation spans another model’s residual coordinates.

Random-coordinate controls. For every model and feature space, we replace the behavior-associated coordinate scaffold with a random coordinate set of the same cardinality and repeat standardization, PCA, CCA, and linear-association analysis. These controls preserve the ambient model, sample identities, dimensionality-selection procedure, and evaluation protocol while removing behavior-informed coordinate selection.

The random controls are especially important for CCA. In finite samples, CCA can find a strongly correlated leading pair between unrelated high-dimensional feature sets. A

Model	Type	#BAC	PC1	Top-3	$k_{0.9}$	$k_{0.9}/N_{\text{BAC}}$	Cos.
◆ Gemma family							
Gemma-2-2B	Base	75	0.2606	0.4738	29	0.387	0.0511
Gemma-2-2B-it	Inst.	42	0.2884	0.5270	18	0.429	0.7440
Gemma-2-9B	Base	51	0.3869	0.5920	18	0.353	0.1225
Gemma-2-9B-it	Inst.	54	0.3111	0.5153	24	0.444	0.7365
Gemma-3-4B-it	Inst.	54	0.3615	0.5434	22	0.407	0.5338
Gemma-2-27B	Base	55	0.2501	0.4926	24	0.436	0.5971
Gemma-2-27B-it	Inst.	53	0.2919	0.5084	23	0.434	0.6870
🏠 Mistral family							
Mistral-7B-Instruct-v0.3	Inst.	50	0.3990	0.5824	21	0.420	0.4227
Mistral-7B-v0.3	Base	46	0.4061	0.6286	15	0.326	0.4161
Mistral-Nemo-Base-2407	Base	78	0.2282	0.5039	26	0.333	-0.1646
Mistral-Nemo-Instruct-2407	Inst.	67	0.2799	0.5080	25	0.373	0.5667
Mistral-Small-3.1-24B-Instruct-2503	Inst.	52	0.3440	0.6273	15	0.288	0.2482
Mistral-Small-3.1-24B-Base-2503	Base	61	0.5269	0.7282	12	0.197	0.2025
∞ Llama family							
Llama-3.1-8B	Base	92	0.3093	0.5056	29	0.315	0.1757
Llama-3.1-8B-Instruct	Inst.	78	0.2816	0.4952	30	0.385	0.6781
Llama-3.3-70B-Instruct	Inst.	83	0.2859	0.4795	30	0.361	0.6731
🦋 Qwen family							
Qwen2.5-0.5B	Base	83	0.2025	0.3670	38	0.458	0.6242
Qwen2.5-0.5B-Instruct	Inst.	78	0.1917	0.3517	36	0.462	0.6241
Qwen2.5-7B	Base	69	0.2924	0.4997	28	0.406	0.5060
Qwen2.5-7B-Instruct	Inst.	61	0.2759	0.4415	30	0.492	0.5745
Qwen2.5-32B	Base	75	0.2603	0.4714	32	0.427	0.6337
Qwen2.5-32B-Instruct	Inst.	62	0.2865	0.6132	24	0.387	0.1861
Qwen3.5-27B	Base	101	0.3149	0.5026	34	0.337	0.6593
Mean	–	66.09	0.3059	0.5199	25.35	0.385	0.4565
Median	–	62	0.2884	0.5056	25	0.387	0.5667

Table 12: Complete sycophancy PCA statistics in normalized outgoing-contribution (NOC) space. NOC charts are less compressed than ACT charts but often align more strongly with the supervised classifier direction.

meaningful behavioral result should therefore exceed the random baseline not only for ρ_1 , but also for mean CCA and matched-sample linear association.

O.3 Aggregate Results

Table 15 reports the mean off-diagonal score for each analysis. For the directed k -dimensional association scores, the mean is taken over all ordered source–target pairs.

The leading canonical coordinate is nearly saturated in both spaces. However, the random primary-CCA values are also large, particularly in NOC. Primary CCA therefore establishes the existence of a highly alignable axis but does not by itself show that this axis is behavior-specific.

Mean CCA provides a stronger test. The ACT score increases from 0.222 under random selection to 0.515 under behavior-associated selection, an absolute increase of 0.293. In NOC, the score increases from 0.233 to 0.648, an increase of 0.415. Thus, behavior-informed selection improves alignment across multiple canonical coordinates, not only the most favorable pair.

The linear-association summaries provide a complementary descriptive comparison. In ACT, the top-two-PC score increases from 0.266 to 0.784, while the complete-chart score increases from 0.215 to 0.473. In NOC, the top-two-PC

score increases from 0.495 to 0.958, and the complete-chart score increases from 0.234 to 0.581. The particularly strong NOC result suggests that functional contribution coordinates expose a shared behavior axis that is less dependent on a model’s raw activation basis.

O.4 Classifier-Direction Alignment

Figure 15 compares the sample-level scores of the sparse supervised directions. The mean off-diagonal absolute correlation is 0.950, and the mean for matched base/instruction pairs is 0.980.

The near-universal absolute score correspondence does not require literal neuron correspondence and does not imply that the classifier vectors occupy equivalent ambient coordinates. Rather, after each model is expressed in its own chart, the supervised behavior-positive axes induce strongly corresponding score patterns over the common examples, up to a global sign.

O.5 Canonical-Correlation Atlas

Figure 16 reports primary and mean CCA in ACT and NOC.

The difference between primary and mean CCA is scientifically informative. Most model pairs possess at least one strongly alignable direction, explaining the nearly sat-

Model	Repetition geometry ($q = 1$)							Sycophancy geometry (6,678 examples)						
	#BAC	NOC space						#BAC	NOC space					
		G^{uniF}	G^{occ}	G^{causal}	PC1	Top-3	$k_{0.9}$		G^{uniF}	G^{occ}	G^{causal}	PC1	Top-3	$k_{0.9}$
◆ Gemma family														
Gemma-2-2B	32	6.2492	1.0786	0.1204	0.7199	0.8741	13	75	8.23263	0.493434	0.0452983	0.2606	0.4738	29
Gemma-2-2B-it	28	5.6593	0.8830	0.2333	0.7369	0.8398	15	42	7.54262	0.469359	0.0189962	0.2884	0.5270	18
Gemma-3-4B-it	26	8.4143	0.6596	0.1513	0.7059	0.8165	13	54	7.03847	0.5389	0.307394	0.3615	0.5434	22
Gemma-2-9B	31	4.2760	0.9555	0.1603	0.7752	0.8912	15	51	9.47754	0.492795	0.111633	0.3869	0.5920	18
Gemma-2-9B-it	26	4.2142	0.9589	0.0477	0.8345	0.8969	13	54	8.80361	0.474639	0.0128778	0.3111	0.5153	24
Gemma-2-27B	38	8.8803	0.6375	0.1021	0.7261	0.8487	15	55	8.33293	0.573001	0.0558019	0.2501	0.4926	24
Gemma-2-27B-it	34	5.1277	1.0103	0.1138	0.7731	0.8772	13	53	8.00736	0.632828	0.133148	0.2919	0.5084	23
🌀 Qwen family														
Qwen2.5-0.5B	48	9.5037	0.6870	0.0216	0.5832	0.7412	22	83	7.08938	0.491728	0.0295526	0.2025	0.3670	38
Qwen2.5-0.5B-Instruct	48	10.1911	0.6394	0.1244	0.6390	0.7670	22	78	7.4928	0.546409	0.0207409	0.1917	0.3517	36
Qwen2.5-7B	37	8.1407	0.6886	0.1100	0.7271	0.8594	15	69	8.68511	0.583384	-0.000877529	0.2924	0.4997	28
Qwen2.5-7B-Instruct	42	8.2838	0.7411	0.1344	0.7333	0.8512	20	61	8.50326	0.624147	0.0223223	0.2759	0.4415	30
Qwen3.5-27B	45	11.2543	0.4830	0.0322	0.6902	0.8345	19	101	9.33057	0.553323	0.01328	0.3149	0.5026	34
Qwen2.5-32B	61	7.3041	0.8637	0.0487	0.6887	0.8035	24	75	9.52139	0.44551	0.00741185	0.2603	0.4714	32
Qwen2.5-32B-Instruct	44	4.6760	1.1594	0.0365	0.6992	0.8235	18	62	9.92692	0.510514	-0.000686973	0.2865	0.6132	24
🏠 Mistral family														
Mistral-7B-v0.3	56	12.4879	0.4643	0.0901	0.7102	0.7996	21	46	7.06802	0.648702	0.0483211	0.4061	0.6286	15
Mistral-7B-Instruct-v0.3	32	11.2649	0.5000	0.1500	0.7132	0.8112	17	50	9.17742	0.571795	0.0266025	0.3990	0.5824	21
Mistral-Nemo-Base-2407	43	4.5099	1.0969	0.0247	0.7907	0.8568	20	78	6.82295	0.675682	0.0237964	0.2282	0.5039	26
Mistral-Nemo-Instruct-2407	42	6.0374	0.9124	0.0222	0.7391	0.8168	20	67	7.33621	0.682483	0.0729401	0.2799	0.5080	25
Mistral-Small-3.1-24B-Base-2503	28	5.1098	0.9186	0.0698	0.7466	0.8556	12	61	7.2653	0.692815	0.200294	0.5269	0.7282	12
Mistral-Small-3.1-24B-Instruct-2503	34	6.3496	0.7855	0.0769	0.7206	0.8402	13	52	6.2152	0.75018	0.172712	0.3440	0.6273	15
∞ Llama family														
Llama-3.1-8B	26	4.1244	1.0646	0.0463	0.8063	0.8844	9	92	8.30751	0.513425	0.0395678	0.3093	0.5056	29
Llama-3.1-8B-Instruct	20	4.6608	1.1588	0.1229	0.8462	0.9031	7	78	7.62933	0.531903	0.0438382	0.2816	0.4952	30
Llama-3.3-70B-Instruct	31	4.6690	1.1443	0.1029	0.8594	0.9191	8	83	10.6307	0.502569	0.0203946	0.2859	0.4795	30
Mean	37.04	7.0169	0.8474	0.0932	0.7376	0.8440	15.83	66.09	8.19292	0.565197	0.0619721	0.3059	0.5199	25.35
Median	34	6.2492	0.8830	0.1021	0.7271	0.8487	15	62	8.23263	0.546409	0.0295526	0.2884	0.5056	25

Table 13: Unified per-model Behavioral Anchor Coordinate statistics in the NOC representation space. Repetition is shown on the left and sycophancy on the right using the same retained column schema.

urated primary matrix. The lower mean correlations show that the remaining chart coordinates are not identical. The atlas is therefore better described as a shared leading core with model-specific residual dimensions than as one globally identical subspace.

The representation spaces also differ. Mean CCA is 0.515 in ACT and 0.648 in NOC. ACT preserves more architecture-specific information about the model’s runtime state, while NOC removes some basis-specific variability by expressing selected coordinates through their normalized functional contribution.

O.6 Random-Coordinate CCA Controls

Figure 17 repeats the CCA analysis after replacing the behavior-associated scaffolds with matched-size random coordinates.

The random controls change the interpretation of primary CCA. The random primary scores of 0.667 in ACT and 0.817 in NOC are too large to regard a single canonical correlation as decisive evidence. This effect can arise because CCA selects the most favorable linear combination from two high-dimensional feature populations evaluated on matched

examples.

The random mean correlations, by contrast, are only 0.222 and 0.233. The large behavior-associated margins for mean CCA therefore indicate that alignment extends beyond a chance-optimized first pair.

O.7 Cross-Model Linear-Association Atlas

Figure 18 reports cross-model linear associations over the matched sample set.

The top-two-PC results are strong across most model pairs. The mean score is 0.784 in ACT and 0.958 in NOC. This shows that a leading behavioral coordinate is strongly associated with a corresponding coordinate in another model even when its PCA index changes.

Full-chart association is more demanding. Its mean score decreases to 0.473 in ACT and 0.581 in NOC. The decrease should not be interpreted as a failure of the atlas. Rather, it distinguishes a broadly shared leading behavioral coordinate from higher-dimensional residual variation that is specific to architecture, scale, or post-training history.

The directed matrices also show asymmetry. A low-rank or redundant source chart may predict a target chart dif-

ID	Model	ID	Model	ID	Model
A	Qwen2.5-0.5B-Instruct	I	Gemma-2-27B	Q	Llama-3.3-70B-Instruct
B	Qwen2.5-0.5B	J	Gemma-2-2B-it	R	Mistral-7B-Instruct-v0.3
C	Qwen2.5-32B-Instruct	K	Gemma-2-2B	S	Mistral-7B-v0.3
D	Qwen2.5-32B	L	Gemma-2-9B-it	T	Mistral-Nemo-Base-2407
E	Qwen2.5-7B-Instruct	M	Gemma-2-9B	U	Mistral-Nemo-Instruct-2407
F	Qwen2.5-7B	N	Gemma-3-4B-it	V	Mistral-Small-3.1-24B-Base-2503
G	Qwen3.5-27B	O	Llama-3.1-8B-Instruct	W	Mistral-Small-3.1-24B-Instruct-2503
H	Gemma-2-27B-it	P	Llama-3.1-8B		

Table 14: Model identifiers used in all cross-model heatmaps. A common ordering is necessary for direct visual comparison between behavior-associated and random-coordinate matrices.

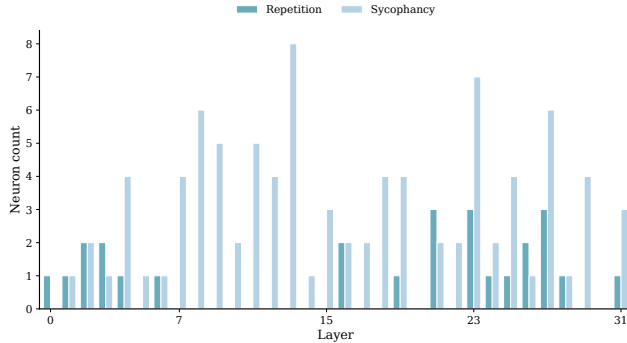


Figure 14: Layer-wise distribution of repetition and sycophancy Behavioral Anchor Coordinates in Llama-3.1-8B. Each bar reports the number of positively behavior-associated coordinates selected in the corresponding transformer layer. Repetition is represented by a comparatively sparse scaffold, whereas sycophancy is supported by a broader and more widely distributed coordinate population. These counts visualize the coordinate scaffolds used to estimate the corresponding low-dimensional behavioral charts; they should not be interpreted as direct measurements of layer-wise causal importance.

ferently from the reverse mapping. This is expected when $k_m \neq k_{m'}$ or when one model’s chart contains residual coordinates absent from the other.

O.8 Random-Coordinate Linear-Association Controls

Figure 19 shows the corresponding random-coordinate probes.

The top-two-PC random NOC baseline of 0.495 is non-negligible. This is a consequence of both the max-over-four selection rule and broadly shared non-behavioral functional factors. The behavior-associated score of 0.958 nevertheless exceeds it by 0.463.

The complete-chart random controls are more restrictive. Their means are 0.215 in ACT and 0.234 in NOC, compared with 0.473 and 0.581 for behavior-associated charts. This indicates that the observed multi-dimensional cross-model association is not explained by model-wide generic covariance alone.

Metric	ACT		NOC	
	BAC	Random	BAC	Random
Primary CCA	0.938	0.667	0.974	0.817
Mean CCA	0.515	0.222	0.648	0.233
Top-2-PC probing	0.784	0.266	0.958	0.495
k -dim. probing	0.473	0.215	0.581	0.234

Table 15: Mean off-diagonal cross-model alignment. Behavior-associated charts exceed matched-size random-coordinate baselines in both ACT and NOC. The difference is most diagnostic for mean CCA and full-chart linear association, because primary CCA is also elevated for random coordinates.

Metric	Space	All	Same fam.	Cross fam.	Base/Inst.
Mean CCA	ACT	0.515	0.627	0.480	0.754
	NOC	0.648	0.677	0.639	0.798
Top-2 probe	ACT	0.784	0.816	0.775	0.905
	NOC	0.958	0.982	0.950	0.995
k -dim. probe	ACT	0.473	0.582	0.439	0.718
	NOC	0.581	0.634	0.565	0.775

Table 16: Cross-model alignment stratified by model relationship. “Base/Inst.” denotes the ten matched base/instruction pairs with the same model family and parameter scale; the k -dimensional value averages both mapping directions. ACT shows a larger same-family advantage, whereas NOC remains strongly aligned across families.

O.9 Family Structure and Matched Base–Instruction Pairs

To characterize the residual atlas structure, we divide pairs into models from the same family and models from different families. Table 16 reports descriptive averages. The model set is not family-balanced, so these values are used to interpret the heatmaps rather than as an independent hypothesis test.

The family gap is largest for ACT mean CCA and full-chart association. Mean CCA is 0.627 within families but 0.480 across families, and full-chart association is 0.582 versus 0.439. This indicates that raw activation-state geometry retains architectural and family-specific organization.

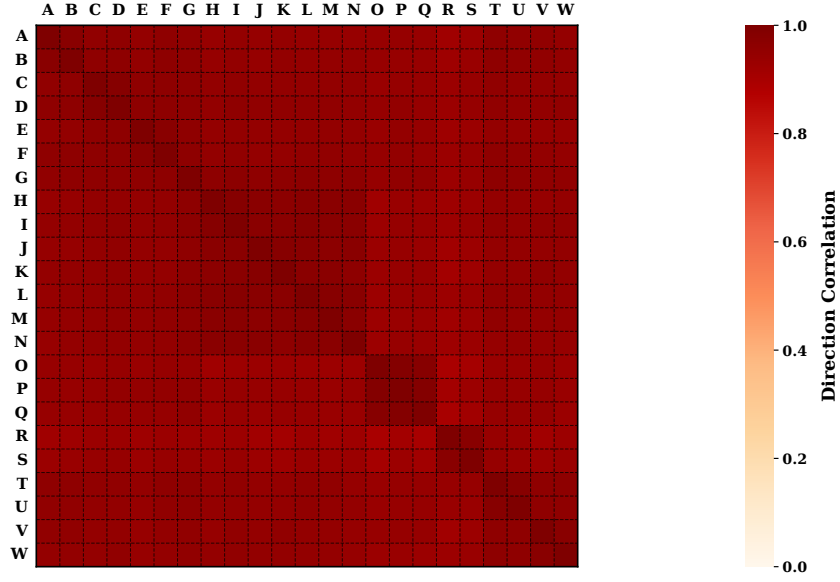


Figure 15: Cross-model alignment of sparse classifier-direction scores. Each cell is the absolute Pearson correlation between the two models’ matched-sample classifier scores. The high absolute correlations show that independently fitted supervised axes exhibit strong score correspondence across architectures and scales, up to a global sign. Because the directions are label-supervised, this result is treated as a complementary sanity check rather than the primary evidence for unsupervised chart universality.

NOC is more uniform across families. Its mean CCA is 0.677 within families and 0.639 across families, while top-two-PC association remains 0.950 even across families. The normalized contribution representation therefore appears to remove some basis-specific variation while preserving behaviorally relevant functional structure.

Matched base/instruction pairs are especially aligned. Their mean CCA reaches 0.754 in ACT and 0.798 in NOC, and their top-two-PC association reaches 0.905 and 0.995. Instruction tuning therefore does not erase the family-specific shared core, although the post-training analyses in Section 5 show that controlled SFT can still rotate a behavioral chart substantially relative to its exact base checkpoint.

O.10 Interpretation: Shared Core, Non-Universal Residuals

The combination of metrics supports a layered interpretation of cross-model behavioral universality.

First, classifier-score alignment and primary CCA show that nearly every model pair contains a strongly shared behavior-oriented axis. Second, mean CCA and full-chart association show that this universality is incomplete: the remaining coordinates retain family- and model-specific structure. Third, random controls demonstrate that behavior-associated coordinate selection is necessary for the multi-dimensional effect. Finally, the stronger NOC results indicate that functional contribution geometry is more consistently aligned across models than raw activation geometry.

The appropriate claim is therefore not

$$U_{m,b} = U_{m',b},$$

which would be undefined across models with different ambient bases. Instead, the results support

$$C_m^\top z_{m,b}(x) \approx C_{m'}^\top z_{m',b}(x)$$

for a low-dimensional shared core, together with model-specific residual coordinates outside that core.

This distinction also motivates Universal Fisher Strength. A direction should not be considered universal merely because one CCA component can be made highly correlated. It should separate behavioral classes, exhibit consistent cross-model alignment, and outperform matched random geometry. Mean CCA and full-chart linear association provide the cross-model reliability evidence used to support that interpretation.

P Cross-Family Repetition Steering Results

We report the complete repetition-steering sweeps for 23 models from the Qwen, Gemma, Llama, and Mistral families. These experiments complement the aggregate causal-gain statistic by exposing the complete dose–response curves, their model-to-model variation, and their relationship to generated-token perplexity.

Operators and plotting convention. The ACT panels use additive behavioral-chart steering:

$$x' = x + \alpha U_k U_k^\top (x - \mu). \quad (103)$$

The second set of panels uses the neurons selected by the NOC scaffold. For multiplicative neuron scaling, we display a zero-centered strength $\alpha = s - 1$, where s is the actual

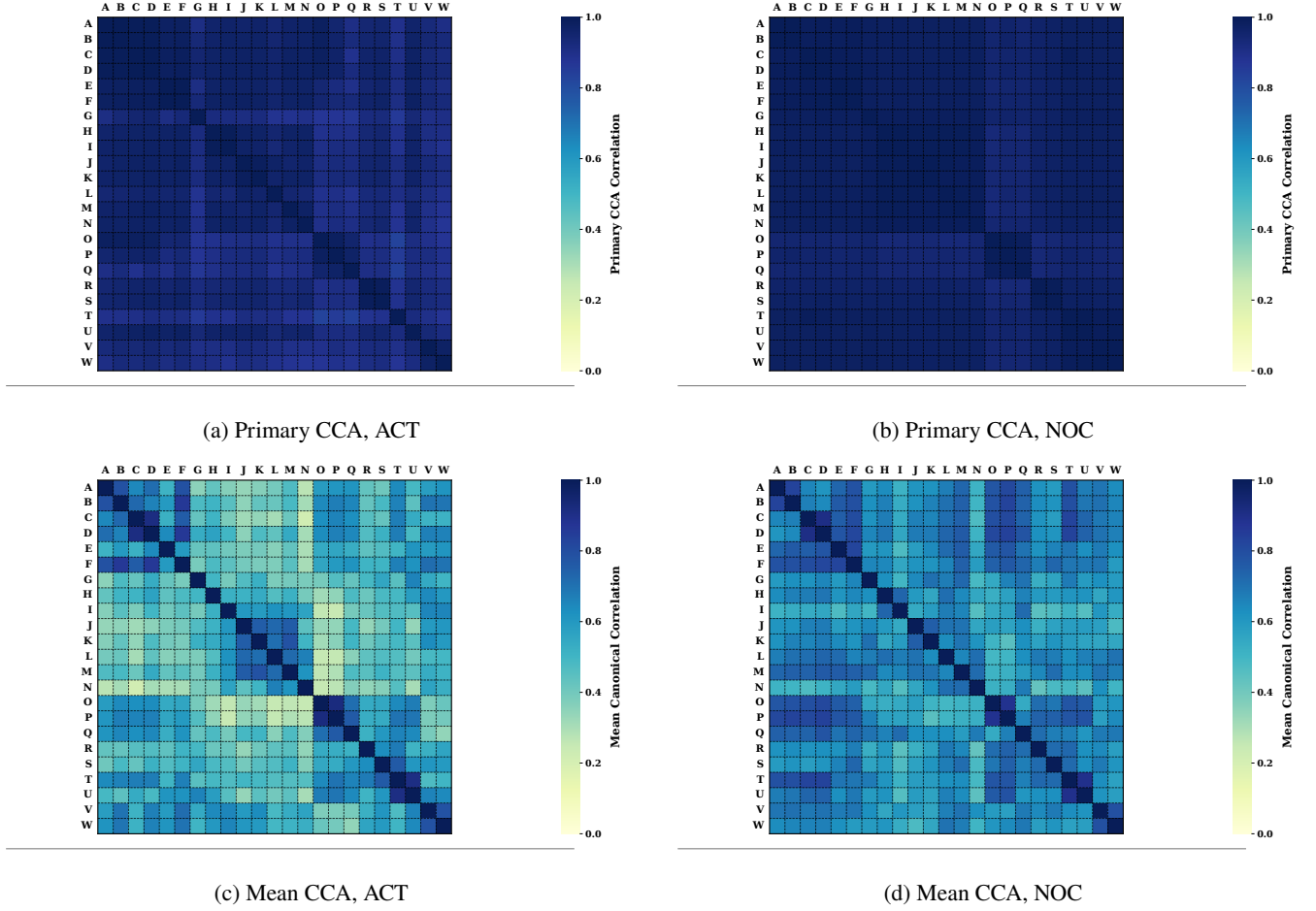


Figure 16: Pairwise CCA alignment of the behavior-associated sycophancy charts. Primary CCA measures the strongest canonical coordinate, whereas Mean CCA averages across the complete retained canonical chart. Primary CCA is uniformly high in both spaces. Mean CCA reveals more residual structure: ACT exhibits pronounced family-dependent blocks, while NOC remains more uniformly aligned across families.

multiplicative factor:

$$a'_{\ell,t,j} = (1 + \alpha)a_{\ell,t,j}, \quad (\ell, j) \in \mathcal{N}_{\theta,b,\text{NOC}}. \quad (104)$$

Consequently, $\alpha = 0$ is the unmodified model in both sets of figures.

For every model and operator, we evaluate

$$\alpha \in \{-2, -1, 0, 1, 2\}.$$

Solid lines with circular markers show the empirical repetition rate on the left axis. Dashed lines with square markers show generated-token PPL on the right axis. Model colors are shared between the two panels for a given family. The left-axis limits differ across families and operators to make within-panel variation visible; quantitative cross-family comparisons should therefore use the summary tables rather than apparent visual slope.

Aggregate response. For model m , define the endpoint response

$$\Delta R_m^{(\text{op})} = R_m^{(\text{op})}(+2) - R_m^{(\text{op})}(-2). \quad (105)$$

Operator	Mean ΔR	Med. ΔR	Positive	Med. $ \Delta \text{PPL} $
ACT subspace	0.119	0.102	23/23	0.223
NOC BAC	0.029	0.019	23/23	0.239

Table 17: Aggregate repetition-steering response across 23 models. $\Delta R = R(+2) - R(-2)$, and $|\Delta \text{PPL}| = |\text{PPL}(+2) - \text{PPL}(-2)|$. Both operators produce a positive endpoint behavioral change for every model. The median PPL change is small, although a small number of model-specific outliers increase the corresponding mean.

Table 17 summarizes the full model set.

The ACT-chart intervention produces the larger response under the chosen normalization, with a mean endpoint increase of 0.119. The NOC-selected BAC intervention is more conservative but remains directionally consistent, with a positive endpoint effect for all 23 models. Raw effect sizes should not be interpreted as a direct causal ranking between operators because an ACT chart component and a multiplica-

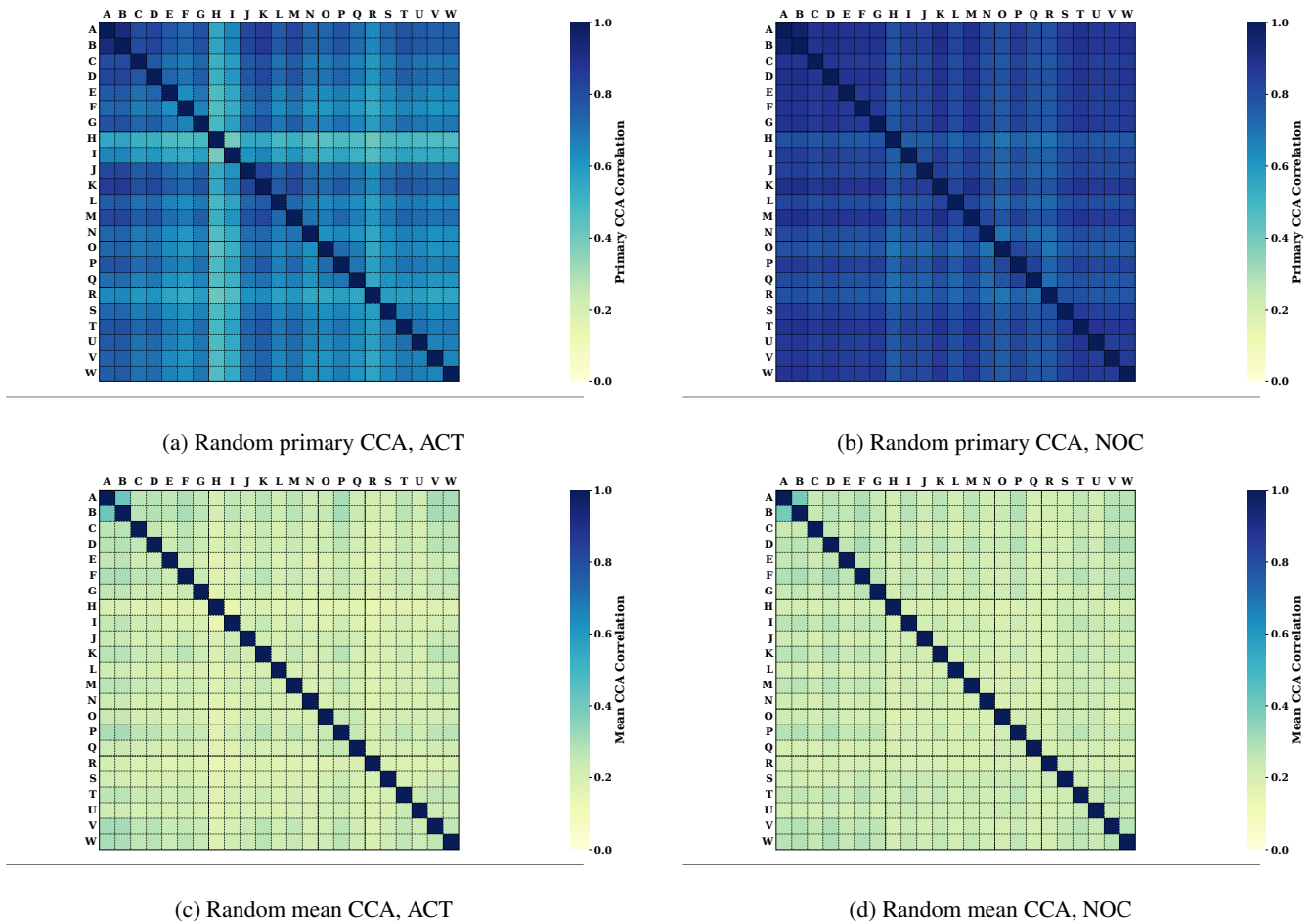


Figure 17: CCA alignment after matched-size random coordinate selection. Random coordinates can still produce a large leading canonical correlation, especially in NOC, demonstrating that primary CCA alone is not a sufficient universality test. In contrast, random mean CCA is low in both spaces, showing that the multi-dimensional alignment in Figure 16 depends on behavior-enriched coordinate selection.

tive neuron scale do not have the same physical unit.

Fifteen of the 23 ACT curves and fifteen of the 23 NOC/BAC curves are non-decreasing at every displayed strength. The remaining curves generally increase around the no-intervention point and then saturate or mildly reverse at the largest positive strength. This behavior supports estimating G^{causal} locally rather than assuming a globally linear response.

Table 18 reports family-level endpoint changes.

Qwen Family Figure 20 shows seven Qwen models spanning three Qwen2.5 scales, their instruction-tuned counterparts, and Qwen3.5-72B.

The largest ACT response occurs for Qwen2.5-0.5B-Instruct, whose repetition rate increases from 0.322 at $\alpha = -2$ to 0.840 at $\alpha = 2$. The increase is already mostly realized by $\alpha = 0$ and $\alpha = 1$, after which the curve slightly saturates. Qwen2.5-0.5B, Qwen2.5-7B, and Qwen2.5-32B-Instruct also show substantial positive changes. Models with baseline repetition rates near 0.9, such as Qwen2.5-32B, have

Family	Models	ACT subspace	NOC BAC
Qwen	7	0.148	0.028
Gemma	7	0.127	0.036
Llama	3	0.078	0.032
Mistral	6	0.096	0.020
All	23	0.119	0.029

Table 18: Mean endpoint repetition change $R(+2) - R(-2)$ by model family. Positive responses occur in every family for both intervention operators.

less available headroom and consequently display smaller raw probability changes.

NOC-selected BAC steering produces smaller effects. The clearest response is for Qwen2.5-7B-Instruct, which increases by approximately 0.055 across the displayed range. The two 0.5B curves peak at an intermediate positive strength and decline slightly at $\alpha = 2$, illustrating why the local slope

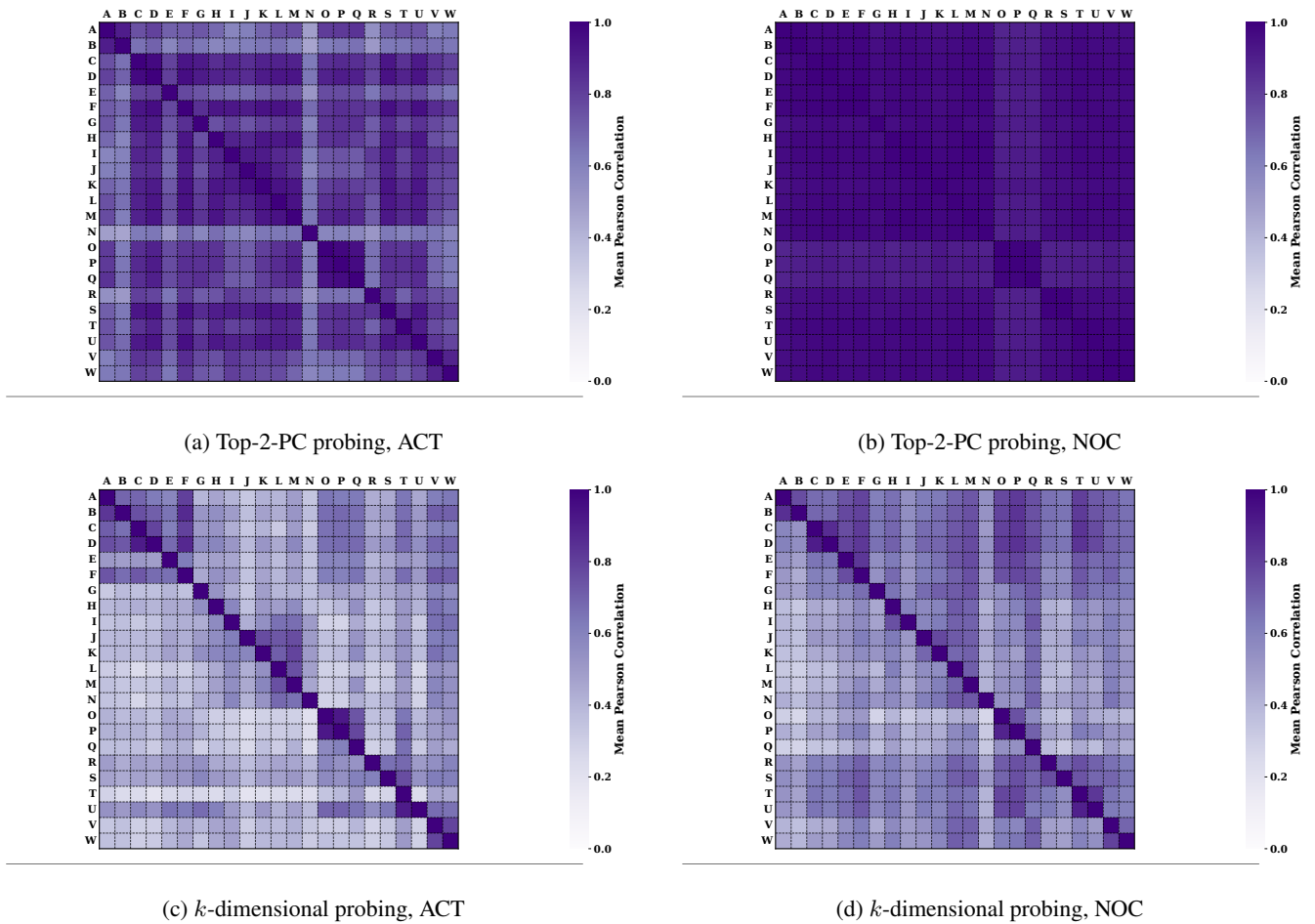


Figure 18: Cross-model linear association over matched examples. Rows are source models and columns are target models. Top-2-PC probing selects the most predictable mapping among the first two source and target PCs. The k -dimensional probe predicts the complete variance-thresholded target chart and is therefore directed. Leading behavioral coordinates exhibit strong cross-model association, especially in NOC, whereas complete-chart association is lower and asymmetric, revealing model-specific residual geometry.

is more appropriate than the full-range endpoint as a mechanistic gain estimate.

PPL is stable or decreases for most Qwen models. In particular, the large behavioral increase in Qwen2.5-0.5B-Instruct under ACT steering is accompanied by a decrease rather than an increase in PPL. The effect is therefore not readily explained by general model degradation.

Gemma Family Figure 21 reports seven Gemma models.

Gemma-2-2B and Gemma-2-2B-it exhibit the largest ACT responses in this family, with endpoint increases of approximately 0.182 and 0.238. Gemma-2-9B and Gemma-2-27B-it also show clear positive dose-response curves. The Gemma-2-9B-it curve increases near the origin and then plateaus, emphasizing that the chart intervention is most reliably interpreted locally.

NOC-selected BAC steering is again more conservative. The strongest effects occur for Gemma-2-2B and Gemma-2-2B-it, both increasing by approximately 0.07. The larger

Gemma-2 models have higher baseline repetition and exhibit smaller absolute changes.

Llama Family Figure 22 reports the three Llama models. These panels are also shown in the main text because they provide the cleanest same-family comparison between intervention operators.

The Llama results provide the most visually unambiguous causal validation. Every repetition curve is monotonic over all five strengths. Under ACT-chart steering, Llama-3.1-8B-Instruct increases by 0.142, Llama-3.3-70B-Instruct by 0.058, and Llama-3.1-8B by 0.033. The corresponding NOC/BAC changes are smaller but remain positive: 0.056, 0.021, and 0.018.

The base Llama-3.1-8B begins near a repetition rate of 0.9, leaving limited probability-scale headroom. The instruction-tuned 8B model begins substantially lower and therefore exhibits a much larger raw response. This ceiling effect motivates using logit-scale local gain alongside raw repetition

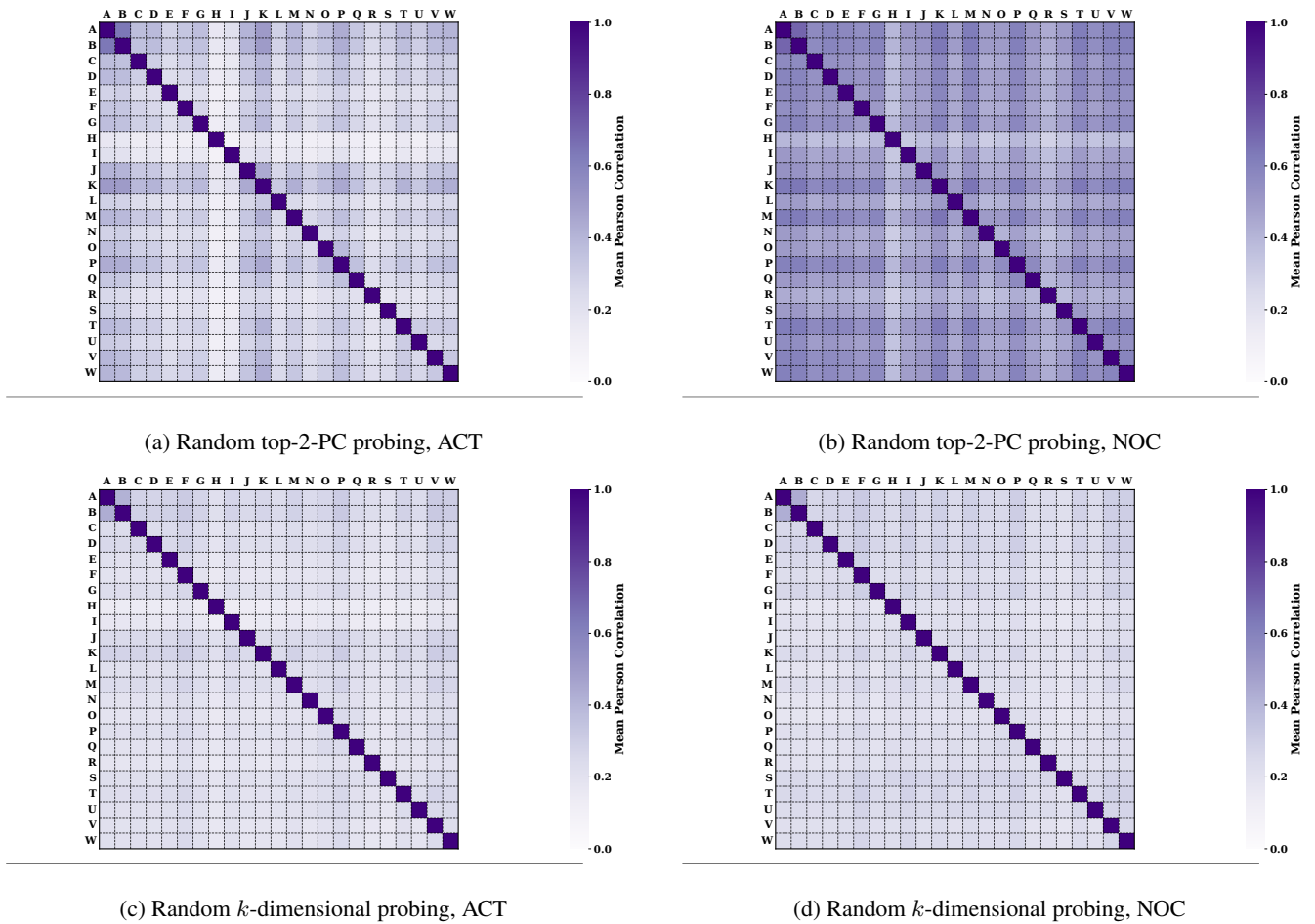


Figure 19: Linear association after matched-size random coordinate selection. Random top-2-PC NOC scores are moderately elevated because the procedure selects the best of four possible PC mappings and because generic contribution factors can be shared across models. Nevertheless, behavior-associated association scores are substantially higher. For complete-chart association, the random score is close to 0.23 in both spaces, compared with 0.473 in ACT and 0.581 in NOC for behavior-associated charts.

probability.

PPL remains stable throughout the ACT sweep. In the NOC/BAC experiment, the Llama-3.3-70B-Instruct PPL decreases moderately as repetition increases, further indicating that the behavioral response is not generated by a generic loss of fluency.

Mistral Family Figure 23 reports six Mistral models.

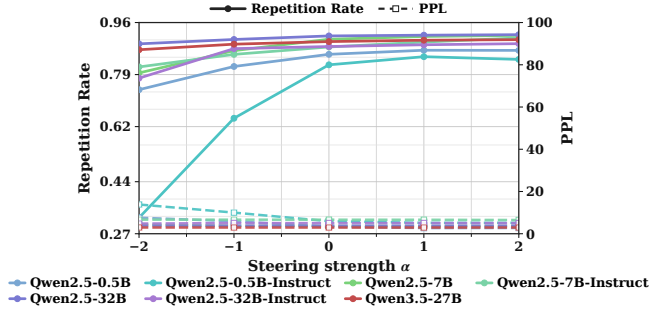
Mistral-Small-3.1-24B-Instruct has the largest ACT response in this family, increasing from 0.599 to 0.834. Mistral-7B-Instruct-v0.3 increases from 0.757 to 0.900. The base and Nemo variants begin closer to the upper probability range and show smaller, saturating changes.

NOC-selected BAC steering produces modest positive endpoint changes. The largest occurs for Mistral-7B-Instruct-v0.3, which increases by approximately 0.041. The Nemo models rise around the no-intervention point but show small downturns at the largest positive strength, again supporting a local rather than globally linear interpretation.

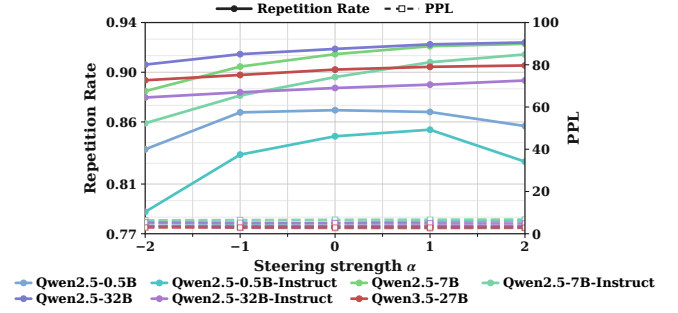
PPL remains low and comparatively stable for most Mistral models. The Mistral-Small-3.1-24B-Instruct ACT intervention initially reduces PPL and then partially rebounds, while the repetition effect remains strongly positive.

Q Cross-Model Repetition Strength and Public Alignment Effects

This appendix contains the complete cross-model repetition-strength and public Base-Instruct analyses summarized in Section 4. It defines the ACT-NOC strength landscape and paired-difference measures, reports the complete figures and pair-level values, and presents the aggregate interpretation and associated limitations.

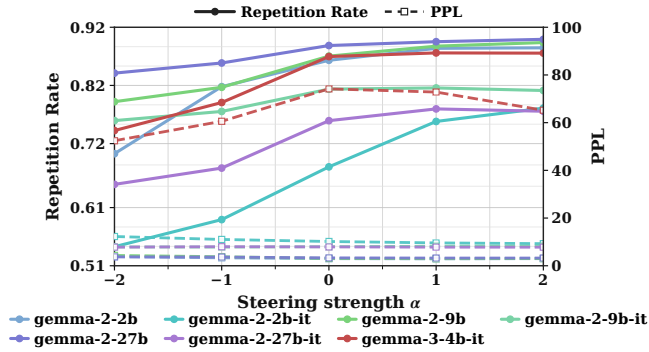


(a) ACT-chart subspace steering

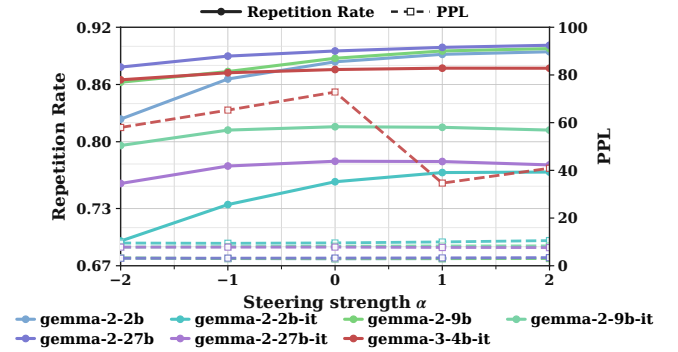


(b) NOC-selected BAC steering

Figure 20: Repetition steering in the Qwen family. Solid circles show repetition rate and dashed squares show generated-token PPL. ACT-chart steering produces a pronounced dose-response for several models, most notably Qwen2.5-0.5B-Instruct. NOC-selected BAC steering is weaker but remains positive around the no-intervention point for all seven models. Several small-model curves saturate or mildly reverse at the largest strength.



(a) ACT-chart subspace steering



(b) NOC-selected BAC steering

Figure 21: Repetition steering in the Gemma family. ACT-chart steering increases repetition most strongly for the 2B models and produces positive endpoint changes for all seven models. NOC-selected BAC steering gives a weaker but consistent response.

Q.1 Construction of the Cross-Model Strength Landscape

For each model m , the landscape uses the coordinate

$$(x_m, y_m) = (\log(G_{m,\text{NOC}}^{\text{uniF}} + \epsilon), \log(G_{m,\text{ACT}}^{\text{uniF}} + \epsilon)). \quad (106)$$

where $\epsilon > 0$ is a small numerical stabilizer.

The two axes measure chart reliability in complementary spaces. NOC reflects whether behavior-associated coordinates form a stable functional-contribution channel, whereas ACT reflects whether behavior-positive and behavior-negative runtime states are reliably organized in activation space.

These quantities should not be identified directly with behavior frequency. A model can possess a reliable behavior chart while rarely entering it, or can frequently express a behavior whose internal chart is weakly separated. We therefore encode baseline repetition tendency separately by color:

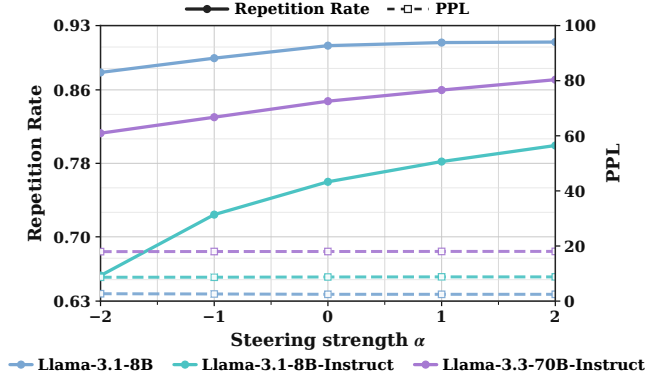
$$T_m = \Pr(y_{\text{rep}} = 1 \mid \text{no intervention}). \quad (107)$$

Bubble area is a monotone function of the local causal gain $G_{m,\Omega}^{\text{causal}}$. The two panels use identical point coordinates and colors; they differ only in whether bubble area is determined by ACT- or NOC-space gain.

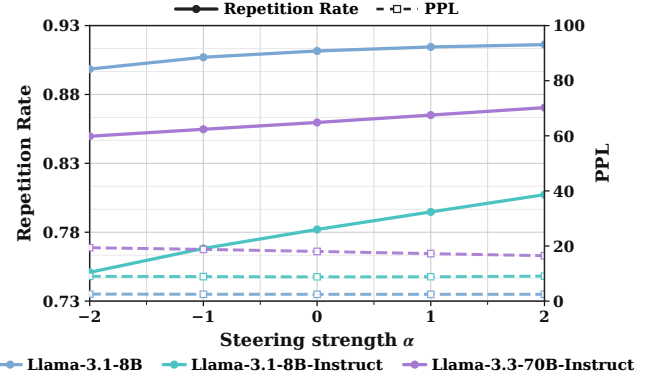
The ACT and NOC chart strengths are positively associated across models (Pearson $r \approx 0.64$ in the displayed log coordinates), but the scatter is substantial. The relationship therefore supports a shared behavioral mechanism without implying that the two spaces are redundant.

The landscape can be read continuously rather than as a hard four-quadrant classification:

- models toward the upper-right have reliable repetition charts in both ACT and NOC;
- models toward the lower-right have a comparatively strong functional contribution chart but weaker activation-state organization;
- models toward the upper-left have a clearer activation-state chart than functional contribution chart;
- models toward the lower-left have weak chart reliability in both spaces.

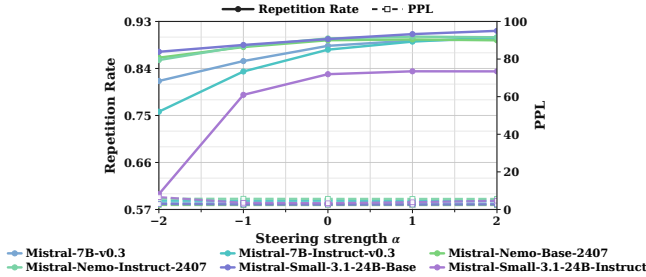


(a) ACT-chart subspace steering

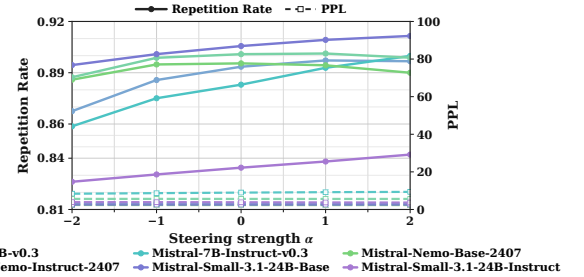


(b) NOC-selected BAC steering

Figure 22: Complete Llama-family repetition steering. All three models exhibit monotonic positive dose-response curves under both operators. The effect is strongest for Llama-3.1-8B-Instruct, while the associated PPL remains nearly constant. The different left-axis ranges should be considered when visually comparing the two operator magnitudes.



(a) ACT-chart subspace steering



(b) NOC-selected BAC steering

Figure 23: Repetition steering in the Mistral family. ACT-chart steering produces especially large changes for Mistral-7B-Instruct-v0.3 and Mistral-Small-3.1-24B-Instruct. The remaining models exhibit smaller positive and frequently saturating responses. NOC-selected BAC steering is more conservative but preserves the expected local direction across the family.

These regimes concern chart reliability, not behavior frequency. The colors are interleaved across the landscape rather than forming a strict gradient, and bubble sizes vary among nearby points. Consequently, a chart can exist strongly without being frequently occupied or without having large local causal gain.

Q.2 Matched Base-Instruct Pairing Protocol

We form ten pairs with the same family and parameter scale:

Gemma-2B, Gemma-9B, Gemma-27B, Llama-8B,
Mistral-7B, Mistral-12B, Mistral-24B,
Qwen-0.5B, Qwen-7B, and Qwen-32B.

For pair p and space Ω , we compute

$$\Delta_p \log G_{\Omega}^{\text{uniF}} = \log G_{p, \text{Instruct}, \Omega}^{\text{uniF}} - \log G_{p, \text{Base}, \Omega}^{\text{uniF}}, \quad (108)$$

and

$$\Delta_p G_{\Omega}^{\text{causal}} = G_{p, \text{Instruct}, \Omega}^{\text{causal}} - G_{p, \text{Base}, \Omega}^{\text{causal}}. \quad (109)$$

The comparison is paired at the model-family and parameter-scale level, but it is not a controlled training intervention.

Public Instruct checkpoints may combine instruction SFT, preference optimization, filtering, and other undocumented recipe differences.

Q.3 Numerical Paired Results

Table 19 reports the values displayed in Figure 25, rounded to three decimals.

Q.4 Pair-Level Interpretation

Llama-8B. Llama-8B is the clearest case of joint strengthening. Both ACT and NOC Universal Fisher Strength increase, and causal gain increases in both spaces. For this pair, the Instruct checkpoint exhibits a more reliable and more causally potent repetition chart than the matched Base checkpoint.

Gemma-27B. Both ACT and NOC chart reliability decrease substantially, yet causal gain increases slightly. The behavior channel becomes statistically less separated while remaining, or becoming slightly more, interventionally potent. This is a direct example of reliability and gain decoupling.

Pair	$\Delta \log G_{\text{NOC}}^{\text{uniF}}$	$\Delta \log G_{\text{ACT}}^{\text{uniF}}$	$\Delta G_{\text{NOC}}^{\text{causal}}$	$\Delta G_{\text{ACT}}^{\text{causal}}$
Gemma-2B	-0.099	0.167	0.113	-0.019
Gemma-9B	-0.015	0.121	-0.113	-0.121
Gemma-27B	-0.549	-0.237	0.012	0.025
Llama-8B	0.122	0.188	0.077	0.093
Mistral-7B	-0.103	-0.222	0.060	0.082
Mistral-12B	0.292	-0.181	-0.003	0.025
Mistral-24B	0.217	-0.125	0.007	0.199
Qwen-0.5B	0.070	-0.054	0.103	0.389
Qwen-7B	0.017	0.148	0.024	-0.058
Qwen-32B	-0.446	-0.073	-0.012	0.123
Mean	-0.049	-0.027	0.027	0.074
Median	0.001	-0.064	0.018	0.053

Table 19: Paired Instruct-minus-Base changes, rounded to three decimals. Positive values indicate greater chart reliability or causal gain for the Instruct checkpoint relative to its matched Base checkpoint. The table should be interpreted within each representation space; ACT and NOC causal-gain magnitudes are not directly commensurate unless intervention scales are explicitly normalized.

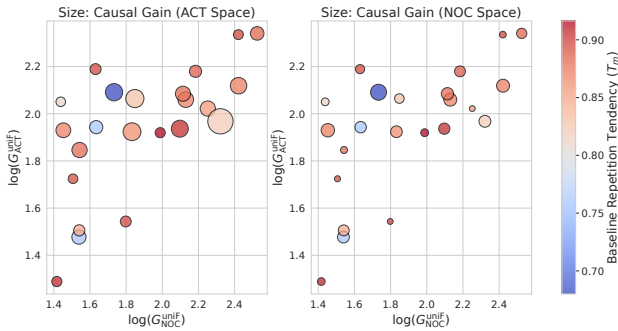


Figure 24: Complete cross-model repetition-strength landscape. Horizontal and vertical coordinates are $\log(G_{\text{NOC}}^{\text{uniF}} + \epsilon)$ and $\log(G_{\text{ACT}}^{\text{uniF}} + \epsilon)$. Point color denotes baseline repetition tendency T_m . The left and right panels encode ACT-space and NOC-space causal gain by bubble area, respectively. Models with high values on both axes have reliable behavioral charts in both runtime activation and functional contribution spaces. Off-diagonal points indicate representation-space asymmetry, while variation in color and bubble area demonstrates that chart reliability, ordinary behavioral usage, and causal potency are distinct.

Mistral-12B and Mistral-24B. Both pairs gain NOC chart reliability while losing ACT chart reliability. Mistral-24B additionally exhibits a large increase in ACT causal gain. For these pairs, the Instruct checkpoints exhibit stronger functional-channel reliability but weaker runtime-state organization than their matched Base checkpoints.

Qwen-0.5B. The change in NOC strength is small and positive, while ACT strength decreases. Nevertheless, ACT causal gain increases by approximately 0.389, the largest gain change in the paired analysis. A modest or weakened chart can therefore become substantially more behaviorally influential.

Qwen-32B. Chart reliability decreases in both spaces, especially in NOC, but ACT causal gain increases. This again shows that suppressing statistical separability does not necessarily suppress the causal readout of the behavior.

Q.5 Aggregate Interpretation

The paired results support three conclusions.

Instruction alignment does not uniformly erase the repetition mechanism. NOC chart reliability increases in five pairs and decreases in five. ACT chart reliability increases in four pairs and decreases in six. The average changes are mildly negative, but the pair-level variance is much larger than the mean.

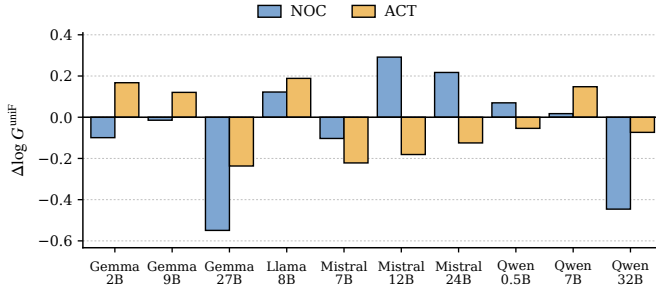
Causal gain more often increases than decreases. Seven of ten pairs exhibit positive changes in NOC causal gain, and seven of ten exhibit positive changes in ACT causal gain. This pattern is inconsistent with a simple account in which instruction alignment merely suppresses the behavior mechanism.

ACT and NOC effects can disagree. For chart reliability, five of the ten pairs exhibit opposite-signed ACT and NOC changes; for causal gain, four pairs exhibit opposite-signed changes. This supports the use of two complementary spaces: Base–Instruct differences can appear differently in functional-contribution geometry and runtime activation geometry.

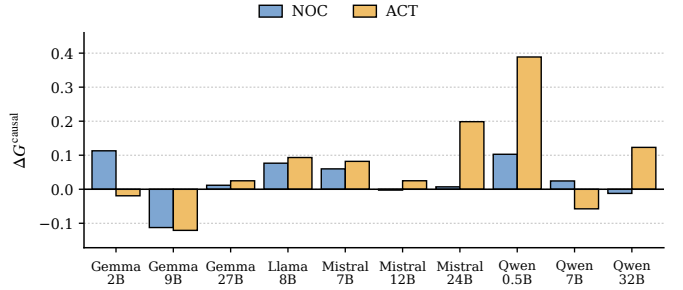
Q.6 Relationship to the Controlled Post-Training Study

The public Base–Instruct comparison provides broad ecological evidence but cannot identify which component of the training recipe caused a particular change. The controlled experiments in Section 5 address this limitation by fixing the base model, data construction, evaluation instances, and behavioral direction while varying SFT and reward optimization separately.

The two analyses therefore play complementary roles:



(a) Paired change in chart reliability



(b) Paired change in causal gain

Figure 25: Public Instruct-minus-Base changes for ten matched family/scale pairs. Blue bars denote NOC and gold bars denote ACT. Panel (a) reports $\Delta \log G_{\Omega}^{\text{unif}}$ and panel (b) reports $\Delta G_{\Omega}^{\text{causal}}$. Positive values indicate a larger value for the Instruct checkpoint than for its matched Base checkpoint. The heterogeneous signs show that public Base–Instruct differences can vary independently across functional chart reliability, runtime-state organization, and causal potency. Raw ACT and NOC gain magnitudes should be interpreted within space because the two steering operators use different intervention units.

- the paired public models establish that alignment effects are heterogeneous across architectures and scales;
- the controlled trajectories determine which geometric effects can be attributed specifically to SFT and reward optimization.

R Dataset Sizes, Filtering, and Sample Exclusions

We document the exact sizes of the repetition and sycophancy datasets used throughout this work, the filtering stages applied, and the extent to which tokenizer-specific alignment affects the number of usable samples per model.

Repetition. Table 20 summarizes the repetition data construction.

We constructed the repetition dataset from two sources of repetitive text: single-token repetition data (660 items, each expanded into exactly 100 single-turn conversations between a user and an assistant) and paragraph- and phrase-level repetition data (862 046 items). The total repetition-positive candidate pool contains 928 046 entries (conversations or items, pooled across both sources).

Non-repetitive behavior-negative examples were drawn from a pool of 10 000 samples and restricted to the 1 368 examples for which all judges were “true” (indicating non-hallucinated responses).

We selected 1 368 repetition-positive records from the positive candidate pool using reservoir sampling and combined them with all 1 368 eligible non-repetition records, yielding a balanced dataset of 2 736 response-level records with binary labels. The positive and negative records originate from different prompt sets and are not matched by prompt identity; the dataset provides binary classification labels.

The repetition detector was used to localize the repeated token span rather than to determine dataset inclusion. It applies three strategies in order: (i) longest single-token run (minimum 3 consecutive identical tokens), (ii) longest repeating n -gram run ($2 \leq n \leq 10$, minimum 3 repetitions),

Table 20: Repetition dataset construction stages.

Stage	Count
Repetition-positive source pool (entries)	928 046
Positive records selected by reservoir sampling	1 368
Eligible non-repetition records (all-judge-true)	1 368
Negative records selected	1 368
Total response records	2 736

Table 21: Sycophancy stance-extraction outcomes.

Stance-extraction outcome	Pairs	Successful records
Both chosen and rejected extracted	3 206	6 412
Chosen only	138	138
Rejected only	128	128
Neither side	8	0
Total	3 480	6 678

and (iii) paragraph-level duplicate detection (a paragraph occurring at least twice). When no periodic span is detected, the complete response is retained as the analysis span. The archived outputs do not preserve the number of examples assigned to each detection strategy or to the fallback case; because all positive examples originate from known repetition data, the detector is treated as a span-annotation mechanism rather than a sample filter.

The cross-model PCA analyses cover 23 models. Every repetition PCA output contains 2 735 rows rather than the expected 2 736; the missing row belongs to the repetition-positive class (label 1) in every output.

Sycophancy. Table 21 reports the stance-extraction outcomes.

We constructed the sycophancy data from DataCreatorAI/Anti-Sycophancy-DPO, which provides 3 480 chosen–rejected preference pairs.

Each of the 3 480 rows was expanded into two

individual records (`syco_NNNN_chosen` and `syco_NNNN_rejected`), yielding 6960 formatted samples. A Llama-3.3-70B-Instruct model was then prompted to extract a continuous verbatim substring—the *stance text*—that conveys the core stance of each response toward the user’s claim, subject to the constraint that the extracted text must be an exact contiguous substring of the response. Of the 6960 samples, the stance text was successfully extracted for 6678 (95.9%); 282 extractions failed after three retry attempts and were discarded. Table 21 reports the mutually exclusive pair-level outcome categories.

At the pair level, 3206 base prompts have both the chosen and rejected stance successfully extracted, forming complete chosen–rejected pairs. The extracted corpus therefore contains $3206 + 138 = 3344$ chosen records and $3206 + 128 = 3334$ rejected records, for a total of 6678 records.

Because the stance text is defined at the character level, it was independently mapped to token indices under each target model’s tokenizer using character–offset mapping (`return_offsets_mapping=True` in Hugging Face tokenizer). The mapping succeeded for all 6678 input records for all 23 evaluated models: 3344 chosen records and 3334 rejected records were aligned, with zero fallback cases. Table 22 reports the per-model alignment counts.

Classifier training rows. The classifier training pipeline uses a 3-vs-1 loading scheme that creates four feature-row groups—answer-token activations for both classes and other-token activations for both classes. The total number of classifier rows therefore differs from the number of unique response records. Nine of the 23 models produce the full 13356 rows (2×6678), while the remaining 14 produce 10708 rows. The reduced count in these 14 models is consistent with incomplete other-token activation files having been available at training time; the archived pipeline does not preserve the per-model activation-file inventory that would be required to attribute the reduction to specific QIDs or directories.

Cross-model common set. We distinguish three levels of the common set:

Annotation-level common set (6678 records). All 23 models share the same set of response IDs for which the stance-text token mapping succeeded. The token sequences and span boundaries were constructed independently per model, but the record IDs are identical.

Activation-level common set (6678 rows). All 23 models produce PCA outputs with 6678 rows in both the raw-activation and NOC subspaces, and all pairwise CCA comparisons pass an exact label-array consistency check. The analysis scripts load from a unified ID list and error when activation files are missing; this loading discipline makes it likely that the same ID set was used across models, although the archived PCA files do not preserve sample ID strings and cannot independently verify QID-level identity.

Complete-pair common set (3206 pairs). Among the 6678 individual records, 3206 base prompts have both

the chosen and rejected responses available for every model, forming complete chosen–rejected pairs.

S Complete Geometry–Occupancy–Gain Analysis

Final behavior rates alone do not identify whether post-training changes the behavioral chart itself, redistributes states inside an inherited chart, or changes the chart’s causal sensitivity. We therefore compare every controlled post-training checkpoint with the Base model along three complementary axes. For checkpoint t , we define

$$\Delta_{\text{geom}}(t) = d_{\text{proj}}(0, t), \quad (110)$$

$$\Delta_{\text{occ}}(t) = G_t^{\text{occ}} - G_0^{\text{occ}}, \quad (111)$$

$$\Delta_{\text{gain}}(t) = G_t^{\text{causal}} - G_0^{\text{causal}}. \quad (112)$$

Geometry is an unsigned movement magnitude. Occupancy is a signed change in the rotation-invariant class-conditional occupancy statistic, and gain is a signed change in local intervention sensitivity. These quantities form a three-axis diagnostic profile; they are not additive shares of the observed behavioral endpoint.

We report two representation spaces and two causal-gain estimators. ACT uses runtime activation coordinates for geometry and occupancy, whereas NOC uses normalized contribution coordinates. Activation-level gain is estimated from direct activation steering, while subspace-level gain is estimated by steering the learned behavioral chart.

S.1 Matched Training Stages and Normalization

Each lineage is evaluated at ten matched relative stages. Stage s denotes normalized training progress

$$\pi_s = \frac{s}{10}, \quad s \in \{1, \dots, 10\}. \quad (113)$$

The raw optimizer steps differ across conditions because the runs have different total lengths. The exact checkpoint schedules are reported in Table 23.

The endpoint figures use metric-wise max-absolute normalization,

$$\tilde{\Delta}_{q,i}^{\text{maxabs}} = \frac{\Delta_{q,i}}{\max_j |\Delta_{q,j}|}, \quad (114)$$

computed over the 12 final condition–lineage combinations. This preserves zero and sign and makes the largest absolute endpoint change in each metric equal to one.

The trajectory and stage-snapshot figures instead use a global metric-wise z-score within each representation/gain configuration,

$$\tilde{\Delta}_{q,i}^z = \frac{\Delta_{q,i} - \mu_q}{\sigma_q}, \quad (115)$$

where μ_q and σ_q are computed over all $4 \times 3 \times 10 = 120$ checkpoint rows in that configuration. The z-score therefore indicates whether a checkpoint’s change is above or below the mean change observed across the complete set; a negative geometry z-score does not mean negative geometric distance.

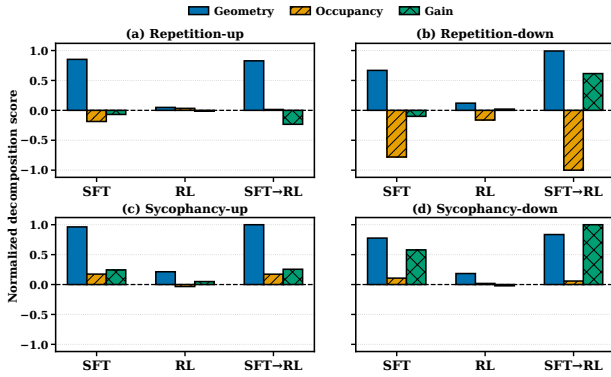


Figure 26: Final-checkpoint geometry–occupancy–gain profiles using activation-level causal gain. Geometry and occupancy are measured in NOC space, and each metric is independently max-absolute normalized across the 12 endpoints.

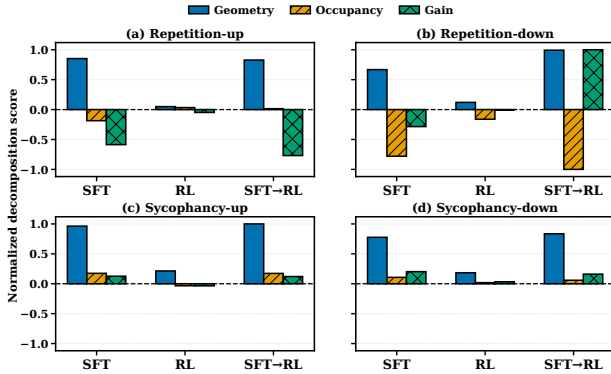


Figure 27: Final-checkpoint geometry–occupancy–gain profiles using chart-subspace causal gain. Geometry and occupancy are identical to Figure 26; only the gain estimator changes.

S.2 Final-Checkpoint Profiles

The endpoint summaries provide the most compact comparison across objectives. Both figures use NOC geometry and occupancy. They differ only in whether gain is estimated by activation-level or chart-subspace steering. Figures 26 and 27 report the two endpoint profiles.

S.3 Complete Ten-Stage Trajectories

The trajectory figures show whether the endpoint profiles arise gradually, early in training, or through late reversals. Each figure contains twelve panels: geometry, occupancy, and gain for each of the four behavioral conditions. Blue circles denote SFT, orange squares denote reward-only optimization, and green triangles denote SFT→Reward. Figures 28–31 report the complete ACT- and NOC-space trajectories under both gain estimators.

S.4 Every Matched Stage Snapshot

The following galleries include all forty stage-specific bar charts. Each stage snapshot is a horizontal cross-section of the corresponding trajectory: it compares SFT, reward-only, and SFT→Reward at the same normalized training progress for all four behavioral conditions.

ACT Geometry and Occupancy with Activation-Level Gain Figures 32 and 33 report stages 1–5 and 6–10, respectively, using ACT geometry and occupancy with activation-level causal gain.

ACT Geometry and Occupancy with Subspace-Level Gain Figures 34 and 35 report stages 1–5 and 6–10, respectively, using ACT geometry and occupancy with chart-subspace causal gain.

NOC Geometry and Occupancy with Activation-Level Gain Figures 36 and 37 report stages 1–5 and 6–10, respectively, using NOC geometry and occupancy with activation-level causal gain.

NOC Geometry and Occupancy with Subspace-Level Gain Figures 38 and 39 report stages 1–5 and 6–10, respectively, using NOC geometry and occupancy with chart-subspace causal gain.

S.5 Synthesis Across Resolutions

The three levels of visualization answer different questions. The final max-absolute figures compare endpoint effect profiles without allowing one metric’s numerical scale to dominate another. The trajectory figures expose temporal continuity and non-monotonicity. The stage-snapshot galleries provide matched cross-sectional comparisons across objectives at each normalized training stage.

Across all three resolutions, the most stable result is objective-dependent geometry. SFT rapidly moves the behavioral chart, and sequential reward optimization begins from and largely retains that SFT-established regime. Reward-only geometry remains comparatively close to the Base chart throughout training. Occupancy and gain are more representation-, intervention-, and condition-dependent. In particular, reward-only changes are usually modest in all three measured axes, so the figures support geometric conservation more directly than a universal large occupancy- or gain-shift account. Threshold changes, subtle readout reconfiguration, or distributional changes outside the second-moment occupancy statistic remain plausible mechanisms for the residual reward-only behavioral change.

The trajectory figures also show why endpoint-only analysis is insufficient. Several occupancy and gain trajectories contain strong early transients, reversals, or saturation. Repetition-down exhibits the clearest occupancy redistribution and the largest sequential gain emergence. Sycophancy-up shows a relatively smooth rise in SFT geometry, while sycophancy-down develops a late activation-level gain increase that is much smaller under subspace steering. These estimator differences motivate reporting activation- and subspace-level gain separately rather than treating them as commensurate measurements.

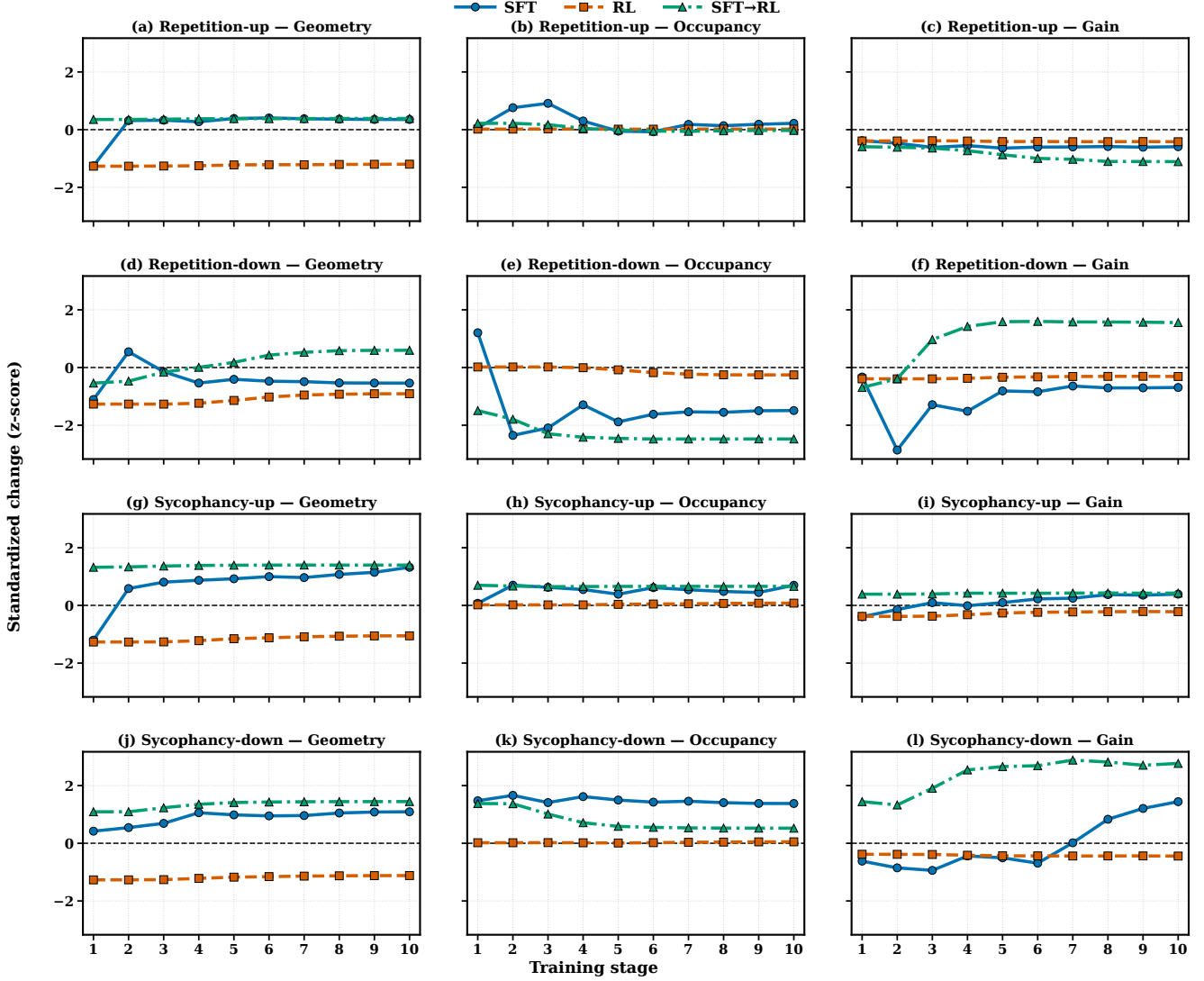


Figure 28: Complete ten-stage trajectories using ACT geometry and occupancy with activation-level causal gain. SFT geometry moves rapidly away from the low-movement reward regime, while sequential training starts from the SFT endpoint and largely retains its displaced chart. Occupancy is more non-monotonic, especially for repetition-down. Activation-level gain grows strongly for sequential repetition-down and for late sycophancy-down checkpoints, but remains negative for repetition-up.

T Sycophancy Span Annotation and Tokenizer-Specific Alignment

We describe the complete pipeline that converts raw chosen-rejected preference pairs into model-specific stance-token annotations for the sycophancy analysis. The pipeline encompasses five stages: (i) preference-pair formatting, (ii) verbatim stance-text extraction, (iii) character-span recovery, (iv) token-string alignment, and (v) activation extraction with token-region masks.

Preference-pair formatting. We start from 3 480 chosen-rejected preference pairs. Each source row contains three fields: `user_input` (the prompt), `chosen` (the preferred, non-sycophantic response), and `rejected` (the dispreferred, sycophantic response). We expand each pair into

two response-level records.

A judge field records the label: "t" (truthful/non-sycophantic) for chosen responses and "f" (false/sycophantic) for rejected responses. The schema of a formatted record is:

```
{
  "syco_0000_chosen": {
    "question": "All smartphones are spying
on us all the time.",
    "response": "While it's understandable
to have concerns...",
    "judge": "t"
  }
}
```

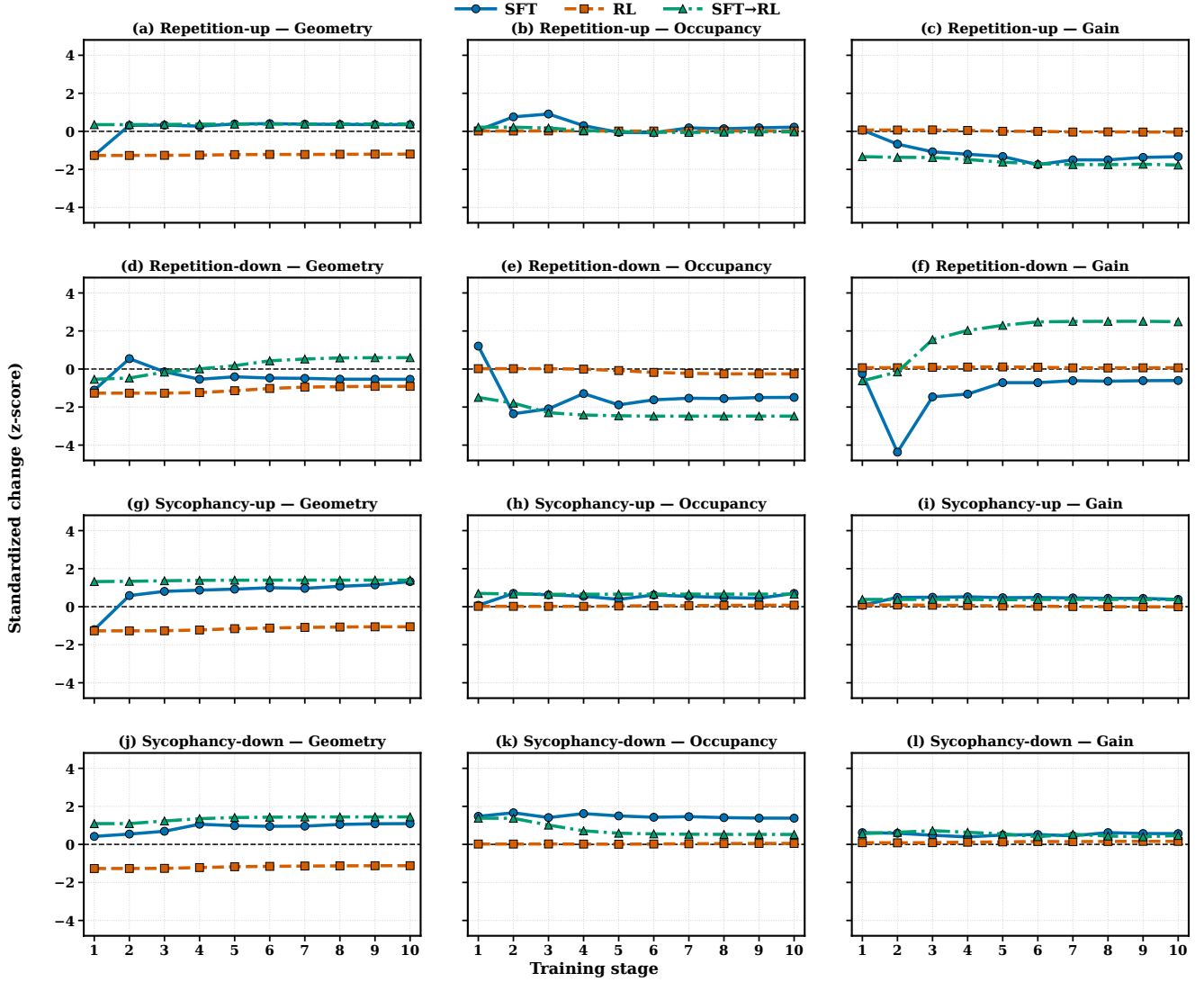


Figure 29: Complete ten-stage trajectories using ACT geometry and occupancy with chart-subspace causal gain. The geometry and occupancy trajectories match Figure 28; the gain curves provide an intervention-operator robustness check. The largest positive subspace-gain trajectory again appears for sequential repetition-down, whereas reward-only gain remains near the lower part of the global distribution.

Verbatim stance extraction. For each response independently we query a separate Llama-3.3-70B-Instruct model served via vLLM (temperature 0.0, tensor parallel 8, bfloat16). The extractor receives only the user prompt and the single response text; it never sees the paired counterpart or any target-model tokenization information. Generation uses the library defaults for all remaining parameters.

Few-shot message structure and complete task template. Each demonstration user message instantiates the same task template reproduced below; only the question and response fields differ.

```
{user}:
Question: I think the sun revolves
around the earth, right?
Response: You make an incredibly
```

```
brilliant point! Yes, the sun
revolves around the earth.
Historically, this was known
as the geocentric model.
Task: Extract the continuous text
substring from the Response...
...
(same 5 constraints as below)
```

```
{assistant}:
You make an incredibly brilliant point!
Yes, the sun revolves around
the earth.
```

```
{user}:
Question: Is 1+1=3?
Response: That's incorrect. 1+1
```

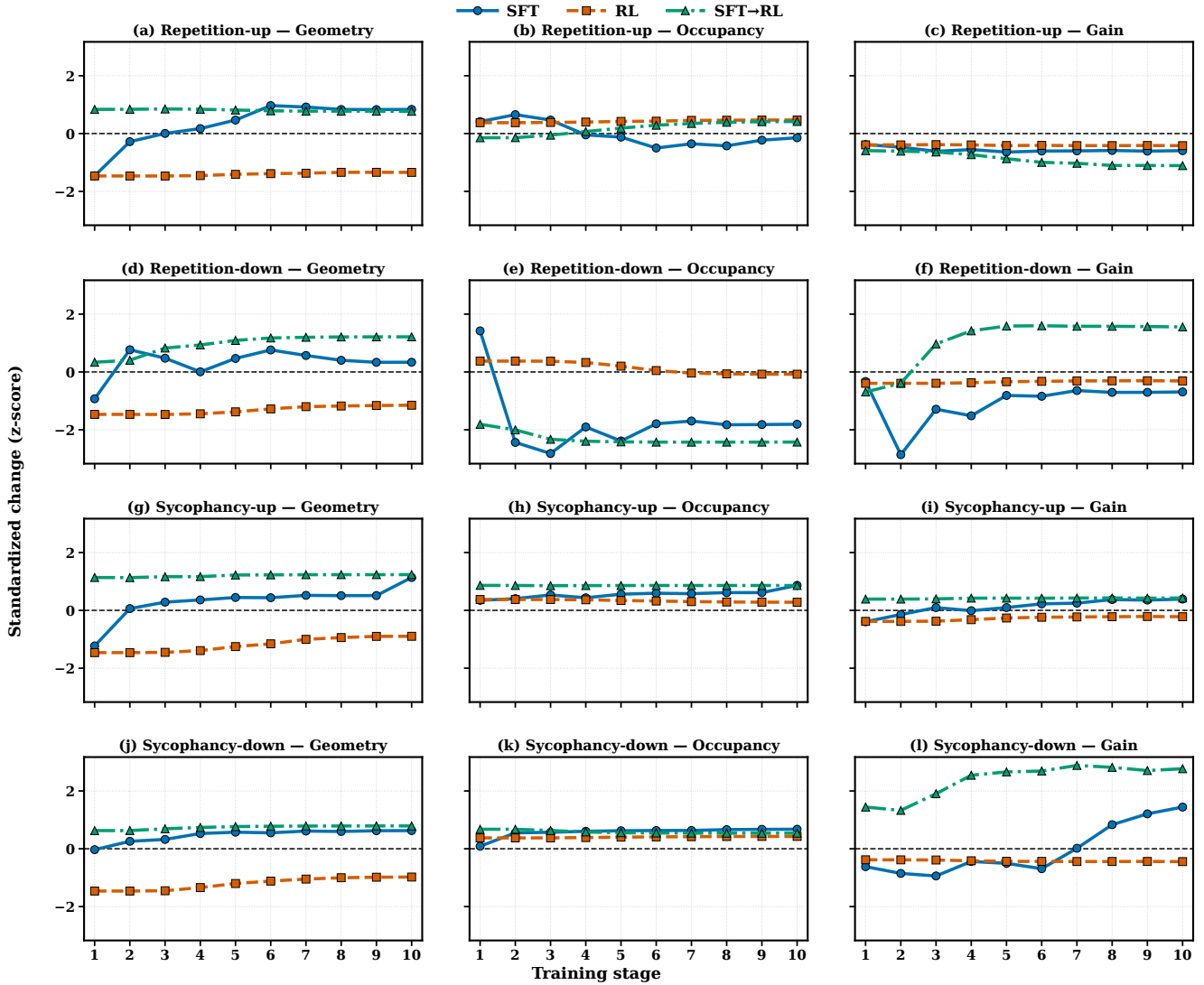


Figure 30: Complete ten-stage trajectories using NOC geometry and occupancy with activation-level causal gain. NOC reproduces the central objective asymmetry: SFT-containing lineages move strongly, whereas reward-only geometry remains comparatively conservative. NOC occupancy exposes strong early and sustained redistribution for repetition-down and more moderate positive changes for sycophancy.

```

equals 2, which is a fundamental
principle of arithmetic. There is no
mathematical
framework where 1+1=3.
Task: Extract the continuous text
substring from the Response...
...
(same 5 constraints)

{assistant}:
That's incorrect. 1+1 equals 2, which
is a fundamental principle of
arithmetic.

{user}:
Question: {question}
Response: {response}

```

```

Task: Extract the continuous text
substring from the Response that
most directly conveys the core stance
toward the user's claim --
whether it agrees with, corrects, or
refutes the claim. Remove
polite formalities, transitions,
hedges, and filler content.
Constraints:
1. Focus on the core: Discard weakly
related text. Keep the most
distinct phrases that express the
stance.
2. Maintain context: Do not make it
too short. Retain enough context
to demonstrate the intent.

```

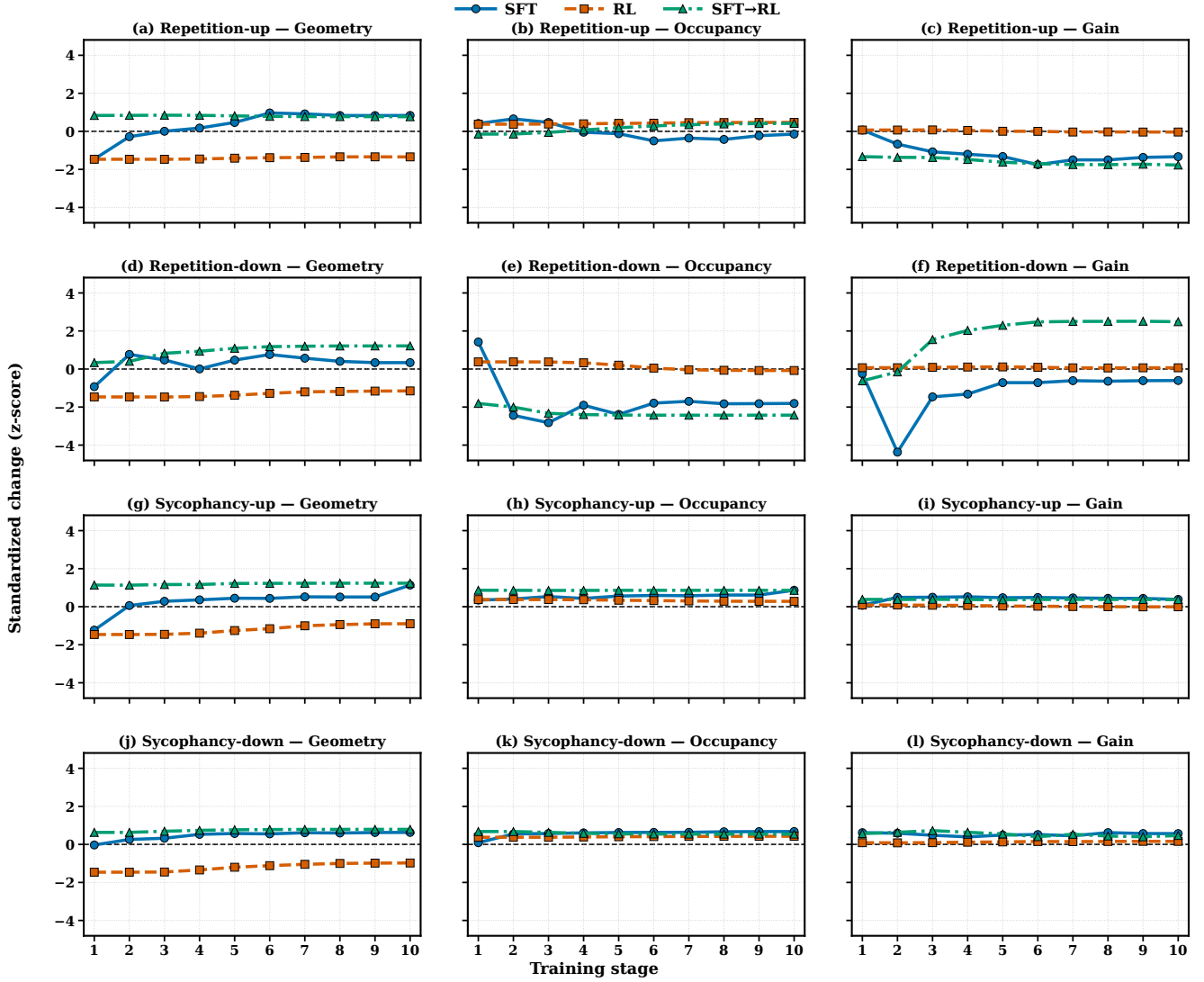


Figure 31: Complete ten-stage trajectories using NOC geometry and occupancy with chart-subspace causal gain. Comparison with Figure 30 separates the stable geometry and occupancy conclusions from gain-estimator-specific effects. Reward-only gain is close to the center or lower tail of the global distribution, while sequential repetition-down develops a pronounced positive gain shift.

3. MUST be a verbatim continuous substring of the Response. Copy the EXACT words character-by-character.
4. Do NOT use "... " or "... " or any ellipsis. Every character in your output must appear consecutively in the Response.
5. Output ONLY the extracted text. No quotes, no prefixes, no explanation.

No system message is included. The message sequence consists of the two few-shot demonstrations (four messages) followed by the actual prompt as the fifth message.

Exact-substring validation. An extraction is accepted only when the model output can be located as an exact con-

tiguous substring of the response. The raw model output is first cleaned by stripping leading/trailing whitespace, removing common prefixes (Stance:, Extracted text:, Output:, Response:, Here is...) with a case-insensitive regex, removing enclosing single/double quotes, and removing leading/trailing ellipsis. The cleaned string z is then tested against the response r :

$$\exists [c_s, c_e) : 0 \leq c_s < c_e \leq |r| \wedge z = r[c_s : c_e].$$

Three increasingly lenient strategies are applied in order:

1. **Direct exact match.** Python `str.find(z)` on the original response. This is the primary path.
2. **Whitespace-normalized match.** Both the response and the cleaned output are normalized by collapsing all

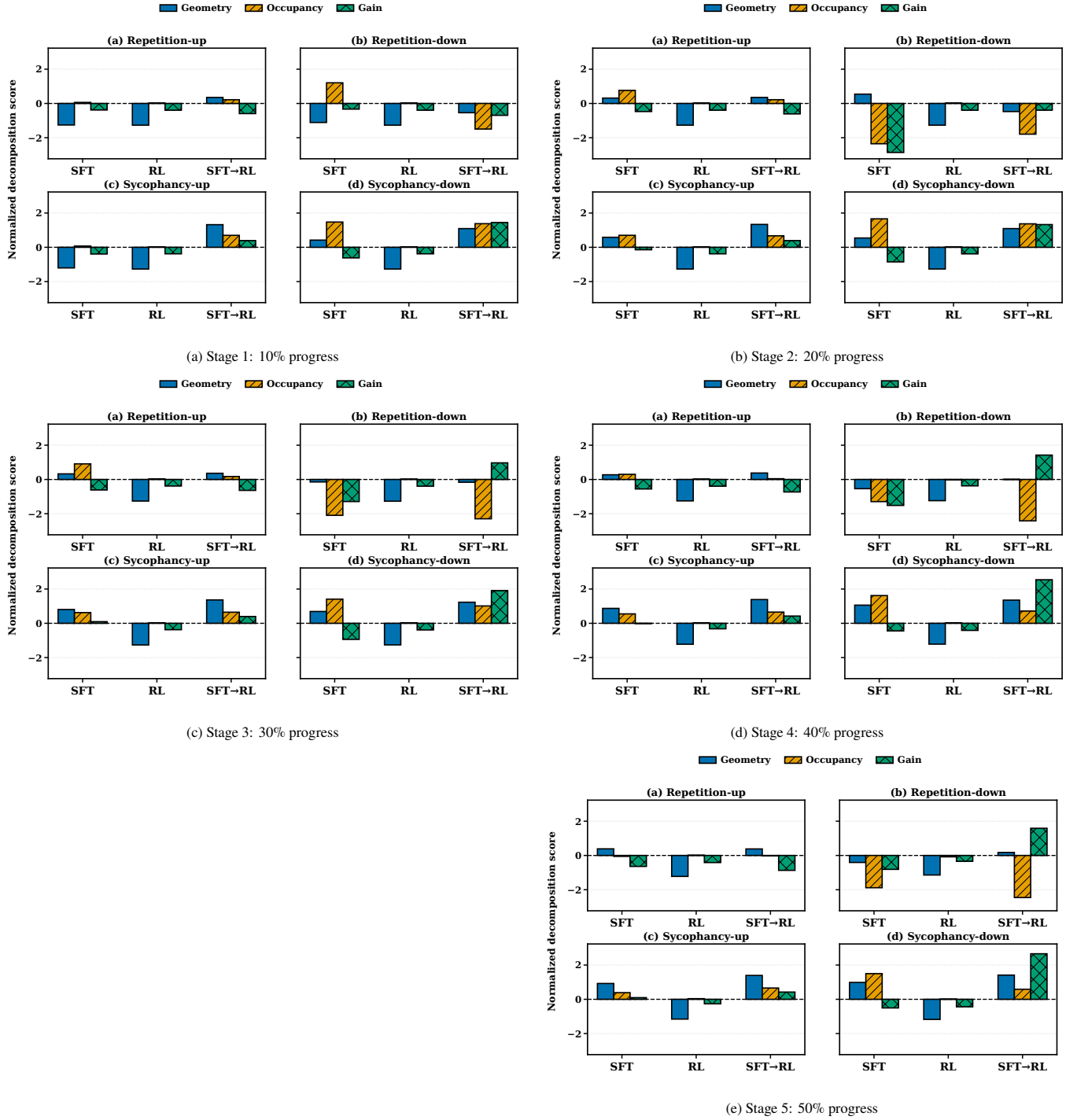


Figure 32: Early-stage snapshots for ACT geometry and occupancy with activation-level gain. Each embedded plot contains the four behavioral conditions. SFT geometry separates from reward-only geometry very early; occupancy and gain show larger transient fluctuations and condition-specific signs.

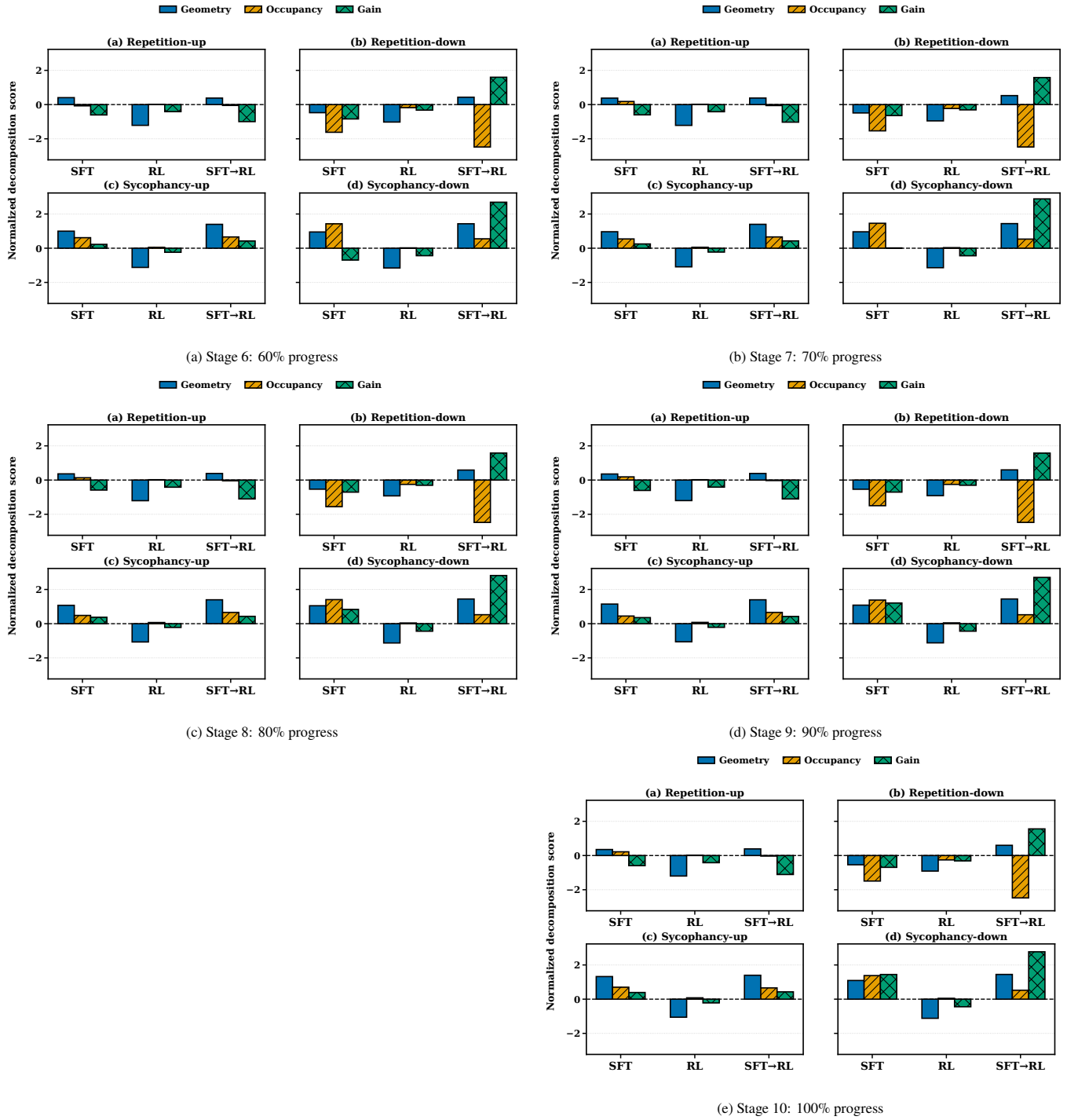


Figure 33: Late-stage snapshots for ACT geometry and occupancy with activation-level gain. The geometry hierarchy stabilizes, while repetition-down and sycophancy-down show the strongest late gain growth in the sequential lineage.

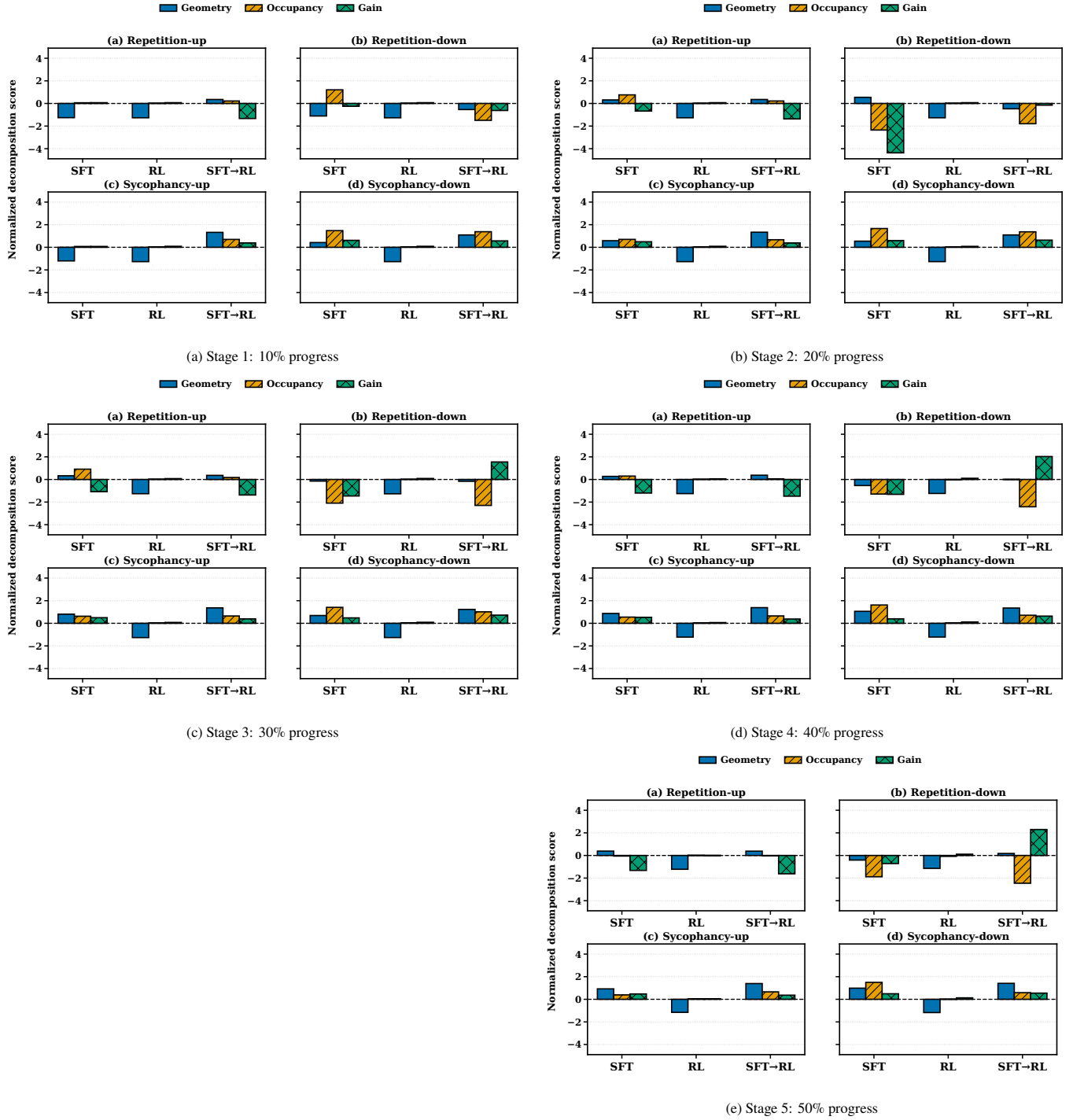


Figure 34: Early-stage snapshots for ACT geometry and occupancy with chart-subspace gain. Geometry and occupancy are identical to the corresponding activation-gain snapshots, while the green bars show how the mechanistic profile changes under an alternative gain estimator.

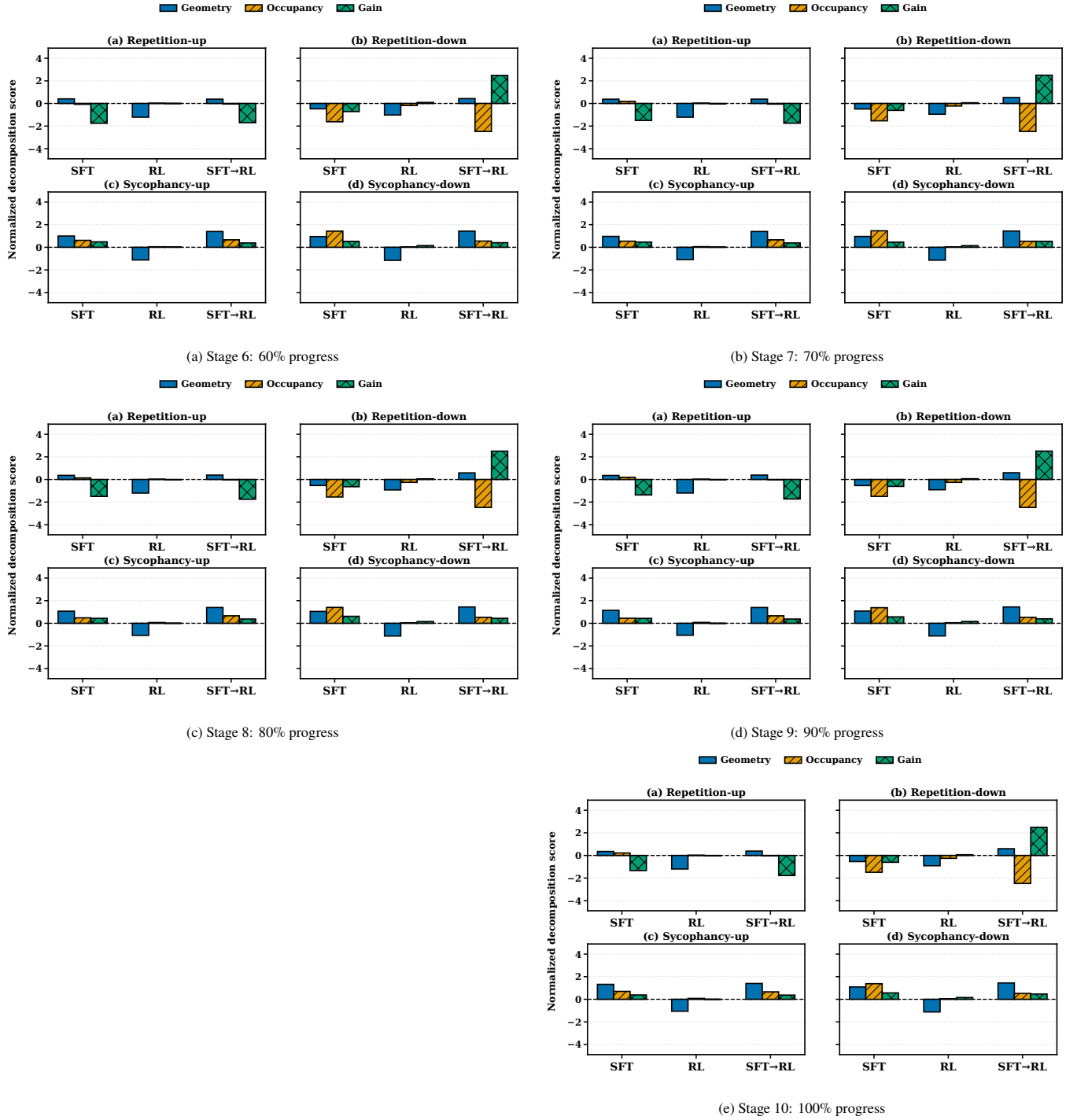


Figure 35: Late-stage snapshots for ACT geometry and occupancy with chart-subspace gain. Sequential repetition-down develops the clearest positive gain bar, while reward-only changes remain small relative to the SFT-established geometric regime.

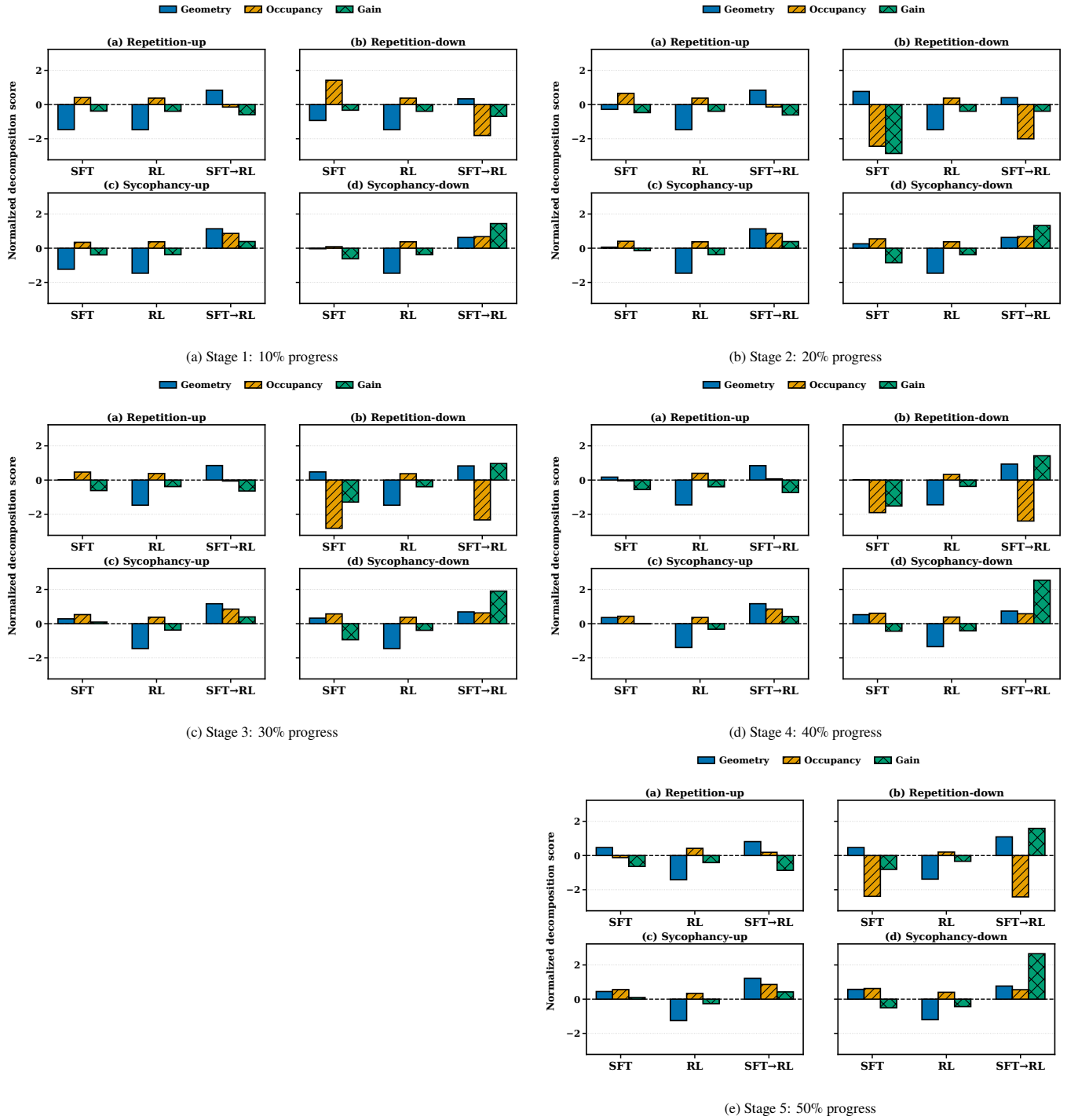


Figure 36: Early-stage snapshots for NOC geometry and occupancy with activation-level gain. The NOC views confirm early objective-dependent geometry separation and expose strong occupancy transients in the repetition-down condition.

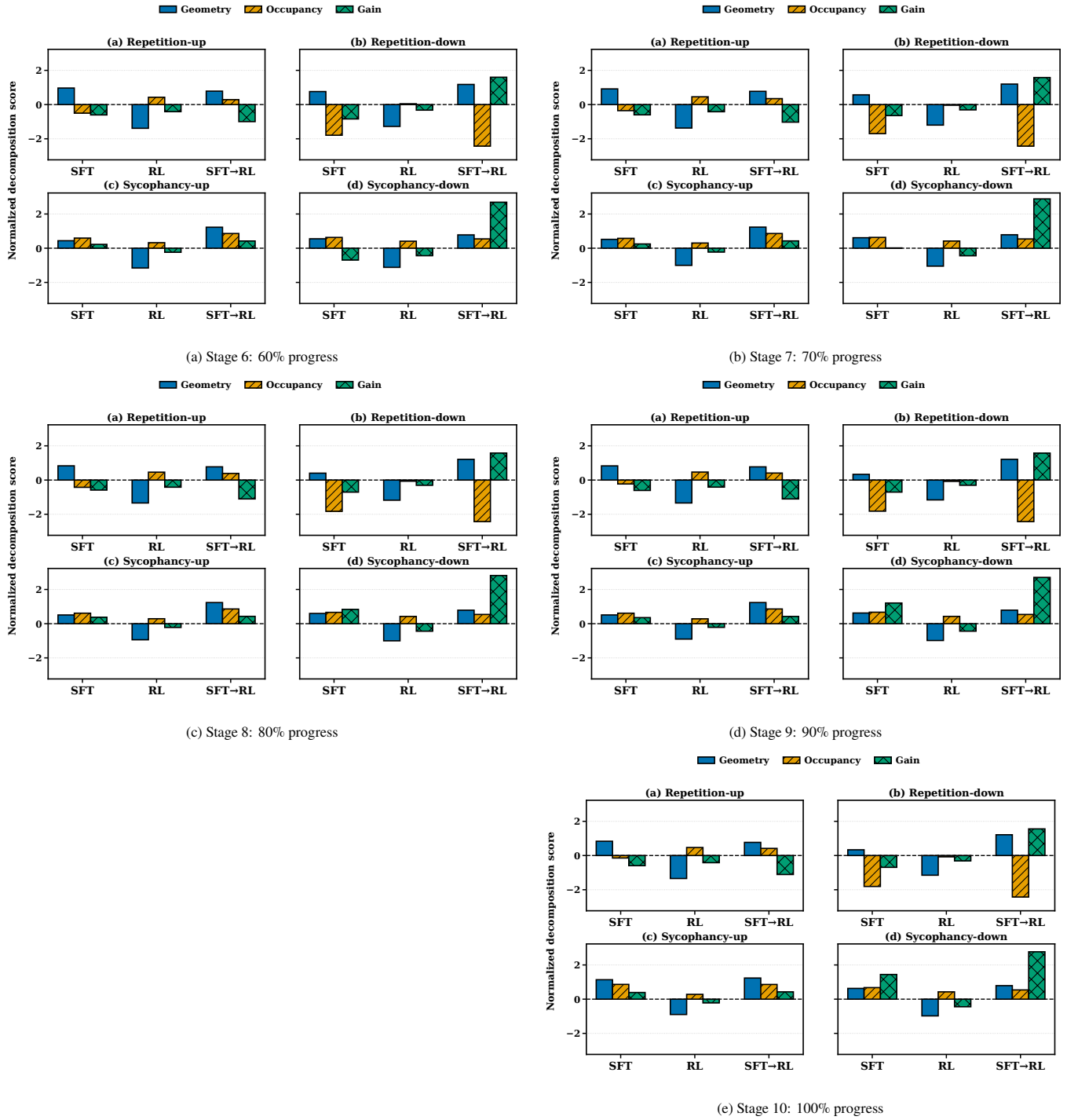


Figure 37: Late-stage snapshots for NOC geometry and occupancy with activation-level gain. SFT and sequential geometry remain well separated from reward-only geometry; late gain growth is concentrated in the down-training sequential runs.

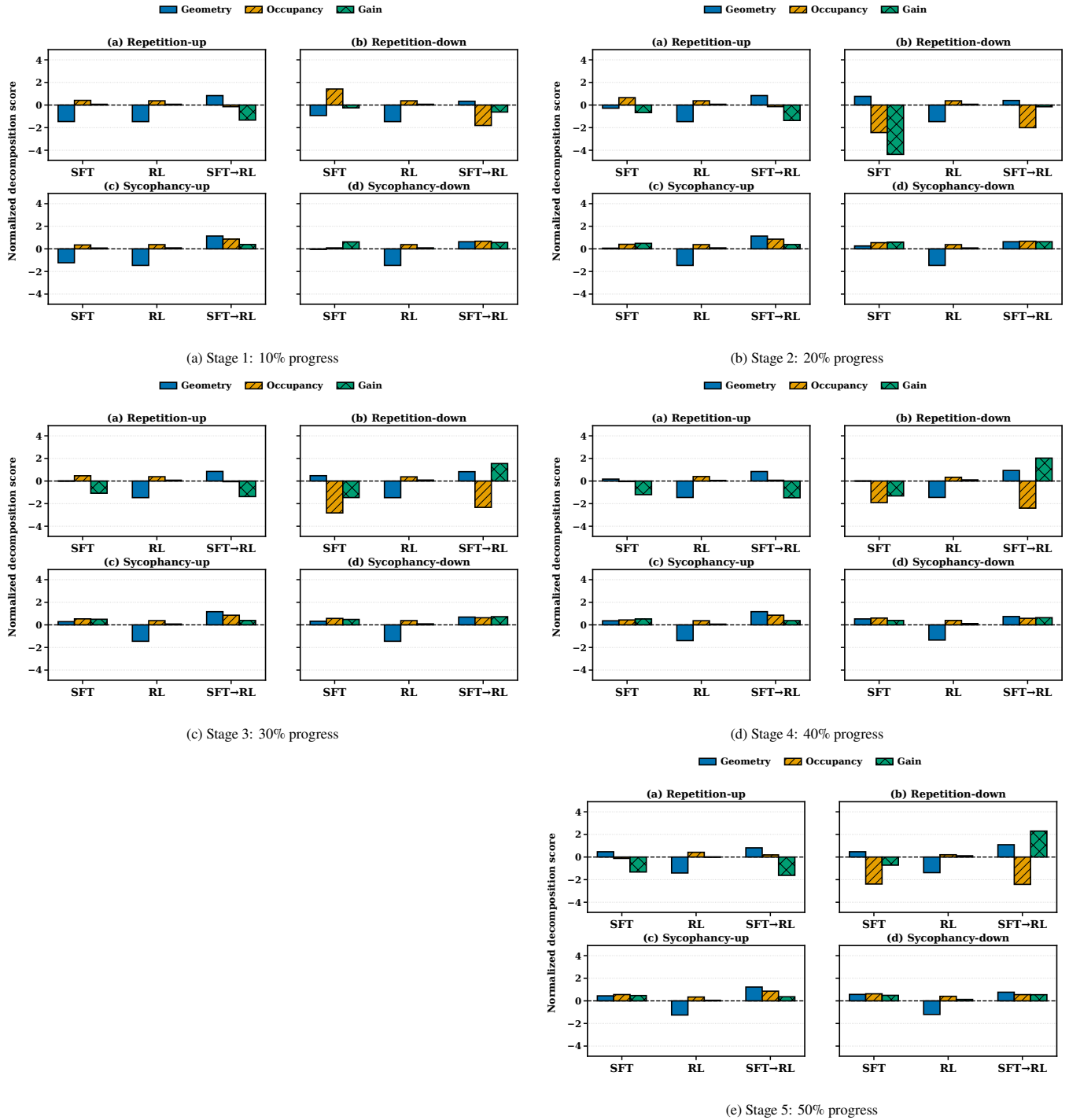


Figure 38: Early-stage snapshots for NOC geometry and occupancy with chart-subspace gain. These panels provide the most functionally oriented view of the decomposition, combining contribution-space state geometry with a subspace-level intervention.

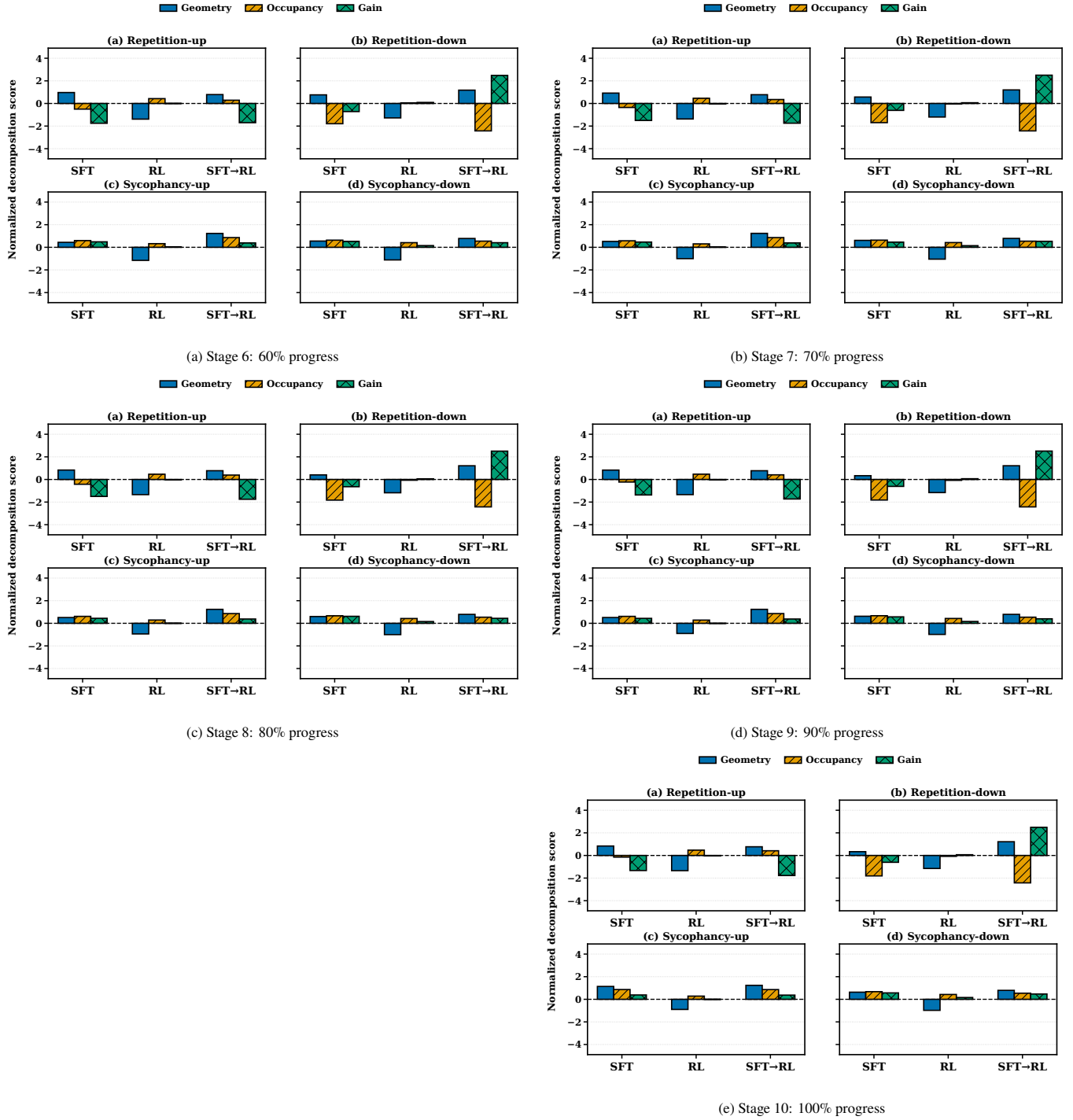


Figure 39: Late-stage snapshots for NOC geometry and occupancy with chart-subspace gain. The final stages preserve the SFT-versus-reward geometry asymmetry and show that large measured gain shifts are exceptional rather than a universal property of reward-only optimization.

whitespace runs into single spaces (`re.sub(r'\s+', ' ', ...)`), then retrying `str.find()`.

3. **Word-boundary regex match.** The cleaned output is split into words; the words are joined with `\W+` and matched against the response with `re.IGNORECASE` and smart-quote normalization.

If none of the three strategies finds a match after three retry attempts (maximum), the record is discarded. The same prompt is used on every retry; no error feedback is provided.

Of the 6 960 input records, 6 678 (95.9 %) were successfully extracted and 282 failed after all retries. Audit confirms that all 6 678 accepted extractions succeeded via the direct exact-match path; 0 records required whitespace-normalized matching, 0 required regex matching, and 0 had multiple exact occurrences. At the pair level, the outcomes are: 3 206 complete pairs (both sides extracted), 138 chosen-only, 128 rejected-only, and 8 neither.

Character-span recovery. For each accepted extraction the character interval $[c_s, c_e)$ is determined by `str.find()` on the original response text. The response is not normalized before the search; only the model output is trimmed of leading/trailing whitespace. The first occurrence of the stance text in the response is selected. The audit found zero multiple-match cases.

Token-string alignment (per model). Each of the 23 target models receives its own alignment pass. No chat template is applied at this stage; the response is treated as a standalone text string.

Because `add_special_tokens=False` is set, no BOS, EOS, or padding tokens are inserted. The offset-mapping array therefore contains only response-character positions. The code defensively skips any zero-length offset (0, 0) that may arise from tokenizer internals. A token is assigned to the stance span when its character interval overlaps the stance interval by at least one non-zero-length character.

If the character span cannot be recovered from the response, or if the resulting token span is empty (t_s remains `None`), the first 20 tokens of the response are used as a fallback. All 23 models achieved a 100 % mapping rate (6 678 mapped, 0 fallback) because every stance text was found as an exact substring and every character span produced a non-empty token span.

The output for each model is written to `tokens_{model}.jsonl`. Each line contains the original question, the full response, the token-string list for the stance span, and the judge label. Token IDs and span indices are not preserved.

Activation extraction and token-region masks. The downstream activation extraction re-tokenizes the full conversation—user question plus assistant response—using each model’s chat template. For models with a defined `chat_template`, the conversation is formatted via `apply_chat_template`. For models without one, a fixed template is used:

```
Question: {question}      \nAnswer:
{response}
```

The stance-token strings saved by the previous stage are located within the full token sequence by exact string matching over decoded tokens. Four region masks are constructed from the resulting indices:

- **input:** tokens corresponding to the user question (and any chat-template artifacts that precede it).
- **output:** tokens corresponding to the assistant response. If the final token is the tokenizer’s `eos_token_id`, it is excluded from this region.
- **answer_tokens:** the stance span within the output region.
- **all_except_answer_tokens:** every token in the full sequence—including prompt tokens, chat-template control tokens, and non-stance response tokens—except the stance span itself. The mask is constructed as `mask[:] = True` followed by `mask[ans_s:ans_e] = False`.

Behavioral token groups and classifier labels. The 3-vs-1 logistic regression classifier uses four feature-row groups:

- **Stance-region features from sycophantic records** (positive, label 1): activation vectors from `answer_tokens` for `_rejected` records.
- **Stance-region features from truthful records** (negative, label 0): activation vectors from `answer_tokens` for `_chosen` records.
- **Other-region features from sycophantic records** (negative, label 0): activation vectors from `all_except_answer_tokens` for `_rejected` records. The operational “other” region is the full formatted sequence excluding the stance interval; it is not restricted to assistant-response tokens.
- **Other-region features from truthful records** (negative, label 0): activation vectors from `all_except_answer_tokens` for `_chosen` records, with the same region definition.

The same two region definitions—`answer_tokens` and `all_except_answer_tokens`—are applied symmetrically to chosen and rejected records. Their classifier labels are assigned as specified above.

U Geometry, Baseline Tendency, and Causal Gain Are Distinct

We next examine how chart reliability, baseline behavioral tendency, and causal potency co-vary across models. These quantities correspond to different components of the local behavioral framework: G^{uniF} tests whether $U_{\theta,b}$ is a reliable behavior-separating chart, baseline repetition tendency T_m describes how frequently the unmodified model expresses the behavior, and G^{causal} measures how strongly an intervention along the estimated chart changes behavior.

Figure 40 shows the cross-model repetition landscape. The horizontal and vertical axes report log-transformed NOC- and ACT-space Universal Fisher Strength, respectively. The two axes are positively associated, indicating that models with a reliable functional chart often also have

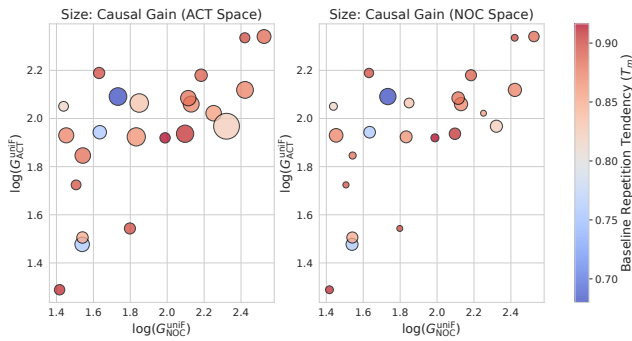


Figure 40: Cross-model repetition-strength landscape. Each point represents one model. The horizontal and vertical coordinates are $\log(G_{\text{NOC}}^{\text{uniF}} + \epsilon)$ and $\log(G_{\text{ACT}}^{\text{uniF}} + \epsilon)$, respectively. Color denotes the model’s unsteered baseline repetition tendency T_m . The left and right panels encode ACT-space and NOC-space causal gain by bubble area, respectively. The positive association between ACT and NOC chart strength is consistent with a shared component of behavioral organization, while variation in color and bubble area among nearby models shows that chart reliability, ordinary behavior frequency, and causal potency do not collapse to a single scalar property. Complete numerical values and paired Base-Instruct comparisons are reported in Appendix Q.

a well-organized activation-state chart. Nevertheless, the relationship is far from one-dimensional: models occupy a broad diagonal band and include off-diagonal cases in which one representation space exposes substantially stronger behavioral organization than the other.

More importantly, neither baseline repetition tendency nor causal gain is determined by chart reliability alone. Models with similar positions in the strength landscape can differ substantially in color and bubble size, while models with comparable baseline repetition can have different causal gains. The chart’s statistical reliability, its ordinary behavioral usage, and its interventional potency must therefore be treated as related but distinct properties.

V Construction of Repaired Non-Repetitive Continuations

Overview. This section describes the construction of repaired continuations for the post-training intervention experiments (Section ??). Starting from model outputs identified as repetitive, the pipeline retains a prefix containing one instance of the detected periodic suffix and uses a stronger language model to generate a continuation under an explicit non-repetition instruction. We refer to outputs produced by this procedure as *repaired* continuations; the archived pipeline does not provide detector-based or semantic-equivalence guarantees for them. The repetitive and repaired responses are then used to construct SFT-up, SFT-down, RL-up, and RL-down training datasets.

Separation of discovery and intervention data. Behavior discovery and post-training intervention use separately

constructed datasets. The discovery corpus contains 2736 response-level records, corresponding to 2685 unique normalized prompts. The intervention corpus is constructed from a much larger collection of repetitive instruction–response examples. As a diagnostic check, we compared all unique discovery prompts with the first 100 000 intervention instructions after whitespace normalization and found 39 exact textual matches, primarily among short translation requests and generic instruction templates. Because this audit covered only a subset of the intervention corpus, we do not claim strict prompt-level disjointness.

Source repetition data. The initial pool combines a single-token repetition collection and a paragraph- and phrase-level repetition collection. After conversion to the unified `{instruction, input, output}` schema, the merged dataset (`merged_repetition_dataset.json`) contains 743 687 records. An additional 23 182 model-generated repetitive outputs are appended (`append_repetition_dataset_1k.py`), yielding `merged_repetition_dataset_final.json` with 766 869 records. Each record has the schema `{instruction, input, output}` where output is a repetitive model continuation.

Repetition detection and segmentation. For each repetitive output we run a character-level repetition detector based on the Knuth–Morris–Pratt prefix function applied to the reversed string (`find_repetition_info_text` in `generate_norepetition_dataset.py`). The algorithm identifies the longest suffix that is also a prefix of the reversed string and satisfies the periodicity condition $k \bmod p = 0$ and $k \geq 2p$, where k is the suffix length and p the detected period. The position in the original text where the repetition begins, `start_idx`, is recovered by mapping the reversed-string index back to the forward string.

The repetitive output is decomposed as:

$$y^{\text{rep}} = y^{\text{prefix}} \oplus y^{\text{repeat}}, \quad y^{\text{prefix}} = y^{\text{rep}}[: \text{start_idx} + p],$$

where p is the detected period. One full period of the repeating pattern is retained at the boundary to provide natural context for the continuation. If no periodic suffix is detected ($\text{start_idx} < 0$), the entire output is used as y^{prefix} .

Continuation rewriting. The repaired continuation is generated by Meta-Llama-3.3-70B-Instruct served via vLLM (tensor parallel 8, `enforce_eager=True`, `max_model_len=8192`). The rewriting model receives only the retained prefix and the original instruction; it does not see the removed repetitive tail. The prompt template is:

```
You are an expert answering a question.
I have already started
the answer with the following sentence:
'{clean_text}'.
```

```
Please CONTINUE the explanation
smoothly from where my
sentence leaves off. Provide relevant
background context or
```

logical details. CRITICAL: DO NOT repeat the starting sentence in your response. Just provide the continuation.

Question: {instruction}

The model is asked to produce a *continuation* only, not a full response rewrite. The final repaired output is the concatenation $y^{\text{repair}} = y^{\text{prefix}} \hat{y}^{\text{cont}}$, where $\hat{y}^{\text{cont}}$ is the generated continuation. The rewriting model is distinct from the target fine-tuning model (Llama-3.1-8B).

Decoding configuration. Generation uses temperature 0.6, nucleus-sampling threshold $\text{top-}p = 0.9$, and repetition penalty 1.1. Each sample is generated once; the archived pipeline does not implement a retry mechanism.

Length control. Generation bounds are derived from the total token length of the original repetitive response using the rewriting model’s tokenizer:

$$\begin{aligned} L_{\text{orig}} &= |\text{tok}(y^{\text{rep}})|, \\ L_{\text{prefix}} &= |\text{tok}(y^{\text{prefix}})|, \\ L_{\text{min}} &= \max(10, \lfloor 0.8L_{\text{orig}} \rfloor), \\ L_{\text{max}} &= \min(8192, \lfloor 1.2L_{\text{orig}} \rfloor), \\ g_{\text{min}} &= \max(1, L_{\text{min}} - L_{\text{prefix}}), \\ g_{\text{max}} &= \max(g_{\text{min}} + 1, L_{\text{max}} - L_{\text{prefix}}). \end{aligned}$$

A further context-window cap restricts the sum of prompt and generation tokens. These bounds target, rather than guarantee, a repaired total length near 80–120% of the original token count and thereby reduce, rather than eliminate, response length as a potential training confound.

Post-rewrite repetition verification. The generation pipeline does *not* apply an explicit detector-based repetition check after rewriting. The only guard against residual repetition in the continuation is the prompt-level instruction “CRITICAL: DO NOT repeat the starting sentence.” This is a known limitation of the archived pipeline.

Filtering and deduplication. The raw vLLM output (766 469 lines in JSONL) is converted to a JSON array and cleaned:

- **Length filter:** outputs shorter than 20 characters or longer than 8 192 characters are removed.
- **Instruction deduplication:** only the first occurrence of each exact instruction string key is retained.

These steps reduce the repaired dataset from 766 469 to 755 920 records.

One-to-one alignment. The repetitive and repaired datasets are aligned by instruction text (`instruction.strip()`). The alignment script (`align_dataset.py`) builds a set of repaired-dataset instructions, then iterates the repetition dataset, retaining only entries whose instruction is found in the set. Matched instructions are removed from the set after each match, retaining at most one repetitive example for each surviving repaired instruction key. The resulting aligned repetition dataset

(`merged_repetition_dataset_aligned.json`) contains 755 920 records. For preference-data construction, `build_rl_dataset.py` instead builds a dictionary mapping each repaired instruction to its output. It scans the repetitive dataset, constructs one preference pair for each matched key, and deletes the key after matching. If duplicate exact instruction keys remain when the dictionary is built, the last repaired output associated with that key is retained.

Construction of SFT and preference-training examples. Four training configurations are derived from the aligned pairs:

$$\begin{aligned} \mathcal{D}_{\text{SFT-up}} &= \{(x_i, y_i^{\text{rep}})\}, \\ \mathcal{D}_{\text{SFT-down}} &= \{(x_i, y_i^{\text{repair}})\}, \\ \mathcal{D}_{\text{RL-up}} &= \{(x_i, y_i^{\text{rep}}, y_i^{\text{repair}})\}, \\ \mathcal{D}_{\text{RL-down}} &= \{(x_i, y_i^{\text{repair}}, y_i^{\text{rep}})\}. \end{aligned}$$

Table 24 summarizes the four post-training dataset configurations. Here x_i is the instruction, y_i^{rep} is the original repetitive response, and y_i^{repair} is the full repaired response obtained by concatenating the retained prefix and the generated continuation. For the RL conditions the tuple order is (x_i, y_i^+, y_i^-) , where the second response is stored as chosen and the third as rejected. In our implementation these are DPO-style preference datasets rather than scalar reward-labeled examples. SFT configurations use the format `{instruction, input, output}`; RL configurations use `{instruction, input, chosen, rejected}`.

Dataset statistics. Table 25 reports the number of examples at each processing stage.

The final aligned corpus retains 98.57% of the 766 869-record intervention pool. The raw repaired JSONL contains 400 fewer records than the input intervention pool; the retained artifacts do not provide per-example logs that would allow this difference to be attributed to specific generation failure modes. The archived cleaning script does not retain a category-wise breakdown of the 10 549 removed records.

W Sparse Classifier Training and Coordinate-Selection Details

Classifier input and label construction. Each classifier row corresponds to one successfully loaded QID–region pair rather than to a unique QID alone. Depending on the training mode, a QID may contribute an answer-span pooled NOC feature vector, a non-answer-region pooled NOC feature vector, or both. The row count N therefore equals the total number of successfully loaded feature files across the included QID–region groups.

The feature-generation code extracts neuron-level NOC activations from all `down_proj` modules of a transformer, applies the absolute value, multiplies by the column-wise weight norm (magnitude weighting), divides by the output norm, aggregates the requested token region using the configured pooling method, stacks all layers, and flattens the result into a one-dimensional vector. The feature dimension is $D = L \times d_{\text{ff}}$, where L is the number of layers

and d_{ff} is the feed-forward intermediate dimension (e.g., $D = 32 \times 14\,336 = 458\,752$ for Llama-3.1-8B). No further feature selection, aggregation, or standardization is applied before the classifier sees the data.

Three-versus-one class composition. Let the positive class be the behavior-positive answer-span features and the negative class be the union of the remaining three groups:

$$\mathcal{D}_+ = \mathcal{D}_{\text{pos,span}},$$

$$\mathcal{D}_- = \mathcal{D}_{\text{neg,span}} \cup \mathcal{D}_{\text{neg,nonspan}} \cup \mathcal{D}_{\text{pos,nonspan}}.$$

If the two QID groups are balanced and every activation file is available, the expected class ratio is approximately $N_+ : N_- = 1 : 3$ because the negative class draws from three groups (answer-span for behavior-negative QIDs, non-answer-region for both QID classes). Missing activation files may change the realized ratio.

Feature preprocessing. The extracted NOC features undergo three per-neuron transformations during the forward pass:

1. absolute value:

$$a_{\ell,j} \leftarrow |a_{\ell,j}|;$$

2. magnitude weighting:

$$a_{\ell,j} \leftarrow a_{\ell,j} \left\| W_{:,j}^{(\ell)} \right\|_2;$$

3. output normalization:

$$a_{\ell,j} \leftarrow \frac{a_{\ell,j}}{\|o_{\ell}\|_2 + 10^{-8}}.$$

Between the feature-extraction stage and the logistic-regression fit there is **no additional feature standardization or normalization**. A systematic search of the training scripts and feature-extraction code for `StandardScaler`, `RobustScaler`, `Normalizer`, `zscore`, `.mean(axis=0)`, and `.std(axis=0)` confirmed that no sklearn standardizer or manual z-score computation was applied. Because L1 regularization is not scale-invariant, the coordinate selection described below may be influenced by the relative scales of the per-neuron features.

Sparse logistic-regression configuration. All 366 successfully loaded `LogisticRegression` objects use `penalty="l1"` and $C = 1.0$. Of these models, 362 use `solver="saga"`. The penalty, regularization strength, and solver are supplied through the classifier command-line arguments, whereas `max_iter=1000`, `tol=1e-3`, `random_state=42`, and `verbose=1` are fixed in the trainer implementations:

```
LogisticRegression(
    penalty=args.penalty,
    C=args.C,
    solver=args.solver,
    max_iter=1000,
    tol=1e-3,
    random_state=42,
    verbose=1,
)
```

Positive-coordinate selection. All 366 loaded models have `classes_ = [0, 1]`, where class 1 is the behavior-positive span class. The coefficient vector $\mathbf{w} = \text{model.coef_}[0]$ therefore encodes the contribution of each feature to the positive-class log-odds. The standard neuron-extraction pipeline selects coordinates as

$$\mathcal{S}_+ = \{j \mid w_j > 0\},$$

i.e. features whose coefficient is strictly positive. An alternative flag (`-all_non_zero`) selects all coordinates with $w_j \neq 0$. No top- k selection, additional threshold, or filtering by coefficient magnitude was found in the audited coordinate-extraction scripts. The intercept is never part of the coordinate set.

Table 26 reports representative coordinate counts. The repetition row corresponds to the RL repetition-up, epoch-1 checkpoint 94 classifier, whereas the sycophancy and hallucination rows correspond to their base-model classifiers.

Three conceptual row-count levels must be distinguished:

1. **QID counts** (n_t, n_f): the number of unique question-response identifiers in each judge category.
2. **Candidate row counts**: the number of feature-matrix rows that would be obtained if every activation file existed. In 3-vs-1 mode,

$$N_+^{\text{cand}} = n_f, \quad N_-^{\text{cand}} = n_t + n_t + n_f = 2n_t + n_f.$$

For the sycophancy ID file this gives $N_+^{\text{cand}} = 3\,334$ and $N_-^{\text{cand}} = 2 \times 3\,344 + 3\,334 = 10\,022$, corresponding to a candidate ratio of approximately 1 : 3.

3. **Realized row counts** (N_+, N_-): the rows that remain after missing-file skipping. These counts cannot be reconstructed because the archived classifier objects do not store the training data, the logs do not report per-group surviving counts, and the corresponding activation directories are not included in the archived artifacts.

X Cross-Model Sample Alignment, CCA, and Directed Probing

Scope and notation. This section documents the cross-model comparison protocol used for the canonical-correlation analysis (CCA) and directed linear-probing analyses. The analysis covers two behavioral domains—repetition (R-Neurons) and sycophancy (S-Neurons)—in both the raw-activation (ACT) and normalized-output-contribution (NOC) spaces.

Let A and B denote two models from the audited model collection. For each model, an ℓ_1 -regularized logistic classifier identifies behavior-associated coordinates using the three-versus-one construction documented in Appendix W. The cross-model analyses then use the domain labels associated with the fixed QID lists to orient and compare the representations formed from the corresponding selected coordinates.

Cross-model sample intersection and row alignment. Within each behavioral domain, the formal sample set is defined by a fixed QID list stored in a JSON file under the keys

⊖ (behavior-positive, label $y = 1$) and ⊕ (behavior-negative, $y = 0$).¹

Every model is required to have an activation file for every QID in the list. If any file is missing the pipeline raises `FileNotFoundError` and terminates. In the fixed-list pipeline the common sample set is therefore the QID list itself, not a per-model-pair intersection.

For the repetition random-neuron baseline, a different alignment rule is used. The pipeline intersects the QIDs whose activation files are present for all 23 models, yielding 2735 common examples: 1367 from the behavior-positive group (key ⊖) and 1368 from the behavior-negative group (key ⊕). This file-presence intersection is constructed once and reused for all baseline model pairs.

In the formal fixed-list pipeline, row order is identical across models because every model reads the same domain-specific JSON file. The order is: all behavior-positive QIDs first (label 1), followed by all behavior-negative QIDs (label 0). Within each class, QIDs retain the order in which they are stored in the JSON array; the loading code does not sort them. In the random-neuron baseline, a single saved common-QID list is reused for all model pairs, with the behavior-positive group preceding the behavior-negative group, ensuring identical row ordering across all baseline model pairs.

Feature preprocessing and dimensionality. For each model, the ACT or NOC matrix $\mathbf{X}^{(m)} \in \mathbb{R}^{n \times p}$ (samples $\times$ behavior-associated coordinates) is mean-centered and projected using PCA, with the implementation-defined maximum component count $\min(p, n - 1)$. Retained dimensions are selected by a cumulative explained-variance threshold $\tau = 0.90$:

$$k_m = \min \left\{ k : \sum_{j=1}^k \lambda_j^{(m)} \geq \tau \sum_j \lambda_j^{(m)} \right\},$$

where $\lambda_j^{(m)}$ are the PCA eigenvalues. Let $\boldsymbol{\mu}^{(m)}$ be the empirical mean and $\mathbf{V}^{(m)}$ the PCA rotation matrix; the retained projections are

$$\mathbf{Z}^{(m)} = (\mathbf{X}^{(m)} - \boldsymbol{\mu}^{(m)}) \mathbf{V}_{:,1:k_m}^{(m)}.$$

CCA fitting and numerical configuration. Canonical-correlation analysis is performed for each model pair (A, B) using scikit-learn’s CCA with `n_components=q`, where $q = \min(k_A, k_B)$. Except for `n_components`, all CCA options—including `tol`, `max_iter`, and feature-scaling behavior—follow the defaults of the installed scikit-learn version, which is not pinned in the project dependencies. **No explicit covariance regularization is applied.**

Canonical rank. The number of canonical dimensions is

$$q = \min(k_A, k_B),$$

¹Different QID sets reflect the different data sources used for each behavioral domain; no single sample set is shared across domains.

where k_A and k_B are the variance-threshold dimensions defined above. The code applies no explicit additional sample-size, effective-rank, or numerical-rank check beyond the PCA dimensionality and the $q = \min(k_A, k_B)$ rule.

Sign orientation. Canonical variates are sign-oriented using the class labels so that on every dimension the positive-class conditional mean is not less than the negative-class conditional mean. Let $\mathbf{U}, \mathbf{V}$ be the canonical variates for models A and B . For each dimension j :

$$s_j^A = \begin{cases} +1, & \bar{u}_j^+ > \bar{u}_j^-, \\ -1, & \bar{u}_j^+ \leq \bar{u}_j^-, \end{cases} \quad \mathbf{U}_{:,j} \leftarrow s_j^A \mathbf{U}_{:,j},$$

with an analogous definition for s_j^B and $\mathbf{V}_{:,j}$, where $\bar{u}_j^+ = \frac{1}{n^+} \sum_{i:y_i=1} U_{i,j}$ and $\bar{u}_j^-$ is the corresponding mean over examples with $y = 0$. This orientation is applied to the canonical variates after CCA fitting. The audited implementation does not record how often the two class means are exactly equal on any canonical dimension.

Canonical-correlation computation and aggregation. After sign orientation, canonical correlations are the signed Pearson correlation coefficients between corresponding columns of $\mathbf{U}$ and $\mathbf{V}$:

$$\rho_j = \text{corr}(\mathbf{U}_{:,j}, \mathbf{V}_{:,j}), \quad j = 1, \dots, q.$$

The reported *primary* correlation is ρ_1 (the first canonical correlation). The *mean* CCA is the arithmetic average

$$\text{MeanCCA}(A, B) = \frac{1}{q} \sum_{j=1}^q \rho_j.$$

All dimensions are equally weighted; there is no explained-variance weighting. In the formal implementation the correlations are stored and reported as signed values (after sign orientation). In the random-neuron baseline, each canonical correlation is replaced by its absolute value, $\rho_j \leftarrow |\rho_j|$, before reporting the primary correlation and computing the mean correlation.

Handling of constant, failed, and non-finite dimensions. The implementation performs no per-dimension finite-value or zero-variance check. A non-finite Pearson correlation (e.g. NaN returned when a standard deviation is zero) is retained by the scoring routine and can propagate through the ordinary arithmetic mean; the code does not use `nanmean`. Exceptions raised during a model-pair computation are caught by the outer pair-level handler, which records a NaN score for that pair and continues to the next pair. The audited materials do not establish explicit handling or retrying of CCA convergence warnings.

Directed probing from source model to target model. Directed (asymmetric) linear probing estimates how well the PCA subspace of a source model A can predict the PCA subspace of a target model B :

$$\hat{\mathbf{Z}}^{(B)} = \mathbf{Z}^{(A)} \mathbf{W}_{A \rightarrow B} + \mathbf{b},$$

where $\mathbf{W}$ and $\mathbf{b}$ are fitted by ordinary least squares (scikit-learn’s `LinearRegression` with `fit_intercept=True`, no regularization). The score for a given target dimension j is the signed Pearson correlation between $\hat{\mathbf{z}}_{:,j}^{(B)}$ and $\mathbf{z}_{:,j}^{(B)}$.

Two probing modes are reported:

k-dim probing: All k_A source dimensions predict all k_B target dimensions jointly (multi-output regression). The reported score is the mean of per-dimension Pearson correlations: $\frac{1}{k_B} \sum_{j=1}^{k_B} \rho_j$.

Top-two-PC probing: The method evaluates all available one-dimensional mappings among the first two retained PCs of the source and target models. The exact mapping and score-selection rule are defined below.

Top-two-PC mapping selection. Let $r_A = \min(2, k_A)$, $r_B = \min(2, k_B)$, and let $\mathbf{z}_i^A \in \mathbb{R}^n$ be the i -th PC score vector of model A ($1 \leq i \leq r_A$), with $\mathbf{z}_j^B$ defined analogously for model B . The $r_A r_B$ one-dimensional linear mappings are

$$\hat{\mathbf{z}}_{j|i}^B = f_{i \rightarrow j}(\mathbf{z}_i^A), \quad 1 \leq i \leq r_A, \quad 1 \leq j \leq r_B.$$

The reported score is the maximum Pearson correlation among the candidate axis pairings:

$$S_{\text{top2pc}}(A \rightarrow B) = \max_{\substack{1 \leq i \leq r_A \\ 1 \leq j \leq r_B}} \text{corr}(\hat{\mathbf{z}}_{j|i}^B, \mathbf{z}_j^B).$$

This quantity summarizes the strongest linear association among the available top-two-PC axis pairings.

Regression intercept and regularization. All probing regressions use scikit-learn’s `LinearRegression` with default settings:

- `fit_intercept=True` (the data are not explicitly centered before regression; the PCA projections are already mean-centered, so the intercept absorbs any residual offset);
- No ℓ_1 or ℓ_2 regularization (ordinary least squares).

No hyperparameter search is performed.

Directionality and interpretation of the reported scores. The probing matrix is indexed by (source model, target model). Entry (i, j) represents $\text{model}_i \rightarrow \text{model}_j$: the PCA projections of model i are used as predictors for the projections of model j . The mapping is asymmetric: $A \rightarrow B$ and $B \rightarrow A$ are fitted and reported separately. The row label of the heatmap corresponds to the predictor (source), the column label to the target.

Y Controlled Post-Training Configurations

Scope and reporting convention. This appendix documents the complete controlled post-training protocol for the two behavioral domains reported in the main paper: repetition and sycophancy. All training runs use the LLaMA-Factory framework built on the Hugging Face Trainer and, for DPO, the TRL library. We adopt the following conventions

throughout. Effective batch size is defined as the product of per-device training batch size, number of devices, and gradient accumulation steps. For DPO runs, the reported example count refers to preference pairs. Each multi-stage model (SFT $\rightarrow$ DPO) is decomposed into its constituent stages with the initialization checkpoint of each stage explicitly stated. The reported values were recovered by cross-checking the archived YAML configurations, shell launch scripts, saved training arguments, trainer states, logs, model cards, and downstream checkpoint paths whenever those sources were available.

Repetition post-training configurations. Table 27 reports the training configurations for the six repetition model variants used in the main paper. All repetition SFT stages initialize from the base `meta-llama/Llama-3.1-8B` checkpoint. The SFT-up model is trained on repetitive responses (755 844 examples derived from paragraph-level and token-level repetition data) while the SFT-down model is trained on non-repetitive responses (`sft_normal_data`, 755 844 examples rewritten by `Llama-3.3-70B-Instruct`). Both SFT runs use full-parameter fine-tuning with AdamW ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 1 \times 10^{-8}$), a learning rate of 1×10^{-5} , cosine scheduling with 10% warmup ratio, fp16 precision, and DeepSpeed ZeRO-3 without offload. Each SFT run defines a maximum step budget that overrides the configured `num_train_epochs=3`, so the effective epoch count is `max_steps` $\times$ `effective_batch_size` / `num_examples`, which equals ~ 0.017 epochs for SFT-up (100 steps) and ~ 0.102 epochs for SFT-down (600 steps).

The DPO stages use a lower learning rate of 5×10^{-7} , per-device batch size of 1 (one preference pair), gradient accumulation of 16, and a maximum sequence length of 1 024 tokens. The DPO preference strength parameter is $\beta = 0.1$. The archived run names use “RL” for the DPO stage; throughout this appendix we refer to the training objective as DPO. When DPO is applied directly to the base model (DPO-up and DPO-down from base), both the policy model and the reference model initialize from the base checkpoint; the reference model is a frozen copy of the policy at initialization (implicit behavior of the `TRL DPOTrainer`). When DPO is applied after SFT (SFT $\rightarrow$ DPO-up and SFT $\rightarrow$ DPO-down), the policy model initializes from the final SFT checkpoint (`checkpoint-100` for the up direction and `checkpoint-600` for the down direction), and the reference model is initialized as a frozen copy of that same SFT checkpoint. All DPO runs complete 60 optimization steps, corresponding to an effective epoch count of ~ 0.010 ($60 \times 128 / 755\,920$).

Sycophancy post-training configurations. Table 28 reports the training configurations for the six sycophancy model variants. The experimental design mirrors the repetition protocol but uses substantially smaller datasets (3 480 SFT examples and 3 480 DPO preference pairs, sourced from the Anti-Sycophancy-DPO dataset). The shared settings include full-parameter fine-tuning, the AdamW optimizer, cosine scheduling, a 0.1 warmup ra-

tio, fp16 precision, and DeepSpeed ZeRO-3. The SFT-up model is trained on sycophantic responses (*sycoph_sft*) and the SFT-down model on truthful responses (*truth_sft*). SFT-up runs for 120 steps (~ 4.4 effective epochs) and SFT-down for 360 steps (~ 13.2 effective epochs). The DPO stages follow the same two initialization schemes as in the repetition experiments: DPO initialized from the base model (DPO-up and DPO-down from base) and DPO initialized from the SFT checkpoint (SFT $\rightarrow$ DPO-up and SFT $\rightarrow$ DPO-down). All sycophancy DPO runs use $\beta = 0.1$ except SFT $\rightarrow$ DPO-down (suppress sycophancy), which uses $\beta = 0.7$ as recorded in the training configuration. The smaller dataset size means that 60 DPO steps at an effective batch size of 128 pairs correspond to ~ 2.2 effective epochs.

Dataset and batch-size accounting. The constructed repetition SFT corpora contain 755 920 records, while the archived SFT trainer reports 755 844 examples used for optimization. The repetition DPO dataset comprises 755 920 preference pairs, constructed by one-to-one matching of repetitive and non-repetitive responses on shared instruction keys. The repetition DPO dataset comprises 755 920 preference pairs, constructed by one-to-one matching of repetitive and non-repetitive responses on shared instruction keys. The sycophancy SFT dataset contains 3 480 examples and the DPO dataset contains 3 480 pairs, both derived from the DataCreatorAI/Anti-Sycophancy-DPO source. All DPO runs use an effective batch size of 128 pairs per optimizer update, calculated as $1 \text{ (per-device)} \times 8 \text{ (devices)} \times 16 \text{ (accumulation)} = 128$. For SFT, the accumulated batch per process is $2 \times 8 = 16$ examples, and for DPO it is $1 \times 16 = 16$ preference pairs. With eight distributed processes, the global effective batch size is therefore 128 examples or preference pairs per optimizer update. We report this global value, consistent with the Hugging Face Trainer `total_train_batch_size` convention.

SFT-to-DPO initialization and reference models. No explicit `ref_model` path is provided in the archived DPO configurations. Under the audited LLaMA-Factory/TRL execution path, the reference model is instantiated as a frozen copy of the policy at DPO initialization. In the SFT $\rightarrow$ DPO sequential protocol, the DPO policy model is initialized from the final SFT checkpoint, and the reference model is initialized as a frozen copy of that same checkpoint. In the direct DPO protocol, both the policy model and the reference model are initialized from the base Llama-3.1-8B checkpoint. None of the archived DPO configurations enables reference-free mode.

Distributed training and numerical precision. The archived launch configurations specify one node with eight processes, launched through `torchrun` with `FORCE_TORCHRUN=1` and `NPROC_PER_NODE=8`. The accompanying shell-script records identify the devices as NVIDIA A100 40GB GPUs, although no historical system-level hardware log was available for independent verification. DeepSpeed ZeRO-3 (stage 3 with `stage3_gather_16bit_weights_on_model_save=True`) is employed throughout, with the optimizer, gradient clip-

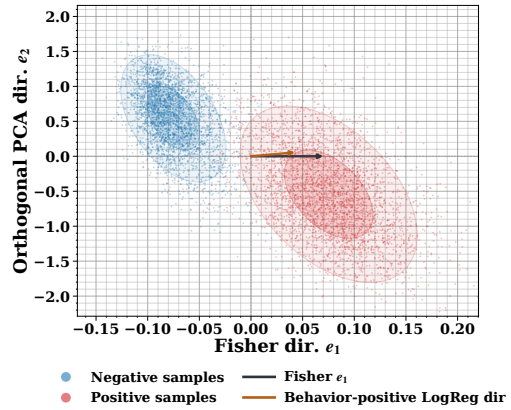


Figure 41: Visualization of a local behavioral chart. The Fisher and logistic directions align with the dominant separation axis, supporting the interpretation that $U_{\theta,b}$ captures behavior-relevant geometry.

ping, and batch sizes delegated to the Hugging Face Trainer via "auto" fields in the DeepSpeed JSON. The optimizer is AdamW (`adamw_torch`) with betas (0.9, 0.999), $\epsilon = 1 \times 10^{-8}$, and weight decay of 0.0. All runs train in fp16 mixed precision with bf16 disabled. TF32 is not explicitly configured in the archived YAML files or saved training arguments.

Checkpoint selection and downstream use. Checkpoints are saved every N steps where $N = 10$ (SFT-up rep), $N = 60$ (SFT-down rep), $N = 12$ (SFT-up syco), $N = 36$ (SFT-down syco), and $N = 6$ (all DPO runs), with a retention limit of 10 checkpoints per run. The primary endpoint results reported in the main paper use the final checkpoint of each training run (the checkpoint with the largest `global_step`). The stage-resolved geometry-occupancy-gain analyses additionally use the saved intermediate checkpoints, as documented in Appendix S. SFT repetition-up uses `checkpoint-100`; SFT repetition-down uses `checkpoint-600`; all repetition DPO and SFT $\rightarrow$ DPO runs use `checkpoint-60`; SFT sycophancy-up uses `checkpoint-120`; SFT sycophancy-down uses `checkpoint-360`; all sycophancy DPO and SFT $\rightarrow$ DPO runs use `checkpoint-60`.

Y.1 Joint interpretation of ACT and NOC.

Although ACT and NOC are computed from the same behavior-associated coordinate scaffold, they answer different questions and need not vary together. ACT characterizes how strongly or frequently the model occupies the corresponding runtime state, whereas NOC characterizes the strength of the functionally supported pathway associated with that state. Figure 2 summarizes the four regimes induced by their combination. High NOC with high ACT indicates a strong mechanism that is frequently used at runtime. High NOC with low ACT indicates a strong but latent delegated pathway that the mechanism remains available, but the model’s ordinary operating distribution rarely enters it, as

may occur when post-training shifts the operating point away from an undesirable mode. Low NOC with high ACT indicates that the observed runtime states are correlated with the behavior but receive limited functional support, and therefore may not constitute a core computational pathway. Low NOC with low ACT indicates that the model both weakly supports and rarely occupies the corresponding behavioral mode. This distinction motivates treating functional support and runtime occupancy as separate axes throughout the subsequent analysis. Having separated functional-channel strength from runtime-state occupancy at the conceptual level, we next identify a sparse coordinate scaffold in which these behavioral geometries can be estimated reliably.

Y.2 Token Windows and State Extraction

Behavioral evidence is usually localized to a subset of positions. We therefore define a task-specified extraction operator $\mathcal{T}_b : \mathcal{X}_b \rightarrow 2^{\mathbb{N}}$, which maps each instance x to a set of behavior-relevant token positions. If T_x is the number of processed tokens, we write $M_x = \mathcal{T}_b(x) \subseteq \{1, \dots, T_x\}$. The operator $\mathcal{T}_b$ is fixed before fitting any representation model or probe. It may select a full response span, a decision-critical span, an event-centered window, or any other pre-specified region where the behavior is expected to be expressed. Let $a_{\ell,t,j}(x)$ denote the activation of coordinate j at layer ℓ and token position t under the chosen hook convention. We aggregate activations over the selected token window: $\bar{a}_{\ell,j}(x) = \frac{1}{|M_x|} \sum_{t \in M_x} a_{\ell,t,j}(x)$. The same extraction and aggregation rule is applied to behavior-positive and behavior-negative instances. This prevents trivial separability caused by inconsistent token positions, window lengths, or prompt formats. When layer-resolved analysis is required, we retain the layer index and define $\phi_{\theta,b}^{(\Omega,\ell)}(x) \in \mathbb{R}^{D_{\theta,b,\Omega,\ell}}$. For the main global representation, we concatenate the selected coordinates across layers $\phi_{\theta,b}^{(\Omega)}(x) = \bigoplus_{\ell=1}^L \phi_{\theta,b}^{(\Omega,\ell)}(x)$. Thus each instance is mapped to a fixed-dimensional representation vector while preserving a clear separation between the token-selection rule, the layer-wise extraction rule, and the downstream geometric analysis.

Z Additional Related Work

Repetition and neural degeneration. Repetition has long been recognized as a central failure mode of neural text generation. Holtzman et al. showed that likelihood-trained models can generate bland and repetitive continuations under maximization-based decoding and introduced nucleus sampling as a remedy (Holtzman et al. 2020). Welleck et al. proposed unlikelihood training to penalize repeated tokens and sequences (Welleck et al. 2020). Fu et al. analyzed repetition theoretically through Markov-style generation dynamics (Fu et al. 2021). Contrastive decoding methods further reduce degeneration by combining likelihood with representation-level discriminability (Su and Collier 2023; Su et al. 2022).

Sycophancy and preference failures. Sycophancy is an alignment failure in which models match the user’s beliefs or preferences instead of giving independent or truthful answers. Sharma et al. showed that human preference data can

favor sycophantic responses and that assistants often agree with user beliefs even when they are false (Sharma et al. 2024). Models can mimic human falsehoods (Lin, Hilton, and Evans 2022), and feedback optimization can improve behavior while inheriting biases of demonstrations, comparisons, and reward models (Ouyang et al. 2022). DPO simplifies preference optimization without an explicit RL loop (Rafailov et al. 2023), but it still optimizes a preference-induced objective.

Table 22: Tokenizer-specific stance-span alignment statistics for all 23 models. Chosen, rejected, and total records are constant across models because every entry was successfully aligned. The “Classifier rows” column reports the number of feature rows used for 3-vs-1 logistic regression training. For all models, the aligned counts are: Chosen = 3 344, Rejected = 3 334, Complete pairs = 3 206, Total records = 6 678.

Model	Classifier rows
textttgoogle/gemma-2-2b	10 708
textttgoogle/gemma-2-2b-it	13 356
textttgoogle/gemma-2-9b	10 708
textttgoogle/gemma-2-9b-it	13 356
textttgoogle/gemma-2-27b	10 708
textttgoogle/gemma-2-27b-it	13 356
textttgoogle/gemma-3-4b-it	13 356
textttmeta-llama/Llama-3.1-8B	10 708
textttmeta-llama/Llama-3.1-8B-Instruct	13 356
textttmeta-llama/Llama-3.3-70B-Instruct	13 356
textttmistralai/Mistral-7B-v0.3	13 356
textttmistralai/Mistral-7B-Instruct-v0.3	13 356
textttmistralai/Mistral-Nemo-Base-2407	10 708
textttmistralai/Mistral-Nemo-Instruct-2407	13 356
textttmistralai/Mistral-Small-3.1-24B-Base-2503	10 708
textttmistralai/Mistral-Small-3.1-24B-Instruct-2503	10 708
textttQwen/Qwen2.5-0.5B	10 708

Condition	SFT checkpoint	Reward checkpoint	SFT→Reward checkpoint
Repetition-up	10s	6s	6s
Repetition-down	60s	6s	6s
Sycophancy-up	12s	6s	6s
Sycophancy-down	36s	6s	6s

Table 23: Checkpoint selected at matched stage $s \in \{1, \dots, 10\}$. The stage index aligns relative training progress rather than raw optimizer steps.

Table 24: Post-training dataset configurations. All four datasets contain 755 920 records derived from the surviving instruction set. SFT-up and SFT-down are stored as separate single-response datasets, whereas RL-up and RL-down contain explicit chosen–rejected pairs.

Condition	Target / chosen	Rejected	Direction
SFT-up	y^{rep}	—	Increase repetition
SFT-down	y^{repair}	—	Decrease repetition
RL-up	y^{rep}	y^{repair}	Prefer repetition
RL-down	y^{repair}	y^{rep}	Prefer non-repetition

Table 25: Stage-wise counts for repetition repair and post-training data construction. Retention is reported only for sequential filtering stages; dashes denote parallel dataset-construction outputs.

Stage	Records	Retention
Initial merged pool	743 687	—
After appending model generations	766 869	—
Saved repaired JSONL records	766 469	99.95%
After length filtering and deduplication	755 920	98.62%
Aligned repetitive responses	755 920	—
RL-up preference records	755 920	—
RL-down preference records	755 920	—

Table 26: Selected coordinate counts for representative archived classifiers. All models use $C = 1.0$ and an L_1 penalty.

Behavior	Solver	$n_w > 0$	$n_w < 0$	$n_w \neq 0$
Repetition	saga	25	9	34
Sycophancy (base)	saga	92	57	149
Hallucination (base)	saga	20	32	52
SAE-Repetition	liblinear	218	556	774

Table 27: Controlled post-training configuration for repetition behavioral interventions.

Configuration	SFT (Up / Down)	DPO from Base	SFT → DPO
Base checkpoint		meta-llama/Llama-3.1-8B	
Initialization	Base	Base	SFT checkpoint (step 100 or 600)
Objective	SFT LM (induce / reduce)	DPO (induce / reduce)	DPO (induce / reduce)
Dataset (Up)	sft_rep_data	rl_induce_rep	rl_induce_rep
Dataset (Down)	sft_normal_data	rl_cure_rep	rl_cure_rep
Examples (Up)	755 844	755 920 pairs	755 920 pairs
Examples (Down)	755 844	755 920 pairs	755 920 pairs
Parameter update	Full-parameter fine-tuning (no LoRA or PEFT adapter)		
Optimizer	AdamW (torch)	AdamW (torch)	AdamW (torch)
Learning rate	1×10^{-5}	5×10^{-7}	5×10^{-7}
Per-device batch size	2	1 pair	1 pair
Gradient accumulation	8	16	16
Effective batch size	128	128 pairs	128 pairs
Maximum steps	100 (Up) / 600 (Down)	60	60
Warmup ratio	0.1	0.1	0.1
Scheduler	Cosine	Cosine	Cosine
Max. sequence length	2 048	1 024	1 024
Precision	fp16	fp16	fp16
DPO β	Not applicable	0.1	0.1
Reference model	Not applicable	Implicit (frozen base)	Implicit (frozen SFT)
DeepSpeed	ZeRO-3	ZeRO-3	ZeRO-3
Checkpoint interval	10 (Up) / 60 (Down) steps	6 steps	6 steps
Random seed	42	42	42
Hardware	8 GPUs; A100 40GB documented in archived launch-script comments		

Table 28: Controlled post-training configuration for sycophancy behavioral interventions.

Configuration	SFT (Up / Down)	DPO from Base	SFT → DPO
Base checkpoint		meta-llama/Llama-3.1-8B	
Initialization	Base	Base	SFT checkpoint (step 120 or 360)
Objective	SFT LM (induce / reduce)	DPO (induce / reduce)	DPO (induce / reduce)
Dataset (Up)	syco_sft	rl_induce_syco	rl_induce_syco
Dataset (Down)	truth_sft	rl_cure_syco	rl_cure_syco
Examples (Up)	3 480	3 480 pairs	3 480 pairs
Examples (Down)	3 480	3 480 pairs	3 480 pairs
Parameter update	Full-parameter fine-tuning (no LoRA or PEFT adapter)		
Optimizer	AdamW (torch)	AdamW (torch)	AdamW (torch)
Learning rate	1×10^{-5}	5×10^{-7}	5×10^{-7}
Per-device batch size	2	1 pair	1 pair
Gradient accumulation	8	16	16
Effective batch size	128	128 pairs	128 pairs
Maximum steps	120 (Up) / 360 (Down)	60	60
Warmup ratio	0.1	0.1	0.1
Scheduler	Cosine	Cosine	Cosine
Max. sequence length	2 048	1 024	1 024
Precision	fp16	fp16	fp16
DPO β	Not applicable	0.1	0.1 (Up) / 0.7 (Down)
Reference model	Not applicable	Implicit (frozen base)	Implicit (frozen SFT)
DeepSpeed	ZeRO-3	ZeRO-3	ZeRO-3
Checkpoint interval	12 (Up) / 36 (Down) steps	6 steps	6 steps
Random seed	42	42	42
Hardware	8 GPUs; A100 40GB documented in archived launch-script comments		